

Global Considerations in Kidney Disease: Near and Middle East

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CHAPTER OUTLINE

ACUTE KIDNEY INJURY, 2245
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MANAGEMENT OF END-STAGE KIDNEY
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KEY POINTS

- Health care disparities do exist among countries of the Middle East (ME).
- Global climate warming and recent disasters in ME countries mandate comprehensive strategies to upgrade health care and logistic systems to minimize casualties, reduce adverse consequences, and optimize the prevention, detection, and treatment of kidney complications.
- Accurate and detailed epidemiologic data regarding acute and chronic kidney diseases (CKDs) in ME countries are still needed.
- Community-based screening programs for CKD, risk factors, and intervention strategies are limited in most countries of the region.
- Diabetes mellitus is the major cause of end-stage kidney disease in most ME countries.
- Recessively inherited kidney diseases are prevalent in the ME, and targeted screening in populations at risk should be encouraged.
- Renal replacement therapies are available in all ME countries, and hemodialysis is mostly used.
- Hepatitis C infection is prevalent among dialysis patients in the ME. Implementation of infection control measures and introduction of new antiviral agents are of immense importance to eradicate it and prevent reinfection.
- Legal kidney transplantation from both living and deceased donors should be expanded, and organ donation encouraged, in view of the positive ethical and religious authorizations.

This chapter is a discussion of the epidemiology, causes, predisposing factors, management, and prevention of kidney diseases. Future strategies for dealing with kidney diseases in the Near and Middle East are proposed. The term *Near and Middle East* is a historical, Eurocentric, and Western term that was used to describe a geographic region whose boundary is imprecise and whose internal borders are constantly changing because of political and historical evolution.¹ Therefore the Near and Middle East, hereafter called the Middle East (ME), is defined for the purpose of this chapter as the region that encompasses the following 19 countries and 1 governing body (in alphabetical order): Algeria, Bahrain, Egypt, Iran, Iraq, Israel, Jordan, Kuwait, Lebanon, Libya, Morocco, Oman, Palestinian Authority (PA), Qatar, Saudi Arabia, Syria, Tunisia, Turkey, the United Arab Emirates (UAE), and Yemen (Fig. 76.1).

The ME has always been important for the following reasons: 1. its strategic location as a tricontinental hub that

links Asia, Africa, and Europe; 2. its economic resources; and 3. its spiritual significance as the birthplace of the world's three major monotheistic religions—Judaism, Christianity, and Islam. Of the adherents to the three religions, Muslims constitute the largest religious population in the ME; the sizes of the Christian and Jewish populations are smaller and vary from country to country.¹ Although the ME has several cultural, linguistic, and geographic associations, considerable disparity exists among the different countries in terms of economy, resources, political systems, health care systems, health expenditure, and disease incidence and prevalence.²

According to data from the World Bank, the estimated population in the ME countries in 2022 was 576,961,848 (Table 76.1).³ The gross national income (GNI) per capita varies across ME countries. According to the classification system of the World Bank,^{3,4} most ME countries are considered “developing countries” and only seven countries



Fig. 76.1 Map of Middle Eastern countries. Countries illustrated in *light green* are also known as “El Maghreb”; those in *dark green*, as “El Mashreq.” (Modified from a public domain image available at http://en.wikipedia.org/wiki/Middle_East) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

are considered high-income countries (see [Table 76.1](#)). Inevitably, disease incidence, prevalence, course, and outcomes are affected by the different socioeconomic factors and health policies in each country.⁵⁻⁷ For example, the infant mortality rate in each ME country is concordant with the socioeconomic, political, and health status of that country,⁸ and life expectancy for men is lowest in the low-income ME countries and in countries affected by man-made conflicts (see [Table 76.1](#)).⁶

In general, the infant mortality rate declined in all ME countries during the past decade. However, in Yemen, the poorest country in the ME, it is still at least five times higher (47/1000 live births) than that in industrialized, high-income ME countries (3–9/1000 live births; see [Table 76.1](#)).^{4,8} Health disparities also exist among minority populations (ethnic groups or expatriates) who live in industrialized ME countries, as well as among individuals of ME origin who live in countries that are not in the ME.⁹⁻²⁰ Unfortunately, about a third of ME countries have been affected by various types of natural disasters (e.g., earthquakes, floods, and droughts) and other disasters (e.g., military conflicts). Casualties, displacement, and migration are significant consequences

of such disasters and adversely affect the socioeconomic strata and health status of a country, as in the recent devastating humanitarian crises in Morocco, Libya, Turkey, Syria, Yemen, PA, and Israel.^{2,21-30} Many ME countries and their communities have insufficient number of trained and experienced local disaster coordinators. Therefore they need assistance to upgrade their existing health care structures to improve their ability to cope with any future disasters (more discussed in [Chapter 82](#)).^{2,21,23,25-35}

ACUTE KIDNEY INJURY

As in many countries worldwide, data on the incidence and prevalence of acute kidney injury (AKI) in many ME countries are scarce and imprecise because of an inconsistent definition of AKI in medical reports, underreporting, seasonal dependency on the occurrence of AKI, and the frequency and location of natural and human-induced disasters.³⁶⁻³⁹ There are large knowledge gaps about the age, number, natural history, and outcome of patients with AKI in both the community and hospitals and about the use of

Table 76.1 Demographics, Health Indicators, Human Development Indices, Wealth and Health Care Expenditures in Middle Eastern and Western Industrialized Countries

Country	Total Population, 2022 ^a	Life Expectancy at Birth (Years), 2021 ^b		Rate of Infant Mortality (per 1000 Live Births, 2021) ^b	Country Ranking in Human Development Index (of 191 Countries, 2021) ^c	World Bank Classification, 2022 ^{a,d}	Total Expenditure on Health (% of GDP, 2021) ^b	Total Expenditure on Health per Capita (current US\$, 2021) ^e
		Birth (Years), 2021 ^b						
		Male	Female					
Middle Eastern Countries								
Algeria	44,903,225	75	78	19	91	Lower-middle income	5.53	204
Bahrain	1,472,233	78	80	6	35	High income	4.27	1146
Egypt	110,990,103	68	73	16	97	Lower-middle income	4.61	179
Iran	88,550,570	71	77	11	76	Lower-middle income	5.77	392
Iraq	44,496,122	68	72	21	121	Upper-middle income	5.25	249
Israel	9,550,600	81	85	3	22	High income	7.9	4339
Jordan	11,285,869	72	77	13	102	Lower-middle income	7.29	299
Kuwait	4,268,873	29.7	81	8	50	High income	5.78	1860
Lebanon	5,489,739	33.7	77	7	112	Lower-middle income	10.06	307
Libya	6,812,341	25.8	74	9	104	Upper-middle income	4.02*	313*
Morocco	37,457,971	72	76	15	123	Lower-middle income	5.74	221
Oman	4,576,298	71	75	9	54	High income	4.37	853
Palestinian Authority	5,043,612	71	76	13	106	Upper-middle income	NA	NA
Qatar	2,695,122	33.7	81	5	42	High income	2.89	1934
Saudi Arabia	36,408,820	20.7	76	6	35	High income	5.97	1442
Syria	22,125,249	23.5	69	18	150	Low income	3.05#	63#
Tunisia	12,356,117	32.7	77	14	97	Lower-middle income	6.97	265
Turkey	85,341,241	32.2	73	8	48	Upper-middle income	4.57	441
United Arab Emirates	9,441,129	38.4	77	5	26	High income	5.31	2,352
Yemen	33,696,614	19.8	67	47	183	Low income	4.24**	63**
Western Industrialized Countries								
Australia	25,978,935	81	85	3	5	High income	10.54	7055
Canada	38,929,902	41.8	85	4	15	High income	12.33	6207
France	67,935,660	41.7	79	3	28	High income	12.31	5381
Germany	84,079,811	47.8	79	3	9	High income	12.93	6191
Italy	58,856,847	46.5	81	2	30	High income	9.38	3066
Japan	125,124,989	48.6	81	88	19	High income	10.82	4347
United Kingdom	66,971,411	40.6	79	83	18	High income	12.36	5138
United States	333,287,557	38.5	74	79	21	High income	17.36	12,473

^aData from World Bank: Indicators. <https://data.worldbank.org/indicator>.^bSource data for country profiles from the World Health Organization: Countries <https://data.who.int/countries>^cData from the 2021-22 Human Development Report, published for the United Nations Development Programme: https://hdr.undp.org/system/files/documents/global-report-document/hdr2021-22pdf_1.pdf.^dWorld Bank country classification according to the gross national income (GNI) per capita per annum, 2022: low income—GNI <\$1135 (USD), lower-middle income—between \$1136 and \$4465, upper-middle income—between \$4466 and \$13,845, high income—more than \$13,846.^e<https://data.worldbank.org/indicator/SH.XPD.CHEX.PC.CD>

Indices from *2011, **2015, and #2012.

GDP, Gross domestic product; NA, not available.

preventive measures in each ME country. Most published reports on AKI in ME countries are short-term studies done in tertiary-level hospitals or single-center studies, and of these studies, only a few were of the prospective type. Moreover, the definition of AKI is not consistent in these reports.⁴⁰⁻⁴⁶ In a case series study of the ME respiratory syndrome (MERS) coronavirus outbreak, AKI requiring kidney replacement therapy (KRT) was reported in 58% of critically ill patients.⁴⁷ This percentage is high compared with that reported in equivalent critically ill patients who were treated during the severe acute respiratory syndrome (SARS) epidemic in Canada (5%).⁴⁸ The Acute Kidney Injury–Epidemiologic Prospective Investigation (AKI-EPI) study was the first multinational and cross-sectional study to investigate the epidemiology of AKI in intensive care unit (ICU) patients worldwide (33 countries) prospectively using the Kidney Disease: Improving Global Outcomes (KDIGO) definition for AKI⁴⁹ and a standardized data collection instrument.³⁸ AKI was found in 57.3% of subjects. Yet only 5 of 97 participating centers were from the ME—3 ICUs from Egypt, 1 from Tunisia, and 1 from Turkey. These are small numbers from which to draw firm conclusions regarding the whole ME region. The Epidemiology of Surgery-associated Acute Kidney Injury (EPIS-AKI) trial, which prospectively evaluated the epidemiology of AKI after major surgery, was designed by investigators of the Department for Anesthesiology, Intensive Care and Pain Medicine at the University Hospital Münster. The study was international with strong representation of middle and low-income countries.⁵⁰ Five medical centers from Algeria, 6 from Egypt, 4 from Iraq, 1 from Jordan, 7 from Libya, 2 from PA, 1 from KSA, and 30 from Turkey were involved, among others, in the EPIS-AKI study. The primary outcome—the incidence of AKI within the first 72 hours after surgery—was observed in 18.4% of patients. Therefore postoperative AKI may represent a significant burden for health care in the ME countries.⁵⁰

CAUSES

The causes of AKI have changed over time in ME countries. A nationwide prospective study, conducted by the Turkish Society of Pediatric Nephrology Acute Kidney Injury Study Group,⁵¹ demonstrated that the cause of AKI in Turkish pediatric patients has greatly changed over the past 2 decades. Acute gastroenteritis and acute poststreptococcal glomerulonephritis have decreased significantly as causes of AKI, and prematurity, malignancy, and congenital heart disease have increased. In adults, obstructive uropathy, unspecified postsurgical complications, and crush injury were the most prevalent causes of AKI in Syria during the 1980s, according to Hadidy and colleagues.⁴⁰ In contrast, Said has reported findings similar to those from industrialized countries in 215 patients with AKI, aged between 12 and 90 years, in 3 Jordanian hospitals over an 18-month study period.⁵² Renal parenchymal disease was the most common cause of AKI (58%), and acute tubular necrosis (ATN) and contrast-induced nephropathy were thought to be the two most prevalent causes of renal parenchymal disease. Prerenal causes (28%) and postrenal causes (14%) accounted for the remaining causes of AKI.⁵² Of note, obstructive uropathy was a common cause of AKI

as a result of the high prevalence of nephrolithiasis in the study patients who originally came from Yemen and Sudan. Interestingly, a case series study from Israel reported severe AKI, accompanied by nausea, vomiting, abdominal pain, and scalp rash, shortly after chemical hair-straightening procedures.⁵³ Kidney biopsies performed in 7 out of 26 patients showed tubular toxicity, associated with oxalate-induced nephropathy and interstitial inflammation, implying kidney toxicity from systemic absorption of topical products containing glycolic acid derivatives.

In another study from one center in Qatar, ATN was reported in 83% of all patients with AKI, who were referred mainly from ICUs.⁵⁴ Al-Homrany⁴² has reported that ATN resulting from sepsis, ischemia, and rhabdomyolysis, as well as distinctive causes, such as malaria and snakebites (4.6% of all cases), were the major causes of AKI in tropical southern Saudi Arabia.

Malaria is rarely reported as a primary cause of AKI in most ME countries, with the exception of Yemen.⁵⁵ In the Hajjah and Sanaa regions of Yemen, malaria caused by *Plasmodium falciparum* was the cause of 29% of all cases of AKI. However, prerenal disorders, such as infectious diarrheal diseases, are still the predominant cause of AKI in Yemen because of the tropical climate and poor hygiene.⁵⁵ Malarial kidney injury is often a consequence of several hemodynamic, immune, and metabolic disturbances, which may also be accompanied by central nervous system sequelae and fluid and electrolyte alterations.⁵⁶⁻⁵⁸ Malarial kidney disease can manifest as AKI in the form of the following: 1. ATN that accompanies or occurs as a complication of severe hemolysis, hemodynamic derangements, and tissue hypoxemia; 2. interstitial nephritis; or 3. glomerular mesangial proliferative lesions with immune complex deposits.⁵⁶ In areas where malaria is not common, a high index of suspicion is required.

Reporting the cause of AKI may, however, be biased by the type of referral hospital that documents the various causes. In a retrospective 18-month study from one cancer hospital in the UAE, sepsis and drug-induced nephrotoxicity were the leading causes of AKI because 30% of the patients were immunocompromised and were receiving chemotherapy.⁵⁹ Preexisting comorbid conditions, such as diabetes, hypertension, and chronic kidney disease (CKD), were documented in approximately one-third of all patients with AKI in the UAE and Jordan^{41,46,59} and in as many as 87% of all such patients in Qatar.⁵⁴

AKI-associated mortality rates range between 12% and 77% in ME countries.^{40-43,46,52,54,59-61} The International Society of Nephrology (ISN) Oby25 AKI initiative aims to prevent all avoidable deaths from AKI by 2025, especially in low- and middle-income countries.⁶² To implement such an initiative, a clear definition of preventable death from AKI and a strategy to regionalize any recommendations for AKI care in terms of health care infrastructure, socioeconomic conditions, and education and training were considered.^{62,63} The first global 10-week snapshot in 2014 included AKI data from 289 sites across 72 countries, mostly academic and referral centers, of which 32 centers represented ME countries—13 centers in Egypt, 4 in Iran, 3 in Saudi Arabia, and 3 in UAE, 2 each in Morocco and Tunisia, and 1 center per country including Israel, Kuwait, Oman, and Syria and the PA.⁶¹ In this study, the median age of patients from

ME countries was 60 years, with a male predominance. Two-thirds of AKI cases were community acquired, and as expected, mortality increased in hospital-acquired AKI and in patients requiring the ICU or dialysis. However, many areas of the ME were underrepresented in this study, and data from Egypt and Tunisia were combined with those from Africa; those from Israel were collected from Western Europe.

Devastating earthquakes have struck some ME countries and are a constant threat because their territory encompasses the Great African Rift Valley. Clinician investigators have described and analyzed the factors that have had major implications for kidney involvement and outcomes in survivors who sustained crush syndrome in catastrophic earthquakes in Turkey and Iran.^{64,65} Crush syndrome often causes profound hypovolemic shock that 1. is complicated and aggravated by gross disorders of acid-base balance and electrolytes, of which hyperkalemia is life-threatening, and 2. increases susceptibility to myoglobinuric AKI. These complications can occur within hours of the initial injury and can lead to early loss of limb or life.^{66,67} This topic is discussed in depth in [Chapter 82](#).

In summary, it is still difficult to draw general conclusions about the epidemiology, causes, and outcomes of AKI in many ME countries because of regional variations and methodologic differences in the few studies that have been performed in the region. Well-conducted regional studies in which the published definition of AKI⁴⁹ is used are needed to clarify the actual occurrence of AKI. Accordingly, the results of these studies can then be used to identify the needed infrastructure and evaluate treatment strategies to prevent AKI and improve clinical outcomes.

CHRONIC KIDNEY DISEASE

EPIDEMIOLOGY

The incidence and prevalence of noncommunicable diseases have been changing rapidly as a result of demographic transition, and the burden of disease has consequently shifted from infections in the pediatric population to chronic diseases of the adult population in many ME countries.^{13,68-76} This demographic shift in the burden of disease is exemplified by the emerging epidemic of diabetes mellitus that is occurring globally and affects people of ME countries, particularly of Arab descent. In 2021, the International Diabetes Federation (IDF) ranked the ME region as having the highest age-adjusted (20–79 years) comparative diabetes prevalence worldwide (18.1%).⁷⁷ The IDF has also predicted that the prevalence of diabetes mellitus in ME countries will increase by 2045. These alarming statistics for ME countries have been attributed to a combination of increasing urbanization, aging populations, increasing obesity, and falling levels of physical activity.^{7,13,70,71,78-89} Accordingly, the direct and indirect medical expenditures associated with diabetes mellitus will become a profound economic burden. As a result, ME countries with limited or scarce resources (see [Table 76.1](#)) will be unable to cope with the social, economic, and public health consequences of complications of diabetes mellitus, one of which is CKD.^{90,91}

In addition, the common practice of consanguineous marriages in ME countries has led to a high incidence of genetic disorders, some of which may lead to CKD and end-stage kidney failure (ESKF), especially in children, and has possibly altered the pattern of renal disease in these countries.^{86,92-100} Genetic renal diseases and their complications in the ME are discussed further later in this chapter.

Obtaining the epidemiologic statistics concerning CKD, ESKF, and kidney replacement therapy (KRT) is a work in progress that is hampered by the lack of well-conducted, cross-sectional, longitudinal cohort studies and a lack of reliable registries that incorporate data on comorbid conditions.¹⁰¹⁻¹⁰⁴ Yet in an attempt to achieve optimal and equitable kidney care worldwide, the ISN launched a global survey project in 2016, the ISN-Global Kidney Health Atlas (ISN-GKHA), to collect information on the current state of readiness, capacity, and competence for kidney health care delivery in each country and region, including the ME.^{91,105-108} The results of the third iteration were released in 2023, which will be useful for defining health policies and the allocation of resources.¹⁰⁸ Additional information could also be extracted from the Chronic Kidney Disease (CKD) Prognosis Consortium and the Global Burden of Diseases (GBD), Injuries, and Risk Factors Studies,^{72,75,76,90,105,109-112} as well as from recent country-specific publications ([Tables 76.2 and 76.3](#)).^{20,113-119}

On the basis of deidentified electronic health records (EHR) data, a large cohort from Maccabi Healthcare Services (1,011,789 adult Israeli individuals without CKD, of whom 72,480 had diabetes) participated in the CKD Prognosis Consortium.¹¹⁰ In those without diabetes, the 5-year cumulative incidence rates of confirmed CKD events at different CKD stages (defined as eGFR 60, ≥ 45 , and 30 mL/min/1.73 m²), were 1.29%, 0.696%, and 0.172%, respectively. In comparison, the cumulative CKD incidence rates were much higher in individuals with diabetes (19.73%, 7.8%, and 2.5% events of CKD for eGFR ≥ 60 , ≥ 45 , and ≥ 30 mL/min/1.73 m², respectively). Furthermore, in a series of long-term, nationwide, population-based, retrospective cohort studies, factors such as overweight, obesity, normal renal function with a history of any childhood kidney disease, or persistent asymptomatic isolated hematuria detected in Israeli adolescents and young adults were found to be strongly associated with an increased cumulative incidence of treated ESKF, with crude hazard ratios of 3, 6.89, 4.19, and 18.5, respectively.¹²⁰⁻¹²²

Community-based screening programs for CKD and risk factors, such as diabetes mellitus, obesity, proteinuria, and hypertension, have been launched worldwide under the auspices of the Research and Prevention Committee of the ISN.^{88,123} These programs are expected to detect CKD in populations in which the people are unaware of such chronic diseases. This initiative includes ongoing programs in Egypt and Morocco. The Egypt Information, Prevention, and Treatment of Chronic Kidney Diseases (EGIPT-CKD) program was the first to report interim results. It investigated the prevalence of microalbuminuria among first-degree relatives of people with ESKF in the city of Damanhur and the surrounding towns in the Al-Buhayrah governorate, in Lower Egypt.¹²⁴ In this study, microalbuminuria was detected in 10.6% of participants. In an adjusted logistic regression

Table 76.2 Prevalence and Deaths for Chronic Kidney Disease in 2017 and Percentage Change of Age-Standardized Rates, Global and by Location in the Middle East, 1990–2017^a

Country	Prevalence (95% UI)			Deaths (95% UI)		
	Count, 2017	Age-Standardized Rate per 100,000, 2017	Percentage Change in Age-Standardized Rates Between 1990 and 2017	Count, 2017	Age-Standardized Rate per 100,000, 2017	Percentage Change in Age-Standardized Rates Between 1990 and 2017
Global	697,509,472 (649,209,403 to 752,050,655)	8724 (8124 to 9403)	1.2% (–1.1 to 3.5)	1,230,168 (1,195,114 to 1,258,829)	15.9 (15.5 to 16.3)	2.8% (–1.5 to 6.3)
Algeria	3,497,207 (3,232,001 to 3,812,516)	9813 (9075 to 10 698)	0.9% (–3.5 to 5.8)	4,577 (4,074 to 5,077)	15.6 (13.8 to 17.2)	–10.6% (–20.5 to 0.4)
Bahrain	132,052 (120,764 to 143,901)	10,427 (9660 to 11,291)	–1.7% (–6.6 to 3.5)	133 (118 to 148)	22.3 (19.7 to 24.6)	–42.6% (–49.3 to –34.6)
Egypt	7,101,539 (6,552,681 to 7,720,219)	10,572 (9754 to 11,521)	5.3% (1.4 to 9.6)	13,115 (11,314 to 14,968)	29.0 (25.0 to 33.1)	9.4% (–5.7 to 25.1)
Iran	8,339,849 (7,708,434 to 9,055,743)	10,924 (10,112 to 11,895)	11.3% (8.2 to 15.0)	10,163 (9,381 to 10,549)	16.6 (15.4 to 17.3)	9.8% (–0.6 to 17.2)
Iraq	3,044,399 (2,826,656 to 3,295,432)	10,991 (10,186 to 11,946)	–9.3% (–13.0 to –5.7)	4706 (4245 to 5123)	21.9 (19.7 to 23.9)	–60.0% (–65.2 to –53.7)
Israel	654,367 (606,819 to 708,940)	6246 (5810 to 6757)	0.6% (–2.1 to 3.3)	2242 (2088 to 2407)	17.7 (16.5 to 19.0)	–20.9% (–26.6 to –14.2)
Jordan	745,402 (691,588 to 808,791)	10,378 (9635 to 11,277)	–5.0% (–8.5 to –1.1)	1281 (1135 to 1451)	27.5 (24.5 to 31.0)	–18.0% (–30.3 to –3.4)
Kuwait	339,577 (313,180 to 369,112)	9,716 (8988 to 10,559)	–0.2% (–6.0 to 6.3)	177 (160 to 201)	8.3 (7.5 to 9.5)	–64.1% (–67.7 to –59.6)
Lebanon	668,003 (618,911 to 726,949)	10,029 (9,284 to 10,928)	–0.3% (–5.9 to 5.6)	594 (533 to 674)	11.2 (10.0 to 12.6)	–26.6% (–37.0 to –13.5)
Libya	594,010 (548,524 to 644,371)	10,963 (10,142 to 11,905)	6.6% (2.0 to 11.5)	1181 (1014 to 1363)	29.6 (25.6 to 33.9)	10.0% (–8.4 to 31.5)
Morocco	3,289,444 (3,046,873 to 3,568,865)	9924 (9201 to 10,750)	1.5% (–3.0 to 6.6)	4544 (3830 to 5334)	16.3 (13.9 to 19.1)	6.3% (–11.7 to 29.1)
Palestine	307,284 (283,987 to 333,626)	10,423 (9590 to 11,333)	–2.2% (–5.8 to 1.7)	624 (580 to 680)	28.5 (26.3 to 31.2)	–21.5% (–33.7 to –6.6)
Oman	324,395 (297,665 to 352,901)	10,611 (9836 to 11,554)	9.2% (4.2 to 14.9)	253 (198 to 298)	16.4 (12.4 to 19.2)	–3.5% (–23.1 to 20.0)
Qatar	188,200 (170,639 to 205,354)	9920 (9199 to 10 757)	–8.9% (–13.3 to –3.6)	115 (93 to 137)	26.5 (19.9 to 31.1)	–47.4% (–57.7 to –36.0)
Saudi Arabia	2,387,872 (2,201,128 to 2,595,630)	9892 (9139 to 10,812)	0.1% (–4.0 to 4.4)	3818 (3111 to 4411)	29.9 (24.2 to 34.1)	–2.6% (–21.7 to 22.7)
Syria	1,384,897 (1,271,191 to 1,505,808)	9881 (9099 to 10,748)	–4.8% (–8.8 to –0.1)	2257 (1951 to 777)	19.7 (17.1 to 24.9)	–35.0% (–44.9 to –21.9)
Tunisia	1,218,223 (1,125,849 to 1,331,895)	9916 (9164 to 10,827)	5.1% (–0.6 to 11.9)	1645 (1358 to 1959)	15.3 (12.7 to 18.2)	–12.6% (–28.3 to 5.7)
Turkey	9,042,506 (8,350,852 to 9,874,836)	10,311 (9532 to 11,248)	–2.5% (–7.4 to 2.6)	15,153 (13,383 to 16,778)	17.8 (15.8 to 19.7)	–32.0% (–40.4 to –21.7)
United Arab Emirates	724,351 (655,751 to 795,089)	9951 (9198 to 10,828)	–4.0% (–7.7 to –0.1)	829 (648 to 1046)	30.8 (25.3 to 36.7)	3.0% (–18.1 to 29.0)
Yemen	1,560,862 (1,446,990 to 1,701,455)	9680 (8964 to 10,522)	1.4% (–3.0 to 6.3)	1964 (1,541 to 2,463)	16.4 (13.1 to 20.2)	–19.4% (–38.8 to 10.4)

^aAdapted from GBD-Chronic Kidney Disease Collaboration.⁷⁶
UI, Uncertainty interval.

analysis, smoking and a personal history of cardiovascular disease were strongly associated with microalbuminuria. The well-conducted, cross-sectional *Maladie Rénale chronique au Maroc* (MAREMAR) study involved 10,524 individuals from two Moroccan towns, El-Jadida (coastal

and industrial) and Khemisset (inland and rural).⁸⁹ It found that the adjusted prevalence of CKD in Morocco is among the lowest in the world (5.1%), including estimated glomerular filtration rate (eGFR) under 60 mL/min/1.73 m² (1.6%), confirmed proteinuria (1.9%), and hematuria

Table 76.3 Epidemiology of Renal Replacement Therapy in Middle Eastern Countries

Country	Treated Kidney Failure		Chronic Dialysis (HD+PD)		Chronic Hemodialysis		Chronic Peritoneal Dialysis		Kidney Transplantation					Date of Country's First KRT Modality#		
	Inci- dence (PMP)	Preva- lence (PMP)	Preva- lence (PMP)	Centers (PMP)	Preva- lence (PMP)	Centers (PMP)	Preva- lence (PMP)	Centers (PMP)	Incidence Overall (PMP)	Prevalence Overall (PMP)	Incidence of deceased donor (PMP)	Incidence of living donor (PMP)	Transplant waitlist N, R, A	HD##	PD##	Transplant
Algeria	—	—	251.4	8.22	240.3	—	11.1	0.31	2.07	—	0	2.07	—	1962	1962	1986
Bahrain	207.5	339.7	250.2	—	—	—	—	—	—	52.7	—	—	—	1971	1998	1995
Egypt	56	624	610	5.53	517.7	0.03	0.1	0.32	15.49	—	0	15.49	R	1958	1962	1976
Iran	81	654	343	7.47	211.4	0.35	17.6	0.3	14.76	311	9.76	5	R	1965	1975	1967
Iraq	—	251	145	0.64	—	0.07	—	0.15	—	106	—	—	R	1967	2004	1973
Israel	198 ^a	1273 ^a	752 ^a	7.74	698.32	2.24	62.35	0.65	56 ^a	522*	16.93	37.16	N	1948	1960	1964
Jordan	—	—	446.3	7.91	428.9	0.55	17.4	1.55	10	—	0.1	9.9	R	1968	1990	1972
Kuwait	151 ^a	935 ^a	509 ^a	3.26	417.12	3.26	52.14	0.33	28 ^a	432*	9.07	11.16	N	1976	1982	1979
Lebanon	—	—	694	15.1	668.5	0.94	25.5	0.94	12.88	—	1.02	11.86	N	1960	1994	1972
Morocco	144	541	185.6	15	185	—	0.6	0.15	0.35	—	0.14	0.22	—	1972	2005	1989
Libya	—	—	360.1	11.43	347.1	0.22	8.3	0.19	0.79	—	0	0.79	N	1980	1980	1990
Oman	120	670	358	6.64	250	2.66	10.9	0.53	2.22	279	0	2.22	N	1983	1992	1988
Palestinian Authority	—	—	—	2.67	—	0.33	—	0.33	—	—	—	—	A	1997	1992	2011
Qatar	164	645	366	2.39	303.15	1.2	64.5	0.4	16.55	279	5.86	10.69	N	1981	1996	1986
Saudi Arabia	153 ^a	855 ^a	657 ^a	5.66	409.5	0.71	35	0.28	31 ^a	198*	2.58	27	N	1971	1980	1979
Syrian Arab Republic	—	—	155.1	2.55	149.1	0.32	6	0.28	17.21	—	0	17.21	A	1974	1984	1976
Tunisia	242	1018	759.6	15.89	722.8	0.76	36.8	0.5	11.06	—	1.59	0.88	N	1963	1963	1986
Turkey	150 ^a	994 ^a	750 ^a	10.86	735.84	1.57	40.32	1.11	40 ^a	244*	2.95	26.68	N	1965	1968	1975
United Arab Emirates	—	152	201.4	1.01	184.1	0.5	17.3	0.61	13	—	6.6	6.4	R	1977	1976	1985
Yemen	—	—	108.4	—	99.1	—	9.3	—	—	52.7	—	—	A	1982	1984	1998

^aData source: U.S. Renal Data System 2023 Annual Data Report: International Comparisons. Available at <https://usrds-adr.niddk.nih.gov/2023/end-stage-renal-disease/11-international-comparisons>.¹⁶⁴

^bData source: References^{20,113,114,117,118,139,152,156}

^cAs therapy for acute kidney injury or end-stage kidney disease.

A, Absent; HD, hemodialysis; KRT, kidney replacement therapy; PD, peritoneal dialysis; PMP, per million population; N, national; R, regional; — data not reported or unavailable. From Bello AK et al., 2023, ISN-Global Kidney Health Atlas, ISN, Brussels. Available at www.theisn.org/global-atlas.

(3.4%). The low rate of CKD in this study may reflect the fact that the diagnosis was made on repeat samples and not on purely cross-sectional data, suggesting possible overestimation of CKD in studies relying on a single GFR estimate. The MAREMAR-CKD cohort was also classified according to the KDIGO eGFR stages:¹²⁵ G1, 17.8%; G2, 17.2%; G3A, 40.2%; G3B, 12.3%; G4, 4.4%; and G5, 7.2%. Furthermore in Morocco, population-adjusted prevalence rates for hypertension, obesity, and diabetes mellitus were reported to be 16.7%, 23.2%, and 13.8%, respectively.⁸⁹

According to the Chronic REnal Disease In Turkey (CREDIT) study,⁸⁷ the population-based estimated prevalence rates for CKD stages 1 to 5 among Turkish adults were 5.4%, 5.2%, 4.7%, 0.3%, and 0.2%, respectively, with an overall prevalence of 15.8%. Moreover, the frequency of concomitant cardiovascular risk factors such as diabetes, hypertension, dyslipidemia, and obesity was increased in patients with CKD, and odds ratios versus the non-CKD population were 3.22, 2.86, 1.60, and 1.65, respectively. On the basis of an analysis of one spot urine specimen per participant, 10.2% of the cohort subjects had microalbuminuria and 2% had macroalbuminuria.⁸⁷ The causes of CKD were not detailed, but the study cohort continues to be followed longitudinally.

A small pilot study, which is part of the Global SEEK Project, demonstrated the feasibility of screening and early detection of CKD in a Saudi population. Using standardized GFR prediction equations, the prevalence of CKD stages 1, 2, and 3 was relatively low—3.5%, 1.6%, and 0.6%, respectively—which may be attributed to the low mean age of the participants (37.4 ± 11.3 years).¹²⁶ In Kuwait, a 4-year prospective study from one referral center by El-Reshaid and associates¹²⁷ has reported a high incidence of CKD among Kuwaiti nationals. Specifically, they reported an average annual incidence of 366 per million population (pmp), with a higher incidence among the older population (≥ 60 years) of 913 pmp. Of patients with CKD who were admitted to the center, 6% presented with uremic syndrome and 40% experienced acute deterioration of their kidney function that resulted mainly from drugs (mostly over-the-counter nonsteroidal antiinflammatory drugs [NSAIDs]), infection, and volume depletion. The authors did not report categories of eGFR or proteinuria in their patients.

In 2007, a study involving 31,999 taxi drivers in Tehran, mostly males, showed that 6.5% of drivers had an eGFR of <60 mL/min/1.73 m².¹²⁸ However, community-based studies revealed considerable differences in CKD prevalence among counties in Iran (10.2%–18.9%).^{85,129}

In a pioneering work, Tohidi and colleagues¹³⁰ examined the incidence of CKD among a subgroup, 20 years or older, in the Tehran Lipid and Glucose Study, a long-term community-based prospective study. During a 10-year follow-up, the cumulative incidence of CKD stages 3 to 5 among women and men was 27.8% and 14.2%, respectively. Older age, hypertension, diabetes, and current smoking were found to be independent predictors of CKD stages 3 to 5 in both sexes. In addition, for men, high-normal blood pressure and, for women, being single, divorced, or widowed were associated with an increased risk of incident CKD.¹³⁰ Interestingly, adherence to Mediterranean or Dietary Approaches to Stop Hypertension (DASH)-style diets was associated with a lower incidence of CKD,¹³¹⁻¹³³

which may ultimately translate into favorable survival and cardiovascular outcomes.⁹⁰

Thus with an increasing number of emerging effective therapies, like sodium-glucose cotransporter-2 (SGLT2) inhibitors, it is hoped that early detection of CKD and its risk factors will increase kidney health awareness and permit early intervention, which in turn may modify the disease course, complexity, and costs of overall therapy.^{101,111,112,134-136}

Estimates of the prevalence and incidence of ESKF are more reliable in countries that have a national database in which each patient who has received KRT is registered (see [Tables 76.3 and 76.4](#)). Nevertheless, patient numbers in a national registry in ME countries may be underestimated for two additional reasons: 1. the treatment of ESKF may be beyond the reach of the average citizen in low-income ME countries, such as Yemen; and 2. a high number of expatriates live in some ME countries who may not be included. [Tables 76.3 and 76.4](#) summarize the available data on ESKF from published articles, books, registries, and GKHA-2023.^{20,78,88,113-119,137-164} Of note, most investigators relied mainly on the results of limited retrospective studies and answers to questionnaires that were sent to leading nephrologists in each country and not on accurate documented statistics, registries, or results of epidemiologic studies.

Allowing international comparison, Israel and Turkey national registries have reported their aggregated ESKF data to both the European Renal Association (ERA) registry¹⁶²⁻¹⁶⁴ and the U.S. Renal Data System,^{164,165} whereas Kuwait and Saudi Arabia have reported only to the U.S. Renal Data System.¹⁶⁴ The registry in Turkey was established in 1990.^{147,166} In 2015/6, the prevalence of end-stage kidney disease (ESKD) in Turkey plateaued at ~933 pmp, in parallel to the observed plateau in incidence rate for new KRT patients (140–147 pmp, between 2012 and 2016).^{113,159,165} However, this trend was not sustained. At the end of 2022, prevalence and incidence of treated ESKD increased and were calculated as 1016.2 and 160.9 pmp, respectively (including pediatric patients).¹⁶⁷ For 2 decades, diabetes and hypertension were the two main causes of ESKD in Turkey.^{159,162,165,166,168}

The Israeli Center for Disease Control (ICDC) reported that the incidence rate of ESKF had increased by 73% in 25 years, from 113 in 1990 to 196 pmp in 2015. The average annual increase was highest in the years 1996 to 1999, but it has stabilized since 2003. The increase in ESKD was observed mainly in the older population (≥ 65 years). Likewise, a 2.8-fold increase in the prevalence of ESKF was observed, from 416 pmp in 1990 to 1171 pmp in 2015. Of note, diabetes mellitus was the cause of ESKF in 11.5% of prevalent patients in 1990 and increased to 45.9% in 2015.^{158,169}

The Tunisian registry was started in 1990 and has also reported an increase in the incidence and prevalence of ESKF.¹⁷⁰ The incidence of ESKF increased from 81.6 pmp in the period 1992 to 1993 to 158.8 pmp from 2000 to 2001; the average annual increase was 9.6%. The incidence of ESKF in older persons, women, and individuals with diabetic nephropathy has risen steeply. However, regional variations were noted among urban and rural districts.^{170,171} Of note, the Sfax region of Tunisia also reports its data to the ERA registry.

A comprehensive 1-year observational study, conducted by Alashek and colleagues,^{78,172-175} shed light on the

Table 76.4 Availability, Funding, Registries, and Advocacy for Kidney Care—Nephrology Workforce Capacity in Middle Eastern Countries

		Availability of KRT		Availability of CKM		Funding for Medications/KRT		Availability and Distribution of Registry		Advocacy Group		Nephrology Workforce (PMP)						
Country		HD	PD	Kidney transplantation	Shared Decision	Choice Restricted (limited)	CKD	Dialysis	Kidney Transplantation	CKD	Dialysis	Kidney transplantation	AKI	CKD	AKI	KRT	Nephrologist	Nephrologist trainees
Algeria	2017																11.3	9.1
	2019																12	1.92
Bahrain	2017																5.9	13.4
	2017																21.7	8.9
Egypt	2019																23.14	3.02
	2023																32.88	3.71
Iran	2017																3.9	0.3
	2019																4.97	0.35
	2023																4.84	0.23
	2017																9.7	1.6
Iraq	2019																2.49	1
	2023																2.72	0.69
Israel	2017																9.7	1.6
	2019																29.67	2.37
	2023																24.68	3.93
	2017																10.2	2.7
Jordan	2019																8.08	1.82
	2023																11.91	2.09
Kuwait	2017																10.8	20.6
	2019																10.29	24
	2023																48.89	0.33
	2017																22.6	1.6
Lebanon	2019																28.28	2.05
	2023																36.81	2.27
Libya	2017																12.5	18.7
	2017																10.5	3.6
Morocco	2019																12.53	2.19
	2023																15.24	2.72
	2017																30.4	1.2
	2019																20.03	2.86
Oman	2023																39.85	1.59
	2017																2.2	NA
Palestinian Authority	2019																3.04	0
	2023																5	0.33
	2017																11.4	2.7
	2019																15.87	2.12
Qatar	2023																23.92	1.99
	2017																17.7	1.8
Saudi Arabia	2019																8.46	15.94
	2023																19.8	1.56
	2017																4.7	1.2
	2019																5.14	1.03
Syria	2023																4.64	0.93
	2017																16.3	5.4
Tunisia	2019																15.63	5.51
	2023																18.49	5.46
	2017																6.3	1.9
	2019																8	1.08
Turkey	2023																9.63	0.6

Continued on following page

Table 76.4 Availability, Funding, Registries, and Advocacy for Kidney Care—Nephrology Workforce Capacity in Middle Eastern Countries (Continued)

		Availability of KRT		Availability of CKM		Funding for Medications/KRT		Availability and Distribution of Registry		Advocacy Group		Nephrology Workforce (PMP)						
Country		HD	PD	Kidney transplantation	Shared Decision	Choice Restricted (limited)	CKD	Dialysis	Kidney Transplantation	CKD	Dialysis	Kidney transplantation	AKI	CKD	AKI	KRT	Nephrologist	Nephrologist trainees
United Arab Emirates	2017																8.7	0.9
	2019																6.7	0.72
	2023																NA	0.5
Yemen	2017																0.22	1.87

Green and red boxes indicate that survey participants answered yes and no, respectively, to GKHA Questionnaire;^{88,105} gray, not available.

In “Funding for Medications/KRT”; *dark green*, publicly funded by government but with some fees at the point of delivery; *green*, publicly funded by government and free at the point of delivery; *light green*, mix of public and private funding systems; *yellow*, multiple systems—programs provided by government, nongovernmental organizations, and communities.

AKI, Acute kidney injury; CKD, chronic kidney disease; CKM, Conservative kidney management; HD, hemodialysis; KRT, kidney replacement therapy; PD, peritoneal dialysis; PMP, per million population.

epidemiology of ESKF and KRT practices in Libya before the conflict. The prevalence and incidence of dialysis-treated ESKF from mid-2009 until August 2010 were 624 and 282 pmp, respectively.⁷⁸

The Saudi Center for Organ Transplantation (SCOT) has established an open-access KRT registry that provides annual information on the epidemiology and treatment of ESKF in Saudi Arabia.¹⁷⁶ In 2021, the incidence and prevalence of KRT were 153 and 881 pmp, respectively, including non-Saudis, who comprise 15% of dialysis patients.^{164,177} Over 2 decades, the average annual increase in dialysis population was 6.2%;¹⁴³ however, the average annual percentage fell to 5% between 2015 and 2020.¹⁷⁶ The Jordanian National Registry of End-Stage Renal Disease was established in 2007. Its annual reports did not include epidemiologic information on kidney transplantation.¹⁶⁰ According to the 13th report, in 2020, the incidence and prevalence rates of dialysis in Jordan were 120.1 and 795 pmp, respectively (excluding non-Jordanians, who comprised 7.9% and 5.9% of incident and prevalent dialysis patients, respectively).

In March 2011, Lebanon launched its national kidney registry, aiming to report reliable data on prevalence, incidence, patient management, practice patterns, clinical outcomes, and survival among CKD patients. In the first annual report of the registry in 2012, the incidence of ESKF and patient survival could not be accurately calculated because of incomplete data. However, the prevalence was estimated to be 855 pmp. Between 2014 and 2015, the estimated prevalence and incidence rates of ESKF requiring dialysis were 777 and 191 pmp, respectively. Since then, no other official report has been published. As of 2021–2022, approximately 5400 ESKF patients were receiving KRT in

Lebanon (4400 dialysis patients and a thousand kidney transplanted patients).¹¹⁷

In Iran, a limited treatment program for ESKF was introduced in 1972, which expanded after the training of qualified physicians and nurses to treat 109 patients in 1976. In this millennium, the incidence/prevalence rates of ESKF were 49.9/238 pmp in 2000, increasing to 63.8/357 pmp and 109.9/892 pmp, respectively, in 2006 and 2018. As in other ME countries, this ESKF burden has been accompanied by a growing number of dialysis centers and kidney transplant programs.^{113,137,165,178,179}

In the ME countries for which detailed data are available, there is a male predominance among patients with ESKF, similar to that reported worldwide, except in Kuwait.^{20,113,114,160,163,164}

Of note, kidney diseases and risk factors or access to therapy in expatriates—who may comprise as much as 50% of the population, especially in the wealthy Gulf countries—have not been carefully investigated. This population needs to be distinguished from the resident population and may require special attention because of their different ethnic, socioeconomic, and environmental backgrounds.^{19,20,113,117,143,160,176,179,180}

CAUSES

The causes of CKD in ME countries are highly influenced by the biocology of a particular region and the ethnic and socioeconomic background of its population. Accordingly, the various causes of CKD are ranked differently in ME countries. The populations of the five ME countries in North Africa—Morocco, Algeria, Tunisia, Libya, and Egypt—have similar ethnicities and socioeconomic backgrounds in that

they are of African descent that have been intermixed with Berber, Arab, and Mediterranean population streams.^{7,181} In the 1990s, interstitial nephritis and glomerulonephritis each accounted for approximately 20% of all cases of CKD in these countries.¹⁸¹ The number of individuals with interstitial nephritis increased during the 2000s, possibly as a result of environmental pollution and inappropriate or excess use of over-the-counter drugs.¹⁸² Most cases of glomerulonephritis are of the proliferative type, whereas immunoglobulin A (IgA) nephropathy is rare. The high prevalence of proliferative glomerulonephritis in these ME countries reflects postinfectious glomerulonephritis caused by viruses, bacteria, and parasites. In Egypt, Libya, and southern Algeria, approximately 7% of patients with CKD suffered from obstructive uropathy because of urinary schistosomiasis caused by *Schistosoma haematobium* or *Schistosoma mansoni*. It is noteworthy that tuberculosis, other bacterial infections, and familial Mediterranean fever (FMF) are the main causes of renal (type AA) amyloidosis in many ME countries.^{181,183}

However, the frequency of causes of ESKF has changed. All ME countries except Algeria and Yemen have reported diabetes mellitus as the most frequent cause of ESKF (20%–74%) in incident patients undergoing KRT, followed by hypertension (11%–30%) and glomerulonephritis (11%–24%).^{7,69,78,138,142,158,160–162,164,167,176,178,179,184–188}

GENETIC DISORDERS

Genetic kidney diseases have received much attention in the fields of pediatric and adult nephrology because the underlying molecular defects in many of these diseases have been elucidated as a result of advances in genetics and molecular biology. The ME population is ethnically and genetically diverse.¹⁸⁹ The primary demographic features of Arab Muslim and Druze communities in the ME include large families, rapid population growth, and high rates of consanguinity. Among Palestinian Arabs, more than 40% of marriages are between relatives and, of these, 50% are between first cousins.^{189,190} In the Bedouin society, 40% of women of childbearing age are married to first cousins.¹⁹¹ Nevertheless, a comprehensive study, conducted in a single Muslim village in Israel, has demonstrated significant sociodemographic changes during a 50-year time period.⁹⁸ A shift from the practice of marrying a first cousin to marrying a remote relative was noted. There was a significant reduction in the mean number of children born per woman and, in parallel, the mean age of first-time mothers has progressively increased. The impact of these changes on the incidence of recessively inherited diseases among this population is of much interest.

In contrast, the consanguinity rate among Israeli Jews is reported to be 2.3% and, of these, first cousin marriages account for 0.8%. The highest consanguinity rate among Israeli Jews (7.1%) is found among Eastern (i.e., Asian, non-Sephardic) Jews.¹⁹²

High consanguinity rates have also been reported in the Saudi Arabian, Kuwaiti, Syrian, Lebanese, and Moroccan populations.^{92,93,95,97,99,100,193–195}

Clinical Relevance

Inherited Kidney Disease

High rates of genetic kidney diseases occur in the ME because of high rates of consanguinity.

Results of an epidemiologic survey from Lebanon revealed that 26% of patients undergoing maintenance hemodialysis are children of consanguineous parents.⁹³ In addition, Barbari and colleagues⁹³ have reported that the risk for a family history of kidney disease is particularly high among patients from consanguineous families who were undergoing hemodialysis. Populations with a high rate of consanguinity also have an increased prevalence of diseases associated with CKD, such as hypertension, metabolic syndrome, and diabetes mellitus.^{94,196} A study of kidney biopsies from three pathology centers in Lebanon has shown that mesangioproliferative glomerulonephritis is significantly more frequent among Muslims and offspring of consanguineous unions, whereas focal segmental glomerulosclerosis (FSGS) is most prevalent in Christians.⁹⁶ Mutation screening in genes known to be responsible for either of these histologic diagnoses had not been performed.

A considerable proportion of genetic kidney diseases is inherited in an autosomal recessive manner. It is therefore not surprising that these diseases occur most frequently in communities with high consanguinity rates.^{100,197} The contribution of large families with many offspring, resulting in high carrier rates, to the augmented prevalence of recessively inherited disorders is also appreciated. In a national-multicenter prospective study of all Israeli pediatric dialysis units, implementing whole-exome sequencing, monogenic etiologies for kidney failure were identified in 45% of the 79 children tested.¹⁹⁸ The genetic diagnostic yield was higher among Arabs compared with Jewish individuals (55% vs. 29%) and in children from consanguineous compared with nonconsanguineous families (63% vs. 29%). In a study aimed at investigating the scope of pediatric kidney diseases among Syrian refugees living in Turkey, parental consanguinity was found in 57.7% of the 633 children tested and 23.3% had a close relative with kidney disease.⁹⁹ Congenital anomalies of the kidney and urinary tract (CAKUT) and glomerular diseases (mostly nephrotic syndrome) were responsible for 50% of the etiologies of kidney disease in this cohort. Genetic analyses were not performed. Consanguinity may enhance allelic and locus heterogeneity.^{94,196,197,199} In Palestinian Arabs, 71 different autosomal recessive kidney diseases have been reported;¹⁹⁰ these include primary kidney diseases (congenital nephrotic syndrome, nephronophthisis), metabolic and tubular defects (cystinuria, Bartter syndrome, renal tubular acidosis, primary hyperoxaluria type 1), and FMF (discussed in detail later in this chapter). The influence of genetic factors is evident in pediatric patients with kidney diseases, particularly in Saudi Arabia and Syria, where increased numbers of congenital and hereditary kidney and urologic diseases are now being reported.^{189,200} The results of an extensive study of Jewish and Bedouin populations in southern Israel have shown that genetic kidney diseases are overrepresented in the pediatric population.⁹⁵

In a large international study cohort, a single-gene cause of steroid-resistant nephrotic syndrome (SRNS) was found in 29.5% of unrelated families, in which SRNS was detected before 25 years of age.¹⁰⁰ Interestingly, the rate of detection of causative mutations ranged from 45.2% in the center with the highest rate of consanguinity (71.4%, Saudi Arabia) to 14.3% in the center with the lowest rate of consanguinity (0%, Ann Arbor, Michigan).¹⁰⁰

By combining whole-exome sequencing with high-resolution metabolomics profiling for a highly consanguineous Arab cohort in Qatar, 21 common variants and 12 functional rare variant metabolomics quantitative trait loci (mQTLs) were discovered, of which 45% were novel altogether.²⁰¹ This project may have wide implications for precision medicine in the ME in various areas including kidney diseases. Interestingly, a genome-wide association study (GWAS) analysis performed in a cohort of Iraqi-born residents of the city of Malmö, Sweden, identified a number of loci that may explain why people from the ME have better kidney function compared with those of European ancestry.²⁰² It may also shed light on the tendency for better kidney and cardiovascular outcomes among Iraqi immigrants with type 2 diabetes despite poor glucose regulation.

Cystic kidney diseases

Autosomal dominant polycystic kidney disease is one of the most common genetic renal diseases in adults and occurs in 1.2% to 10% of ESKD patients in several ME communities.^{158,185,203-205} Autosomal recessive polycystic disease (ARPKD; Online Mendelian Inheritance in Man [OMIM] catalogue number 263200) occurs with a proposed incidence of 1/20,000 to 1/40,000. Mortality during the first month of life is particularly high among those born with severe oligohydramnios leading to pulmonary hypoplasia. Its principal manifestations include fusiform dilation of the renal collecting ducts and distal tubules and dysgenesis of the hepatic portal triad. ARPKD is associated with mutations in the *PKHD1* gene on chromosome 6p21.1-p12, encoding fibrocystin, a membrane protein located on the primary cilium.²⁰⁶ More than 700 mutations in the *PKHD1* gene have been reported.²⁰⁷ Several mutations were reported in Turkish and Israeli children who participated in a large international study.^{208,209} One of these variants (c.3761_3762delCCinsG) was found to be a founder mutation among Ashkenazi Jews.²¹⁰

Meckel-Gruber syndrome (MKS) is a lethal, recessively inherited ciliopathy characterized by multiple congenital anomalies, including polycystic kidneys, occipital encephalocele, and polydactyly. At least 13 genes have been reported to date to underlie MKS. The incidence of this relatively rare disorder in Qatar has been reported as 1 in 500 live births for a population with over 40% consanguineous marriages.²¹¹ The underlying genetic cause was identified in 11 of 12 Arab families residing in a distinct region in Israel, comprising mutations in 7 different genes and revealing high genetic heterogeneity.²¹² Likewise, other ciliopathies including various forms of nephronophthisis (including the infantile type) with mutations in *NPHP1*, *NPHP2* (*INVS*), and *NPHP3*, encoding nephrocystin 1, inversin, and nephrocystin 3, respectively, have been reported in several ME countries²¹³⁻²¹⁵ (Table 76.5). Cases of Joubert syndrome, characterized by cerebellar vermis hypoplasia, developmental delay, seizures, retinal dystrophy, and kidney involvement, have been reported in the ME. There is genetic heterogeneity with pathogenic mutations in 33 distinct genes responsible for this disorder. Founder mutations have been identified in Iranian families and in Ashkenazi Jews.²¹⁶

Genetic glomerular diseases

A substantial number of patients in the ME with kidney disease have familial glomerular diseases whose spectrum includes familial hematuria, Alport syndrome, IgA nephropathy, and familial SRNS with histologic findings consistent with FSGS.^{95,100,200,217} A recessive form of SRNS has been shown to be associated with mutations in the *NPHS2* gene, which encodes the glomerular barrier protein podocin.²¹⁸ The most common phenotype is a nephrotic syndrome that is resistant to immunosuppressive treatment in early childhood, and most patients will develop ESKD by the end of their second decade of life. In a group of children from two consanguineous families of Israeli Arab descent, mutation analysis of the *NPHS2* gene revealed homozygosity for the C412T nonsense mutation (R138X).²¹⁹ The same mutation was subsequently found in additional children who presented with SRNS from the same ethnic background, which points to the possibility of a founder effect.²¹⁹ Interestingly, cardiac anomalies, especially left ventricular hypertrophy, pulmonary stenosis, and discrete subaortic stenosis, were detected in a high proportion of the children with *NPHS2* mutations.²²⁰ Frishberg and associates²²⁰ have speculated that podocin may have a role in normal cardiac development because podocin messenger RNA is reported to be expressed in the human fetal heart.

At least 53 *NPHS2* mutations have also been found in Turkish children with familial and “sporadic” SRNS.^{221,222} In this context, the term *sporadic* is a misnomer. As a recessively inherited disease, both parents are obligatory carriers of pathogenic mutations, and the lack of positive family history may merely represent the first case in a given nuclear family. Among Turkish patients with *NPHS2* mutations, the proportion of patients with CKD or ESKD was significantly higher (19 of 73) than that of patients without these mutations (28 of 222). Furthermore, the mean time for progression to ESKD was significantly shorter in patients with mutations compared with those without mutations.^{221,222} The prevalence of *NPHS2* mutations among 49 children with SRNS and 50 children with steroid-sensitive nephrotic syndrome from Iran was 31% and 4%, respectively.²²³ Of these children, 33% experienced recurrence of primary disease after kidney transplantation, none of whom had an *NPHS2* mutation.

The *NPHS1* gene encodes nephrin, an essential protein for maintaining the normal structure and function of the slit diaphragm of the visceral glomerular epithelial cell. Three mutations in the *NPHS1* gene were reported in 12 children with congenital nephrotic syndrome of the Finnish type from a large consanguineous Israeli Arab family.¹⁹⁹ Steroid-sensitive nephrotic syndrome is rarely reported to have a familial pattern. However, a familial pattern associated with this condition has been reported in Israeli Bedouin families with a high rate of consanguinity, and the authors of the report proposed that the increased incidence of steroid-sensitive nephrotic syndrome resulted from selective enrichment of susceptibility genes in this population.²²⁴ Whereas multiple genes, in their mutated form, have been associated with SRNS, there has not been a single monogenic disease leading to steroid-sensitive nephrotic syndrome.

A study has evaluated causative mutations related to childhood nephrotic syndrome in 49 families from Saudi

Table 76.5 Rare Genetic Diseases With Renal Involvement Reported in Middle Eastern Communities

Country, Community	Disease	OMIM No.	Phenotype	Defect, Mutation	References
Saudi Arabs, Turks	Alström syndrome ^a	203800	Retinal degeneration, obesity, cardiomyopathy, sensorineural hearing loss, insulin resistance, renal impairment	<i>ALMS1</i> gene, ubiquitously expressed, encodes a protein of unknown function	273,302
Israeli Jews of Iraqi origin	Renal hypouricemia type I	220150	Hypouricemia, hyperuricosuria, nephrolithiasis, exercise-induced acute kidney injury	<i>SLC22A12</i> gene, encodes uric acid transporter URAT1	297
Israeli Arabs	Renal hypouricemia type II	612076	Increased renal clearance of uric acid, hypouricemia, nephrolithiasis, exercise-induced acute kidney injury	<i>SLC2A9</i> gene, encodes glucose transporter 9 (GLUT9)	288,292,298
Israeli Jews, Turks	Dent disease ^a	300008	Low-molecular-weight proteinuria, hypercalciuria, nephrocalcinosis, nephrolithiasis, rickets, renal failure, hypokalemic metabolic alkalosis, focal glomerulosclerosis	<i>CLCN5</i> , encodes the chloride/proton ClC-5 antiporter	289,291,299
Egyptian and Saudi Arabs, Turks, Iranians, Jordanians	Cystinosis	219750 (adult) 219900 (juvenile) 219800 (infantile)	Failure to thrive, polydipsia, polyuria, Fanconi-like syndrome; corneal, conjunctival, and retinal deposition	<i>CTNS</i> , encodes the lysosomal cystine transporter	287,295,303, 304,306
Israeli Druze	Autosomal recessive proximal tubulopathy with hypercalciuria	138160	Proximal tubulopathy, hypercalciuria, normal or slightly elevated urinary phosphate excretion	<i>SLC2A2</i> (<i>GLUT2</i>), encodes the glucose transporter 2 ^b	301
Israeli Arabs	Autosomal recessive Fanconi syndrome and hypophosphatemic rickets	613388	Proximal tubulopathy, renal phosphate wasting, normocalciuria, bone mineral deficiency, decreased glomerular filtration rate	<i>SLC34A1</i> , encodes the renal sodium phosphate cotransporter IIa	300
Israeli Arabs	Familial renal glycosuria and aminoaciduria ^c	233100	Glycosuria, aminoaciduria	<i>SLC5A2</i> , encodes the kidney-specific Na ⁺ /glucose cotransporter	294
Israeli Jews	Proximal renal tubular acidosis and glaucoma	604278 and 603345	Short stature, deformed teeth, bilateral glaucoma, blindness, metabolic acidosis	<i>SLC4A4</i> , encodes the sodium bicarbonate cotransporter (NBCe1)	290
Israeli Arabs and Jews	Familial autosomal recessive renal tubular acidosis	267300	Distal renal tubular acidosis, deafness	<i>ATP6V1B1</i> , encodes the B1-subunit of H ⁺ -ATPase	305
Israeli Arabs	Familial autosomal recessive renal tubular acidosis	259730	Proximal and distal renal tubular acidosis, osteopetrosis, mental retardation	<i>CA2</i> , encoding carbonic anhydrase	305
Egyptian Arabs	Familial hypomagnesemia with hypercalciuria and nephrocalcinosis	248250	Hypocalcemia, hypomagnesemia, hypercalciuria, nephrocalcinosis, congenital cataracts	<i>CLDN16</i> , the claudin-16 gene	296
Iranians	Familial lecithin-cholesterol acyltransferase deficiency ^d	245900	Lower extremity edema, proteinuria, corneal opacities, hypercholesterolemia, hemolytic anemia	Lecithin-cholesterol Acyltransferase (<i>LCAT</i>) gene	293

^aX-linked.^bHomozygous mutations in *GLUT2* also cause Fanconi-Bickel syndrome.^cAutosomal recessive.^dDiagnosis made on the basis of family history and electron microscopic findings on kidney biopsy.OMIM, Online Mendelian Inheritance in Man catalogue (<http://www.ncbi.nlm.nih.gov/omim>).

Arabia;²²⁵ 62 patients were screened for mutations in genes associated with congenital, infantile, or childhood nephrotic syndrome, such as *NPHS1*, *NPHS2*, *LAMB2* (laminin- β_2), *PLCE1* (phospholipase $C_{\epsilon 1}$), *CD2AP* (CD2-associated protein), *MYO1E* (myosin 1E), *WT1* (Wilms' tumor suppressor gene 1), *PTPRO* (protein tyrosine phosphatase receptor type O), and *NEIL1* (Nei endonuclease VII-like 1).²²⁵ A homozygous mutation in the *NPHS2* gene was found in 11 families (22%) and was the most common cause of nephrotic syndrome. Other mutations were found in *NPHS1* (12%), *PLCE1* (8%), and *MYO1E* (6%) genes. FSGS was the most common pattern of histopathologic injury and a significant proportion of patients developed kidney failure requiring KRT.²²⁵

The Podonet Consortium, an international registry for congenital nephrotic syndrome and childhood SRNS, has analyzed the data retrieved from 1655 patients, 9.2% of whom were from the ME (Syria, Lebanon, Iran, and the UAE). Pathogenic mutations were detected in 23.6% of the screened patients.²²⁶ A cohort of children with SRNS from a single center in Saudi Arabia was tested for causal mutations in *NPHS1*, *NPHS2*, and *WT1*. All 44 children were older than 1 year, and presumably pathogenic mutations were identified in 5 children (11.4%), 3 in *NPHS2* and the remaining 2 in *NPHS1*.²²⁷ None of those with an identified genetic cause showed any response to immunosuppressive treatment.

In Turkey, 25 children with SRNS underwent mutation screening in *TRPC6*, a dominantly inherited form of SRNS. Positive family history was found in nine children.²²⁸ The results showed a variant in one patient, intronic nucleotide substitution in six patients, and previously described missense and synonymous amino acid variants in nine patients. Although the pathogenicity of all these genetic variants remains to be proven, the authors concluded that *TRPC6* mutations may play a significant role in childhood SRNS.

Galloway-Mowat syndrome (GAMOS) is an autosomal recessive disease characterized by the combination of early-onset nephrotic syndrome (SRNS) and microcephaly with brain anomalies. Recessive mutations in *OSGEP*, *TP53RK*, *TPRKB*, and X-linked mutations in *LAGE3* genes encoding the four subunits of the KEOPS complex have been detected in 37 individuals from 32 families with GAMOS, 6 of whom are from the ME (Iran, Turkey, Jordan, and Egypt).²²⁹

Coenzyme Q10 (CoQ10), also known as ubiquinone, is a lipid-soluble component of virtually all cell membranes and plays a critical role in transporting electrons along the respiratory chain of the mitochondrial inner membrane. *ADCK4* encoding CoQ8B plays a role in CoQ10 biosynthesis; mutations in *ADCK4* may cause SRNS and/or CKD of presumably unknown cause.²³⁰ A total of 146 index patients aged 10 to 18 years from Turkey, with newly diagnosed non-nephrotic proteinuria, nephrotic syndrome, or CKD of varying severity, were screened for *ADCK4* mutations.²³¹ In this study, 28 individuals with a biallelic mutation were identified, and CoQ10 supplementation appeared efficacious in reducing proteinuria in a subgroup of this cohort.

The Rat sarcoma homolog gene family of small guanosine triphosphate (GTP)-ases (Rho GTPases) is associated with actin remodeling and podocyte migratory ability. Alterations in Rho GTPases signaling interfere with podocyte motility

and cause proteinuria.²³² Rho guanosine diphosphate (GDP) dissociation inhibitor 1, a regulator of Rho GTPases, is a protein that is encoded in humans by the *ARHGDI* gene and is also expressed in podocytes. *ARHADIA* mutations, associated with SRNS progressing to ESKD during the first decade of life, have been reported in three siblings from a family of Ashkenazi Jews and in a Moroccan infant.²³³

Genetic metabolic diseases and inherited tubular disorders

The primary hyperoxalurias (PHs) are a group of rare autosomal recessive inborn errors of glyoxylate metabolism that result in the excessive production of oxalate.^{234,235} Currently, three types of PH have been characterized genetically and phenotypically.²³⁶ The more frequently identified PH type 1 (OMIM number 259900) is caused by the absence, deficiency, or mistargeting to the mitochondria of the liver-specific peroxisomal enzyme alanine-glyoxylate aminotransferase (AGT), which catalyzes the conversion of glyoxylate to glycine. Accumulating glyoxylate diffuses from the peroxisome into the cytosol, where it is oxidized to oxalate, a reaction catalyzed by lactate dehydrogenase (LDH). Excessive oxalate synthesis and massive urinary excretion of the insoluble calcium oxalate crystals result in nephrocalcinosis, kidney stones, and declining kidney function. As the glomerular filtration rate (GFR) declines, the oxalate load, produced by the liver, can no longer be cleared by the kidneys, plasma oxalate level increases, and calcium oxalate crystals are deposited in almost all tissues in a devastating process termed "systemic oxalosis." PH type 2 (OMIM number 260000) is caused by a deficiency of the mitochondrial enzyme glyoxylate reductase/hydroxypyruvate reductase (GRHPR), which catalyzes the reduction of glyoxylate to glycolate. In 2010, PH type 3 (OMIM number 613616) was described in a cohort of patients with calcium oxalate nephrolithiasis due to hyperoxaluria in whom PH1 and PH2 were excluded.²³⁷ In affected members of nine unrelated families, Belostotsky and colleagues^{237,238} found biallelic loss-of-function mutations in *HOGA1*, formerly *DHDPSL*, which encodes a mitochondrial 4-hydroxy-2-oxoglutarate aldolase that catalyzes the fourth step in the hydroxyproline pathway.

PH1 has the most severe phenotype and therefore is more likely to be diagnosed as reflected in voluntary patients' registries.^{239,240} However, the prevalence calculated on the basis of publicly available data regarding allele frequency by populations, suggests that all three types of PH are much more common.²⁴¹ It is especially underestimated for the less severe type—PH3, which was predicted to be as prevalent as PH1 and twice as common as PH2. Of note, analysis of the gnomAD database reveals a carrier frequency, of one of 4 pathogenic mutations in *HOGA1*, of 1:36 among Ashkenazi Jews.²⁴² There is an additional group of patients with undetected mutations in either of the three known genes.

PH types 1 to 3 are relatively prevalent in the ME. Cases of PH1 and isolated cases of type 2 have been reported in Israeli families.²⁴³⁻²⁴⁵ In 22 Israeli Arab families with type 1 hyperoxaluria, at least 15 different mutations in the AGT-encoding *AGXT* gene have been detected.^{243,245} Marked intrafamilial phenotypic variability, with no definite genotype-phenotype correlation, was noted in these families, and the prevalent phenotype was of early onset of disease

with progression to ESKF in the first decade of life.²⁴³ The diagnosis of PH1 was confirmed in 15 of 19 children tested in a single tertiary center in Egypt.²⁴⁶ Two-thirds of this cohort reached ESKD, with one-third during infancy. A similar phenotype among PH1 patients, namely early onset with a high portion of individuals presenting with kidney failure, has been described in a more recent report from Egypt and in a cohort from Syria.^{247,248} A large European study of 155 patients from 129 families with type 1 hyperoxaluria, including a large proportion of ME individuals, has shown that the most common mutation, among European families, is *p.Gly170Arg* (allelic frequency, 21.5%). This mutation results in mistargeting of AGT into mitochondria while the catalytic activity of the enzyme remains intact. A subgroup of these patients will respond to vitamin B₆ treatment, a precursor of pyridoxal phosphate, which is a known active cofactor in all aminotransferase reactions and therefore may have a better long-term prognosis.²⁴⁹ Unfortunately, mutations in *AGXT* that are amenable to pyridoxine treatment are quite rare in the ME.

Hyperoxaluria manifesting as kidney stone disease and/or progressive CKD has been described in the following reports from the ME. Although systematic genetic analysis has not been performed, it is likely that PH accounts for the disease in a substantial number of individuals. For example, among 260 adult patients with recurrent nephrolithiasis from a tertiary center in Israel, hyperoxaluria was detected in 24.2%.²⁵⁰ Hyperoxaluria was found among 10.7% of 84 patients with CKD, treated conservatively or by hemodialysis, in the Jenin district of the PA.¹⁸⁵ PH1 is the probable cause of hyperoxaluria in most of these cases. In a retrospective report from a single center in Amman, Jordan, 70 children with suspected PH were identified.²⁵¹ At the time of initial presentation, 15.7% were in ESKF and an additional 25% had impaired kidney function. Although not tested at the time of this report, PH1 may be responsible for hyperoxaluria in many of these children, particularly in those with compromised kidney function. Similarly, a high prevalence of hyperoxaluria was demonstrated among Turkish children with kidney stone disease.²⁵² Hyperoxaluria causing nephrolithiasis, nephrocalcinosis, and kidney failure was reported in children from other ME countries in western North Africa, Saudi Arabia, and Kuwait.^{181,253,254} Primary hyperoxaluria is probably underdiagnosed because, by the time of diagnosis, advanced CKD or early graft dysfunction in isolated kidney transplant recipients may have already developed.²³⁵

Until recently, there was no approved treatment for patients with PH, and conservative management including increased water intake and crystallization inhibitors such as citrate were recommended. Intensive hemodialysis was implemented once they reached kidney failure, and the only cure has been combined liver-kidney transplantation.

Lumasiran (Oxlumo) is the first approved drug for patients with PH1. It was approved by the FDA and EMA in November 2020 for treating PH1 patients of all age groups and subsequently in many sites around the globe including Israel. Lumasiran is a small interfering RNA therapeutic (siRNA) targeting glycolate oxidase (GO), harnessing a naturally occurring mechanism for regulating gene expression by selective depletion of mRNA molecules that encode for target proteins.²⁵⁵ GO is exclusively expressed

in hepatocytes and is likely playing a role in glyoxylate metabolism only. GO catalyzes the oxidation of glycolate to glyoxylate, which is the substrate for oxalate synthesis. Following preclinical studies, the safety of GO suppression in humans was supported by the incidental discovery of GO deficiency in an 8-year-old boy with asymptomatic glycolic aciduria, associated with normal kidney and hepatic functions.²⁵⁶ Targeted delivery of siRNAs to the liver was obtained by conjugation to N-acetylgalactosamine (GalNAc). This allows direct uptake of the drug by asialoglycoprotein receptors, which are almost exclusively expressed by hepatocytes.²⁵⁷ The 6-month double-blind phase of a randomized, placebo-controlled phase 3 study evaluating lumasiran in 39 patients with PH1 enrolling children older than age 6 and adults, with genetically PH1 and eGFR >30 mL/min per 1.73 m².²⁵⁸ Lumasiran was administered subcutaneously at 3 mg/kg monthly for 3 months, followed by quarterly doses. The least squares mean change from baseline in 24-hour urinary oxalate (Uox) excretion was 65.4% in the lumasiran group and 11.8% in the placebo group. The proportion of lumasiran-treated patients who achieved normalization or near normalization of 24-hour Uox excretion at 6 months was 84%, compared with 0% in placebo-treated patients. There were no serious or severe adverse events. The most common adverse event was a mild and transient injection site reaction in 38% of lumasiran-treated patients. Significant decrease in Uox levels associated with a decline in the rate of stone events and improvement in nephrocalcinosis, assessed by sequential kidney sonographic studies, have been shown in long-term studies (12 months) of all age groups (including infants and toddlers).²⁵⁹⁻²⁶¹ In PH1 patients with CKD G3b-5 and those treated with hemodialysis, plasma oxalate was used as an endpoint since urinary oxalate excretion decreases with declining GFR.²⁶² Similar trend of a decrease in plasma oxalate was recorded in patients in all age groups treated with lumasiran. Small case series showed that isolated kidney transplantation combined with lumasiran therapy may be a fair substitution for combined liver-kidney transplantation, although larger studies with long-term follow-up are required.²⁶³

Nedosiran, a siRNA targeting the lactate dehydrogenase A (LDHA) enzyme, has been developed. Similarly to lumasiran, conjugation with GalNAc allows selective silencing of LDHA in the liver, preventing off-target effects in other tissues, especially in muscles.

Since LDHA mediates the final enzymatic reaction of oxalate production, nedosiran can potentially be beneficial in all types of PH. Whereas nedosiran has been shown to be beneficial in patients with PH1 by lowering Uox levels with a favorable safety profile, preliminary results in patients with PH2 and PH3 did not demonstrate the expected beneficial effect.^{264,265}

Cystinuria is a recessively inherited kidney stone disease, prevalent in various ethnic groups in ME countries.^{181,253,266} Cystinuria is caused by biallelic mutations in either one or both of two genes:

1. The *SLC3A1* gene on chromosome 2p16.3-p21 encodes rBAT, a protein required for normal trafficking of the amino acid transporter to its appropriate location in the apical membrane of affected epithelia. In its mutated

form, it leads to impairment of cystine and dibasic amino acid (e.g., ornithine, lysine, and arginine) transport.

2. The *SLC7A9* gene located on chromosome 19q12-13.1 encodes an amino acid transporter, b⁰+AT. When this protein is defective, its appearance at the apical membrane, assuming that rBAT is normal, does not lead to normal amino acid reabsorption.

Both proteins together form a heterodimer that mediates sodium-independent transport of cystine and dibasic amino acids (ornithine, arginine, and lysine) in the apical membrane of the proximal tubule and small intestine.

Because parents of patients with cystinuria due to *SLC7A9* mutations have increased levels of urinary cystine excretion, they may sometimes form stones if the urine volume is low or if animal protein intake is high. Otherwise, disease manifestations caused by either mutation are similar.²⁶⁷⁻²⁶⁹ Some patients may bear heterozygous digenic mutations, and they do not tend to form cystine stones unless there are mutations in two copies of a single gene. Mutations in *SCL3A1* have been detected in Turks, Muslim Arabs, Druze, Ashkenazi, and Sephardic Jews of Persian and Yemenite origin. The disease is also common among Libyan Jews, in whom the estimated prevalence is 1/2500 and the corresponding carrier rate is 1/25. In this population, the disease is caused by a single founder mutation, V170M, in the *SLC7A9* gene.²⁶⁷⁻²⁷³

Cystinosis is a recessively inherited lysosomal transport defect leading to the accumulation of cystine. It presents with renal Fanconi syndrome and results in CKD with various extrarenal manifestations (see Table 76.5). The prevalent causes of mutations in a few countries of the ME have been summarized.²⁷⁴ A founder mutation among Jews of Moroccan ancestry has been identified (Y. Frishberg, personal communication).

Fabry disease is a rare X-linked inborn error of the glycosphingolipid metabolic pathway caused by deficiency of α -galactosidase A. A mutation in the gene encoding this enzyme results in insufficient breakdown of lipids, which accumulate to harmful levels in the eyes, kidneys, autonomic nervous system, and cardiovascular system. In untreated patients, the accumulation of globotriaosylceramide (Gb3) in lysosomes may result in multiple organ damage that includes the development of kidney failure between the fourth and fifth decades of life. Histologic evidence of Fabry disease has been detected in graft biopsy samples many years after successful kidney transplantation.^{275,276} Although most disease features have been reported in adults, a pediatric disease phenotype that includes acroparesthesias, skin manifestations, and glomerular alterations has been described.^{276,277} Fabry disease has been diagnosed in families in Israel and Turkey.²⁷⁵⁻²⁷⁹ The disease may be underdiagnosed in other ME countries because of phenotypic variability of the disease, with often nonspecific signs and symptoms or insufficient awareness. However, screening of adult patients with kidney failure undergoing dialysis in Turkey and in Lebanon has yielded an extremely low prevalence rate of Fabry disease.^{280,281} Similarly, screening adults from Turkey with left ventricular hypertrophy yielded 1.58% with genetically proven Fabry disease, mostly males with severe hypertrophy.²⁸²

Bartter syndrome and Gitelman syndrome belong to a group of inherited salt-losing tubulopathies with distinct phenotypes; they are caused by inherited defects in ion transporters in the loop of Henle and distal convoluted tubule, respectively. Most cases of the variants of Bartter syndrome in the ME have been reported in Israeli Arabs, in large Bedouin communities living in southern and northern Israel, and in Kuwaiti children.^{95,283,284}

Congenital anomalies of the kidney and urinary tract (CAKUT) comprise the most common cause of CKD in childhood. To understand the underlying pathogenetic mechanisms better, Vivante and associates²⁸⁵ have identified 20 Israeli pedigrees with isolated nonsyndromic renal hypodysplasia and screened for mutations in genes known to be involved in kidney development. Two brothers were found to have a heterozygous *PAX2* nonsense mutation, responsible for renal coloboma syndrome, which includes kidney dysplasia, eye coloboma, and hearing impairment. Nine affected subjects from two unrelated families were found to harbor heterozygous *HNF1B* mutations, encoding the hepatocyte nuclear factor-1 β . These mutations are associated with variable renal phenotypes, maturity-onset diabetes of the young (MODY5), hypomagnesemia, and hyperuricemia. In one family, two affected brothers were heterozygous for a missense mutation in *WNT4*. Functional analysis of this variant in different cell lines revealed both agonistic and antagonistic canonical WNT stimuli. In primary cultures of human fetal kidney cells, this mutation causes loss of function, resulting in diminished canonical WNT/ β -catenin signaling. These findings were interpreted as suggestive of a role for heterozygous *WNT4* variants in renal hypodysplasia.²⁸⁵ A three-generation Yemenite Jewish family with seven individuals who had CAKUT was investigated by whole-exome sequencing. Renal hypoplasia-dysplasia was the predominant phenotype (six individuals), together with vesicoureteral reflux (VUR; four individuals) and/or ectopia (two individuals).²⁸⁶ A heterozygous truncating mutation of the nuclear receptor interacting protein 1 gene (*NRIP1*) was identified in all seven affected members. *NRIP1* encodes a nuclear receptor transcriptional cofactor that directly interacts with the retinoic acid receptors (RARs) to modulate retinoic acid transcriptional activity.

Some rare genetic kidney diseases that have been reported in ME communities are listed in Table 76.5.²⁸⁷⁻³⁰⁶ Other nonrenal genetic diseases may cause glomerular and tubulointerstitial complications; an example is sickle cell anemia, a hemoglobinopathy that is prevalent in the ME countries in western North Africa and the Arabian Peninsula.^{181,189,268,307,308} Inherited diseases of the kidney are discussed further in Chapters 44, 45, and 71.

Familial Mediterranean fever

FMF is the most common of the hereditary periodic fever syndromes and is a prevalent recessively inherited disease in the ME. FMF is an autoinflammatory disease that is characterized by recurrent attacks of fever, serositis, arthritis, and erysipelas-like skin lesions. The disease affects several ethnic groups in the ME, including Sephardic Jews, Armenians, Turks, and Arabs.³⁰⁹ The most significant complication of FMF is renal amyloidosis that progresses to nephrotic syndrome and kidney failure. Renal amyloidosis is

occasionally diagnosed in patients without a typical history of FMF attacks.

In the 1990s, two groups identified the *MEFV* gene by positional cloning as the underlying genetic cause of FMF.^{310,311} At least 370 mutations in the *MEFV* gene have been reported, and more than 195 of these mutations have been found to be associated with FMF.³¹² The *MEFV* gene encodes the protein pyrin (or marenostin). The gene is located on the short arm of chromosome 16 (16p13.3) and includes 10 exons that encode 781 amino acids. Pyrin is expressed predominantly in polymorphonuclear cells (PMNCs), eosinophils, and monocytes, but not in lymphocytes. It is also expressed in dendritic cells and fibroblasts from the synovium, peritoneum, and skin. Pyrin has an important role in the innate immune system, which constitutes a primary defense against external pathogens and other noxious agents.³¹³ Its exact role and mechanism of action appear to be in sensing changes in the Rho GTPases.³¹⁴ Rho proteins are involved in controlling GTPase activity, which in turn are important in the regulation of actin and tubulin dynamics. Actin-tubulin interactions contribute to neutrophil motility and chemotaxis. Bacterial toxins (e.g., *Clostridium difficile* toxin) may modify the effect of Rho proteins on GTPases, thereby inhibiting actin-tubulin activity and neutrophil chemotaxis. Pyrin does not directly recognize Rho modification but probably senses an event downstream of Rho modification in the actin cytoskeleton pathway and forms a caspase-1-activating inflammasome.^{315,316} Therefore pyrin-mediated innate immunity is unique in that it senses bacterial virulence rather than microbial molecules.³¹⁵

In patients with FMF, the mutations in the *MEFV* gene lead to the production of pyrin protein, which is capable of constructing the inflammasome, even in the absence of a known external trigger such as a toxin or infectious agent. This possibility suggests that the mutations in *MEFV* gene cause gain of function of pyrin protein so that it acts without a known provocation. The outcome of this process is the secretion of interleukin (IL)-1, IL-18, and other mediators of inflammation, which enhance chemotaxis and neutrophilia and induce an attack of FMF.

The carrier frequency of mutant alleles is high, as much as 1:5 to 1:3, in certain populations of the ME countries (Armenians, Jews, and Turks). The most commonly reported mutations are *M694V*, *V726A*, *M680I*, and *M694I* in exon 10 and *E148Q* in exon 2.^{317,318} The first four mutations are believed to be pathogenic, with the *M694V* homozygotes having the most severe phenotype. Although the *E1489Q* homozygotes are asymptomatic in 50% of cases, this variant may be associated with other systemic inflammatory diseases, including Behçet disease, vasculitis, ulcerative colitis, rheumatoid arthritis, and multiple sclerosis.³¹⁹ The most frequent *MEFV* mutations in the ME populations are listed in Table 76.6.³¹⁹⁻³⁴⁷ Data from five databases encompassing whole-genome and whole-exome datasets for 2115 individuals from multiple subpopulations in the ME were available for analysis.³⁴⁸ A compendium for *MEFV* genetic variants was also created. The carrier frequency for genetic variants in *MEFV* was found to be 8%, with differences in allele frequency among subpopulations. Also, three pathogenic variants were added to the four popular studied variants. The former include c.1261-11T>G, Y688X, and E148V. Mutations in the *MEFV* gene have also been reported

in Spanish, Italian, Greek, Portuguese, Indian, Chinese, and Japanese populations.^{349,350} Most *MEFV* mutations are single amino acid substitutions (missense), and some patients with FMF have a single *MEFV* mutation. The latter may imply a dominantly inherited trait, which may go along with a gain of function of the gene product. Alternatively, additional genes or immune factors may modulate the innate immune response in FMF.

The clinical spectrum of FMF and the elucidation of the molecular biology, structure, and regulation of pyrin and its role during inflammation have been reviewed.^{316,351-354} Pyrin belongs to the large family of proteins sharing a conserved domain structure with the tripartite motif (TRIM) consisting of an N-terminal RING domain, B-box domain(s), and C-terminal coiled-coil domain. In pyrin, the RING domain with the ubiquitin ligase activity is replaced by the PYD domain, which belongs to the death domain. It also carries an additional approximately 200-amino acid C-terminal B30.2/rfp/PRY/SPRY domain. Most of the disease-associated mutations are located in the C-terminal B30.2 domain. The B30.2 domain is present in primate and human pyrin, but not in rodent pyrin.³⁵⁴ It includes a SPRY domain that is located in front of the C-terminal region and a SPRY extension at the N terminus. The SPRY domain is a protein interaction module, which is implicated in several biologic pathways, including those that regulate innate and adaptive immunity.³⁵⁵

During the past decade, the functions of pyrin have been the subject of intensive research. Advances in our understanding of its functionality are the result of identifying its interaction with different proteins and oligomers.³¹⁶ In particular, the PYD domain, via cognate pyrin domain association, interacts with an adapter protein termed *apoptosis-associated specklike protein with a caspase recruitment domain* (ASC)³⁵⁶ and thus participates in the regulation of apoptosis, inflammation, and IL-1 β processing.³⁵⁷ The B30.2/rfp/PRY/SPRY domain-mediated protein interactions indicate the role of this domain as an adaptor module to assemble macromolecular complexes.³⁵⁸ As a part of the pyrin inflammasome, pyrin also acts as a pattern recognition receptor, sensing pathogen modification and inactivation of Rho GTPases (see earlier).³¹⁴

Paradoxically, pyrin also acts as a component directing the inflammasome components NLRP1, NLRP3, and procaspase 1 to selective autophagic degradation.³⁵⁹ On its own, the dimerization of B30.2 domains by means of the CHS domain appears to be crucial for the recognition of higher-order oligomers.³⁶⁰ Inflammasomes contain members of the nucleotide-binding oligomerization domain, leucine-rich repeat, and PYRIN domain-containing subfamily of proteins, now known as NLRP proteins. Mutations in the gene expression of the NLRP protein, *NLRP3* (cryopyrin), are associated with monogenic autoinflammatory diseases—cryopyrin-associated periodic syndromes.³⁶¹ Adults with FMF, evaluated in Israel, are at increased risk of stroke and venous thromboembolism.^{362,363}

Clinical spectrum and kidney disease. The wide clinical spectrum and possible genotype-phenotype correlations in FMF are influenced by genotype (extent and position of the *MEFV* mutation), ethnicity, and environmental factors. Although mutations are found throughout the entire *MEFV*

Table 76.6 Genotype-Phenotype Correlations in Several Patient Populations With Familial Mediterranean Fever

Population, Community	Frequent Mutations	Phenotype	Amyloidosis (% of Patients)	Associated Syndromes and Diseases	Reference
Israeli Jews and Arabs	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i>	Arthritis, fever, serositis, vasculitis	1.4% (associated with mutation in <i>M694V</i>)	NR	Brik et al ³³¹
Israeli Jews and Arabs	<i>M694V</i> (very common in Jews), <i>M694I</i> (exclusive in Arabs), <i>M680I</i> , <i>V726A</i> , <i>E148Q</i>	Arthritis, fever, serositis, vasculitis	NR	NR	Ben-Chetrit et al ³²⁹
Israeli Arabs	<i>M694V</i> (associated with severe disease), <i>V726A</i> (most common)	Arthritis, fever, serositis, vasculitis	NR	NR	Shinawi et al ³⁴⁴
Israeli Jews of North African origin and Arabs ^a	<i>M694I</i> , <i>M694V</i> (common in North African Jews), <i>E148Q</i>	Synovitis, pleuritis, abdominal pain, skin rash	95% ^a	Focal glomerulosclerosis	Ben-Chetrit and Backenroth ³²⁸
Israeli Jews of North African origin, Ashkenazi Jews, Jews of Iraqi origin, Israeli Arabs, and Druze	<i>E148Q</i> , <i>M694V</i> (common in North African Jews), <i>V726A</i>	FMF criteria ³³⁷	4.6% (most common in <i>M694V</i> homozygous)	NR	Zaks et al ³⁴⁷
Israeli Jews (Ashkenazi and non-Ashkenazi), Arabs, and Druze	<i>M694V</i> (common in Jews), <i>E148Q</i> , <i>M694V</i> , <i>V726A</i> (equally common in Arabs), <i>E148Q</i> (most common in Druze)	Fever, serositis	NR	NR	Sharkia et al ³⁴³
Turks	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, chest pain, skin rash	3% (mostly associated with <i>M694V</i>)	NR	Solak et al ³⁴⁵
Turks	<i>M680I</i> , <i>M694V</i> , <i>V726A</i>	Abdominal pain, fever, arthralgia, pleuritis, muscle pain, skin rash	12.9%; 0.9% as the main disease manifestation (phenotype II, associated with <i>M694V</i>)	Nonamyloid renal disease, Henoch-Schönlein purpura, polyarteritis nodosa, Behçet syndrome, rheumatic fever, uveitis, inflammatory bowel disease	Tunca et al ³⁴⁶
Turks ^b	<i>M680I</i> , <i>M694V</i> (most common), <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, chest pain, erysipelas-like lesion, vomiting, family history of renal failure	0%	NR	Caglayan et al ³³²
Turks	<i>M680I</i> , <i>M694V</i> (most common), <i>M694I</i> , <i>V726A</i> , <i>E148Q</i> (common)	Abdominal pain, fever, arthritis, pleuritis, erysipelas-like erythema, peritonitis	NR	NR	Ozdemir et al ³²³
Turks	<i>M680I</i> , <i>M694V</i> (most common), <i>V726A</i> , <i>E148Q</i>	Fever, arthritis, pleuritis, erysipelas-like erythema, peritonitis, vasculitis	8.6% (most common in <i>M694V</i> homozygous)	NR	Kasifoglu et al ³²¹
Azeri Turks	<i>M680I</i> , <i>M694V</i> (most common), <i>M694I</i> , <i>V726A</i> , <i>E148Q</i>	Fever, serositis, synovitis, kidney failure	NR	NR	Mohammadnejad and Farajnia ³²²
Jordanian Arabs Palestinian Arabs	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, myalgia, skin rash	1% (associated with <i>M694V</i>)	Protracted fever, myalgia syndrome; 42% homozygous for <i>M694V</i>	Langevitz et al; ³³⁶ Majeed et al ³³⁸

Table 76.6 Genotype-Phenotype Correlations in Several Patient Populations With Familial Mediterranean Fever (Continued)

Population, Community	Frequent Mutations	Phenotype	Amyloidosis (% of Patients)	Associated Syndromes and Diseases	Reference
Jordanian Arabs	<i>M680I</i> , <i>M694V</i> , <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia	9% (<i>M694V</i> in 1 patient, <i>V726A</i> / <i>M680I</i> in 2 patients)	Celiac disease, folliculitis	Medlej-Hashim et al ³⁴⁰
Palestinian Arabs	<i>M680I</i> , <i>M694V</i> , <i>V726A</i> , <i>E148Q</i>	NR	NR	NR	Ayesh et al ³²⁶
Arabs from Jordan, Egypt, Syria, Iraq, and Saudi Arabia	<i>M694V</i> , <i>V726A</i> , <i>E148Q</i>	NR	NR	NR	Al-Alami et al ³²⁵
Iranian Azeris who live in Turkey	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, pleuritis, skin rash	7%	NR	Esmaeili et al ³³⁵
Iranians (72% Azeri)	<i>M680I</i> (most common), <i>M694V</i> , <i>V726A</i>	Fever, peritonitis, arthralgia, pleuritis, skin rash	5.6%	NR	Bidari et al ³³⁰
Egyptian Arabs	<i>M680I</i> , <i>M694V</i> , <i>V726A</i>	Abdominal pain, chest pain, fever, arthritis, myalgia	NR	NR	Settin et al ³⁴²
Egyptian Arabs	<i>M680I</i> , <i>M694I</i> (most common), <i>M694V</i> , <i>V726A</i> , <i>E148Q</i>	Fever, serositis	Mostly associated with <i>M694V</i>	NR	el Gezery et al ³³³
Egyptian Arabs	<i>V726A</i> (most common), <i>M694V</i> (common), <i>M680I</i> , <i>M694I</i> , <i>E148Q</i>	Abdominal pain, fever, arthralgia, pleuritis, skin rash, myalgia	2.9% (associated with <i>M694V</i>)	NR	el Garf et al ³²⁰
Syrian Arabs	<i>M680I</i> , <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i> , <i>A744S</i> , <i>R761H</i>	Serositis, fever, arthritis, pleuritis	5%	NR	Mattit et al ³³⁹
Lebanese Arabs	<i>M680I</i> , <i>M694V</i> (frequent), <i>M694I</i> , <i>V726A</i> , <i>E148Q</i> (frequent); minor alleles also detected	Serositis, fever, arthritis, chest pain	NR	NR	Sabbagh et al ³⁴¹
Algerian, Moroccan, and Tunisian Arabs	<i>M694V</i> , <i>M694I</i> (most common), <i>M680I</i> , <i>M680I</i> , <i>A744S</i> , <i>V726A</i> , <i>E148Q</i>	Serositis, fever, arthritis, chest pain	NR	NR	Belmahi et al ³²⁷
Algerians	<i>M694I</i> (most common), <i>M694V</i> , <i>E148Q</i> , <i>A744S</i> , <i>M680I</i>	Abdominal pain, fever, arthritis, chest pain, erythema	8%	NR	Ait-Idir et al ³²⁴
Tunisian Arabs	<i>M680I</i> (most common), <i>M694V</i> , <i>M694I</i> , <i>V726A</i> , <i>E148Q</i> , <i>A744S</i> , <i>R761H</i> , <i>1692del</i>	FMF criteria ³³⁷	3.5%	NR	Chaabouni et al ³³⁴

^aStudy performed in patients with end-stage kidney disease.

^bStudy performed in compound heterozygous patients.

FMF, Familial Mediterranean fever; NR, not reported.

gene in patients with FMF, the mutations with the most severe forms of disease are clustered in exon 10, which encodes a motif known as the B30.2/SPRY domain at the C terminus of the protein. Genetic variants found in exons 2

(e.g., *E148Q*) and 3 are usually associated with less severe clinical presentations of FMF or even just mild nonspecific inflammatory manifestations. Overall, *M694V* homozygotes have a severe phenotype and are more likely to have arthritis,

renal amyloidosis, erysipelas-like skin lesions, high fever, splenomegaly, and more frequent attacks as compared with individuals with other *MEFV* mutations (see Table 76.6).³⁶⁴ In addition, patients with these mutations require higher doses of colchicine to prevent attacks as compared with patients with other genotypes. Among Arab patients with FMF, the worst disease severity is associated with alleles carrying the mutations *M694V/M694V* and *M694V/M726A*, whereas the *M694I/M694I* mutation is associated with a mild form of the disease.³⁶⁵ In Turkish patients with FMF, the common *M694V* mutation is associated with more severe disease but not with amyloidosis.^{346,353} In contrast, the homozygous *M694V* mutation was reported to be associated with amyloidosis in Jews from North Africa, in Arabs, and in Turks and also with the protracted febrile myalgia syndrome, which is a type of myalgia that is associated with FMF in Arabs.^{321,366}

FMF has traditionally been considered an autosomal recessive genetic disease, but as many as 25% of patients with clinical FMF have only one *MEFV* mutation.^{345,346,352} This finding could explain the vertical transmission of the disease in some families. In patients with FMF who have a single *MEFV* mutation, an additional but less common mutation may account for the disease, but this has not yet been confirmed in carefully performed studies. A dominant segregation of a heterozygous *MEFV* mutation was reported in some families. Therefore it was suggested that complex alleles can lead to a more severe form of disease.³⁶⁷ One study has reported that heterozygous mutations at amino acid position 577 of pyrin (T577) can induce autosomal dominant autoinflammatory disease. The T577 is located in front of the C-terminal B30.2/SPRY domain and is crucial for pyrin function.^{364,368} The T577 mutations were found in a family of Turkish descent with autosomal dominant FMF phenotype and also in European patients with other periodic syndromes. The impact of *MEFV* mutations has been shown in other inflammatory diseases, such as Henoch-Schönlein purpura, polyarteritis nodosa, Behçet disease, rheumatic heart disease, and rheumatoid arthritis (reviewed by Guz and associates).³⁵³ Of interest, individuals who carry a single mutated allele in the *MEFV* gene can also suffer from the syndrome of periodic fever, aphthosis, pharyngitis, adenitis (PFAPA), ankylosing spondylitis, and Crohn's disease.³⁶⁹

Secondary or reactive AA amyloidosis is the most severe complication of FMF. Before the advent of colchicine treatment, amyloidosis was reported to occur in approximately 75% of patients with FMF who were older than 40 years.³⁷⁰ The disease is caused by the extracellular deposition of amyloid A fibrils. These fibrils consist of β -pleated sheet polymers of the N-terminal fragments, the products of incomplete proteolytic digestion of the acute-phase precursor serum amyloid A (SAA) protein, whose production is markedly increased in chronic inflammatory processes.³⁷¹ Amyloidosis is frequently found in Jews of North African descent, in Turks, and in Armenians.³⁶⁶

The *M694V* mutation has also been found in patients with FMF phenotype II, in which amyloidosis is detected before the onset of clinical symptoms of FMF.³⁴⁶ Phenotype III is defined as the presence of two *MEFV* mutations (homozygous or compound heterozygote state) without clinical manifestation of FMF or amyloidosis. This condition, however, like phenotype II, could predispose to the development of amyloidosis in families in which one sibling

is afflicted with FMF.³⁷² A screening study of patients with FMF from 14 countries worldwide—which included Israel, Turkey, Qatar, Jordan, and Lebanon—has revealed that amyloid nephropathy is present in 11.4% of patients with FMF.³⁷³ In this study, however, the country of recruitment, rather than *MEFV* genotype, was found to be the leading risk factor for amyloidosis.³⁷³ These findings underscore the relative contribution of environmental factors to the genetic background in defining the phenotypic variation and severity of the disease.³⁷⁴

Male gender, a positive family history of amyloidosis, pain flares in joints, and the SAA1 α/α genotype have been suggested as additional risk factors for amyloidosis.³⁶⁷ Albuminuria is usually the first manifestation of renal involvement in patients with FMF and amyloidosis, and in untreated patients it may appear early in the course of FMF.³⁷⁵ In one study, the serum cystatin C level was found to be significantly increased in patients with FMF and secondary amyloidosis. It was suggested that the cystatin C serum level may be an early marker of renal impairment, even before the onset of albuminuria.³⁷⁶ Proteinuria, mostly represented by albumin, can reach the nephrotic range, with urinary protein excretion rates higher than 20 g/day. If untreated, AA amyloidosis progresses to kidney failure, which necessitates KRT. Renal disease due to amyloidosis accounts for 35% and 60% of deaths in men and women, respectively, with FMF.³⁷⁷ Extrarenal deposits of AA amyloid can be found in the liver, spleen, lung, thyroid, heart, adrenal glands, stomach, and testes. These deposits can be clinically significant and are associated with intestinal malabsorption and adrenal insufficiency, which are often diagnosed after the initiation of chronic hemodialysis or after kidney transplantation.^{375,378,379}

The diagnosis of amyloidosis can be confirmed by tissue biopsy; the sensitivities of the rectal or bone marrow biopsy and renal biopsy are 79.5% and 88%, respectively, whereas the sensitivity of biopsy of abdominal fat is lower.³⁸⁰⁻³⁸² The presence of amyloid deposits is demonstrated histologically with Congo red staining of the biopsy tissue sample. Congo red-stained amyloid has an orange appearance when viewed under a light microscope and an apple green birefringence when viewed under polarized light. Another amyloid-specific stain, thioflavin T, is used less frequently than Congo red. Amyloidosis is diagnosed definitively on electron microscopy by the demonstration of characteristic amyloid fibrils. The AA protein in tissue can also be detected with antibodies against AA protein.³⁸⁰

In addition to AA amyloidosis, other kidney diseases have been diagnosed in patients with FMF. It is worth mentioning that many of the reports of systemic and inflammatory diseases in patients with FMF contain descriptions of kidney involvement.³⁴⁶ In fact, the frequency of the reported nonamyloid glomerular lesions reflects the number of kidney biopsies performed in patients with abnormal findings on urinalysis or kidney function tests. Recurrent focal and proliferative glomerulonephritis and polyarteritis nodosa were described in an early report of the disease.³⁸³ Rapidly progressive glomerulonephritis, mesangial proliferative glomerulonephritis, IgA nephropathy, and membranous nephropathy have also been reported.^{346,352,384-387}

Treatment of familial Mediterranean fever. Daily oral colchicine is the most effective therapy for patients with FMF, and its use since the 1980s has dramatically changed the course of the disease. Colchicine prevents both acute attacks of FMF and SAA amyloidosis.^{388,389} The mechanism of colchicine action is not completely understood. The pathophysiology of FMF involves the recruitment and activation of neutrophils at serosal surfaces. Colchicine affects the motility of neutrophils by reducing their deformability and elasticity. These properties are crucial for their extravasation in response to inflammatory stimuli. On a cellular level, it is believed that one of its major actions is to interact with cytoskeletal structures, such as microtubules. It has been shown that colchicine accumulates in PMNCs, where it depolymerizes the microtubules and suppresses microtubule dynamics. Pyrin is also expressed in PMNCs and associates with microtubules.³⁹⁰ Because colchicine also has an inhibitory effect on chemotaxis and reduces serum levels of IL-6, IL-8, and tumor necrosis factor- α (TNF- α),³⁹¹ its anti-inflammatory action is thought to be related to its ability to suppress nuclear factor-kappa B (NF- κ B) activation by attenuating calpain-mediated I κ B kinase α degradation, which is enhanced by N-terminal cleaved pyrin.³⁹² Suppression of caspase-1 expression in PMNCs was also suggested to be a potential mechanism of the action of colchicine.³⁹³

In vitro, colchicine has been shown to have a specific effect on pyrin and pyrin-interacting proteins, such as proline-serine-threonine phosphatase interacting protein 1 (PSTPIP1) and ASC. In THP-1 cells, colchicine reorganized cytoskeleton and downregulated MEFV expression. This effect, which may result in a reduction of the level of a pro-inflammatory mutant pyrin, may explain the suppressive mechanism of colchicine on FMF attacks.³⁹⁴

The most common adverse effects of colchicine are gastrointestinal, especially abdominal pain and diarrhea. Colchicine toxicity from overdosing is associated with hepatic, renal, muscle, and cerebral effects.³⁹⁵ Approximately 5% to 10% of FMF patients are nonresponders, and 2% to 5% do not tolerate the drug, mainly due to gastrointestinal side effects. In M694V homozygotes, a significant portion of patients show a partial or minimal response to colchicine, despite being treated with appropriate doses.³⁹⁶ In contrast, prolonged colchicine-free remission has been noted in some individuals, none of whom were homozygous for this mutation.³⁹⁷

One study has revealed the existence of an association between polymorphism of the adenosine triphosphate-binding cassette, subfamily B, member 1 (*ABCB1*) gene, and the response to colchicine in Turkish patients with FMF. The *ABCB1* gene encodes p170—also known as multidrug resistance 1 (MDR1)—a glycoprotein that functions as a drug transport pump, which can cause a variety of drugs including colchicine to become extruded from cells. In this study, patients with FMF who had the TT genotype for the c.3435C>T variant of *ABCB1* responded better to colchicine in terms of treatment efficacy and lower dose requirements in comparison with patients who had the CT and CC genotypes.³⁹⁸ These results warrant validation in further studies in patients with FMF from different ethnic populations.

Thalidomide, NSAIDs, corticosteroids, azathioprine, prazosin, and eprodisate disodium, an inhibitor of fibril

formation, had occasionally been used to treat patients with FMF whose disease was refractory to colchicine.^{380,399-401} Interferon- α (IFN- α) was one of the first drugs used to treat patients with FMF who were refractory to colchicine.⁴⁰² Two studies of patients with FMF who were treated with IFN- α have shown that the duration and pain intensity of most of the attacks were reduced by more than 50%.^{403,404} In addition, clinical improvement has been reported in patients with FMF who were treated with the TNF- α antagonists etanercept or infliximab.^{400,405,406}

Because most inflammatory manifestations of FMF are believed to be associated with the induction of IL-1 β production by mutations in the C-terminal B30.2 domain of pyrin, it has been proposed that IL-1 β antagonists may be an effective therapy for FMF. In fact, more studies have suggested that IL-1 inhibition is a safe and effective alternative for patients with FMF who do not respond to or cannot tolerate colchicine. The IL-1 β receptor antagonist anakinra has been shown to have a beneficial effect in patients with FMF and with cryopyrin-associated periodic syndromes.^{399,401} Indeed, significant improvement and resolution of FMF symptoms have been reported in patients with colchicine-resistant FMF (crFMF) who were treated with anakinra.^{401,407,408} Anakinra elicited a beneficial therapeutic response in individuals with FMF and CKD and even in a patient on maintenance hemodialysis and after kidney transplantation.^{408,409} The main limitation of anakinra treatment in patients with FMF is probably the need for daily subcutaneous injections. The administration of canakinumab, a human monoclonal antibody against IL-1 β , every 8 weeks has been associated with rapid remission of symptoms in FMF and cryopyrin-associated periodic syndromes.⁴¹⁰ In a systematic review of 8 studies that included 40 patients with crFMF, rates of complete and partial response to canakinumab were 68% and 32%, respectively.⁴¹¹

Anecdotal reports suggest that IL-1 inhibition has a beneficial effect on amyloidosis, but more data are needed to confirm this observation. Thus for patients who are colchicine resistant and on IL-1 inhibitors, concomitant colchicine at a tolerable dose is recommended to prevent amyloidosis.

Rilonacept is a dimeric fusion protein consisting of the extracellular portions of the human IL-1 receptor and the Fc region of human IgG1 that binds and neutralizes IL-1. A relatively small, randomized controlled study has shown a decrease in the attack rate, but the duration of attacks did not differ between placebo and rilonacept.³⁵⁵ Similarly, a beneficial effect has been noted with the use of rilonacept in cryopyrin-associated periodic syndromes.⁴¹²

Familial Mediterranean fever and kidney failure. In contrast to the poor prognosis of patients with primary (AL) amyloidosis who require KRT, most patients with FMF and secondary (AA) amyloidosis do relatively well with chronic dialysis.⁴¹³ However, the issue of which dialysis modality is ideal for the treatment of FMF-associated ESKF is still unresolved,⁹⁷ and prospective studies that compare the different modalities are still needed. Of the dialysis modalities, hemodialysis is more widely used in patients with FMF. Continuous ambulatory/cycling peritoneal dialysis (CAPD/CCPD), however, has been performed successfully in selected patients. The effects of CAPD on azotemia and overall survival

are similar in patients with nonamyloid diseases and patients with FMF. However, serum albumin levels are reported to be lower, and the rate of peritonitis and requirements for erythropoietin-stimulating agents (ESAs) are higher, in patients with FMF who underwent CAPD than in patients with FMF who underwent hemodialysis.^{414,415}

A considerable number of patients with FMF and ESKF have undergone kidney transplantation in the past few decades. The rates of survival of patients and allografts are reported to be worse or similar to those in the general population of kidney transplant recipients.^{97,346,416-418} Analysis of the data from a single center in Israel has shown that graft survival is comparable between patients with FMF and controls.⁴¹⁹ However, FMF was associated with more than a twofold increased risk for death following transplantation. Infections and cardiovascular disease were the leading causes of death. Maintenance of colchicine therapy is obligatory after kidney transplantation because it may prevent the recurrence both of FMF symptoms and of amyloidosis. However, recurrence of AA amyloidosis has been frequently reported 8 to 10 years after kidney transplantation,⁴²⁰ and the rate of recurrence may be as high as 71%.⁴²¹ Adverse drug effects are seen in kidney transplant recipients with FMF who are taking cyclosporine. In addition, increased gastrointestinal complications can be observed in patients who are treated with a combination of mycophenolate mofetil and colchicine.⁴²²

MANAGEMENT OF END-STAGE KIDNEY FAILURE IN THE MIDDLE EAST

Overall, KRT is available in all ME countries. Dialysis and kidney transplantation are accessible to every patient with ESKF, regardless of the patient's socioeconomic status or whether the patient has health insurance (see [Tables 76.3 and 76.4](#)).^{20,78,88,91,113-119,137-164}

The number of patients offered KRT in some developing ME countries may, however, be affected by late diagnosis and referral, the presence of comorbid conditions, the country's health system, reimbursement, and the availability of dialysis facilities.^{2,23,30,105,117,423,424}

DIALYSIS

PERITONEAL DIALYSIS

Throughout the ME, hemodialysis is the predominant treatment modality for ESKF. Peritoneal dialysis is an underused treatment modality for ESKF, although it has been increasing to include more than 10% of all dialysis patients in Kuwait, Oman, and Qatar (see [Tables 76.3 and 76.4](#)).^{20,91,105,113,116,138,425-427}

Several medical and nonmedical factors play an important role in inhibiting the widespread use of peritoneal dialysis throughout the ME, including the following: 1. Health care providers generally offer low or no reimbursement for peritoneal dialysis; 2. The number of peritoneal dialysis training programs, qualified nephrologists, and skilled dialysis nurses is limited, and the salaries of attending physicians for peritoneal dialysis are low; 3. Many patients have poor education and poor hygiene practices; and 4.

Patients and caregivers have concerns about the high rates of peritonitis.

At the end of 2006, 38,824 patients were treated with dialysis (87.4% with hemodialysis, 12.6% with peritoneal dialysis) in Turkey and 99.7% of all patients undergoing dialysis were covered by the social security system.⁴²⁸ The Turkish experience in establishing a peritoneal dialysis program has been outstanding and could be used as an example by other ME countries. The Turkish Multicenter Peritoneal Dialysis Study Group (TULIP) has established a platform for organizing peritoneal dialysis facilities and units. The group helped standardize patient records and various peritoneal dialysis treatments and maximized the number of patients with ESKF who can benefit from this treatment modality. TULIP has published clinical research that has affected peritoneal dialysis practice locally and worldwide.⁴²⁸ The Turkish peritoneal dialysis program has good rates of survival of patients and technique efficacy, which are comparable with those in Western industrialized countries.⁴²⁹ The peritoneal dialysis dropout rate was 21%, and the incidence of peritonitis was one episode per 35.5 patient-months in 2007. Dyslipidemia was the most common noninfectious complication;¹⁶⁸ cardiovascular diseases (42.3%), followed by infections (19.9%) and cerebrovascular events (13.6%), were the major causes of death among Turkish patients undergoing peritoneal dialysis.^{168,429} Moreover, peritoneal dialysis has been suggested as a means of reducing the seroconversion rate for hepatitis C virus (HCV) in populations with ESKF and a high prevalence of infection;⁴³⁰ such treatment thereby confers an advantage to kidney transplantation candidates.⁴³¹

Peritoneal dialysis was started in Iran in 1978 and in Saudi Arabia and Kuwait during the 1980s with imported peritoneal dialysis solutions. The costs of this treatment are low in Turkey and Iran because peritoneal dialysis solutions began to be produced locally in Turkey in 1994 and in Iran in 1995.¹⁵⁶ Establishment of a peritoneal dialysis registry and a multidisciplinary approach has dramatically increased the numbers of patients undergoing peritoneal dialysis in Iran since 2001.^{138,156} In 2010, the 1-, 3-, and 5-year patient survival rates in Iran were reported as 89%, 64%, and 49%, respectively, whereas death- and transplantation-censored technique survival rates were 90%, 73%, and 58%.¹³⁸ The main causes of death among Iranian patients undergoing peritoneal dialysis were cardiac events (46%), cerebral stroke (10%), and infection (8%). More recently, the rate of peritonitis has improved from one episode in 15.2 to 19.4 patient-months to one episode in 25 patient-months (in 2010).^{138,156}

The use of peritoneal dialysis has been growing in Qatar since its introduction in 1997, and peritoneal dialysis is offered to approximately 23% of patients with ESKF. Most of these patients are non-Qatari male expatriates, a group that favors a mode of dialysis that can be self-administered, secures the ability to work or travel, requires less dietary restrictions, and potentially decreases the hospital and medical costs.⁴²⁵ These advantages of peritoneal dialysis, which provides flexibility and ease of adjustment for lifestyle and daily schedule, have encouraged some clinically stable Muslim patients on peritoneal dialysis to fast during the holy month of Ramadan.⁴³² However, patients who intend to fast need to be comprehensively educated and followed up by their peritoneal dialysis units.

In contrast, with the expansion of community-based hemodialysis centers in Israel, the percentage of PD from all dialysis has dropped from 34% in 1990 to around 6% to 8% in recent years. PD is only available in Israel in public hospitals.^{107,113,116,433} A limited national survey showed that the major barrier to PD utilization in Israel is the inability of elderly patients to perform self-care dialysis or the absence of a caregiver, among other perceived barriers.⁴³³

HEMODIALYSIS

Although there are limited published detailed data on hemodialysis in each ME country regarding practice patterns and outcomes, global and regional initiatives have filled some gaps in our knowledge.^{20,91,105,107,113-117,186,187,434-438}

In general, quality is ensured according to best medical practice guidelines in industrialized ME countries with more resources, easily accessible health care systems, and availability of advanced medical supplies.^{142,143,159,177,439-443} In ME countries that lack resources, the quality of dialysis is dictated by nonmedical financial considerations. In these countries, dialysis may be offered only once or twice weekly because of recent conflicts, lack of skilled personnel, distance of the dialysis centers from the patient's residence, and inability of the patient to pay for more frequent treatment.^{23,30,102,117,159,172,444} The number of patients who withdraw from hemodialysis therapy because of financial problems has not been documented, however. A small number of patients do seek pre-ESKF care in most ME countries.^{159,445-447}

Practice guidelines for hemodialysis have been developed in Egypt. These included five main domains: personnel, patient care practices, infection prevention and control, facility, and documentation or records. Before distribution of the guidelines, one study had disclosed flawed compliance and variability in adoption of evidence-based hemodialysis guidelines among 16 Egyptian facilities affiliated with the Ministry of Health and Population.⁴⁴⁸ Therefore it is mandatory to distribute these guidelines to all hemodialysis facilities in Egypt, along with the organization of workshops to educate staff and follow-up to ensure implementation of the guidelines. In Libya, the provision of dialysis was adequate in 2010. However, several areas for improvement were identified by Alashek and associates,¹⁷² such as lack of national dialysis practice guidelines, and few clinical practitioners were familiar with KDOQI guidelines. Because of continuous drought, many dialysis centers were obliged to dig their own wells to maintain the water supply. However, testing of water quality was not regularly practiced. In addition, wide discrepancy in methods applied to manage and monitor patients was observed among centers.^{172,174} Likewise, dialysis water quality, reported from dialysis units in Iraq, was not compatible with international standards.⁴⁴⁹

According to the 2016 SCOT report,¹⁴³ 16,315 patients undergoing hemodialysis were treated in 243 centers in Saudi Arabia. Of these patients, approximately 17% were on active waiting lists for kidney transplantation. Bicarbonate dialysis solutions were used in all centers, and only a few centers (6.3%) still use acetate dialysis solutions.⁴⁵⁰ The predominant type of vascular access for hemodialysis varies among the Gulf Cooperation Council (GCC) countries.⁴⁴² In Saudi Arabia, arteriovenous fistulas (71.2% of all patients) are mostly used, followed by jugular catheters (12.6%) and

arteriovenous grafts (7.6%),⁴⁵⁰ whereas in Kuwait 56% of patients undergo dialysis via catheters.⁴⁴² In Tehran, arteriovenous fistula is the type of vascular access used most in patients receiving hemodialysis (91%); only 3% of such patients have arteriovenous grafts, and 4% have permanent catheters.¹⁷⁹ Arteriovenous fistula was also preferred in Turkey. In 2016, it was used in 79.1% of Turkish patients undergoing routine hemodialysis, whereas permanent tunneled catheters and temporary catheters were used in 19.4% of patients.¹⁵⁹

Reuse of dialyzers is not practiced in Saudi Arabia, Libya, and Iran^{172,451} and is prohibited by law in Egypt.⁴⁴⁵ Approximately 87% of Saudi patients undergoing hemodialysis are treated with ESAs and, of these patients, approximately 23% have hemoglobin levels lower than 10 g/dL. In Turkey, ESAs are administered to 61.8% of patients undergoing hemodialysis. In comparison, 48.2% of patients undergoing hemodialysis in Tehran have anemia as a result of low-dose treatment with ESAs, and dialysis may be inadequate because the mean measured treatment adequacy for these patients was 0.97 ± 0.25 .¹⁷⁹ Thus common clinical practice for hemodialysis in Iran does not meet all the targets recommended by international guidelines.^{179,452}

Limited studies have been conducted to describe the patterns, prevalence, or therapy of CKD-associated mineral bone disorder (CKD-MBD) in patients undergoing dialysis in ME countries.^{159,440} In developing ME countries, the use of aluminum-based phosphate binders, presence of high amounts of strontium in the soil, and use of dialysis acetate concentrates that are contaminated with strontium have been implicated as causes of osteomalacia in adult patients undergoing hemodialysis and of rickets in pediatric patients undergoing hemodialysis.^{453,454} Hyperphosphatemia and its complications are serious problems in patients receiving dialysis in developing ME countries; hyperphosphatemia results in part because of limited patient knowledge of, or access to, an appropriate diet. In a large cross-sectional multicenter study that included 1005 patients in Egypt who were undergoing hemodialysis, two-thirds of the patients had hyperphosphatemia and one-third had an elevated calcium-phosphate product. These patients used predominantly calcium-based phosphate binders.⁴⁵⁵ In most developing ME countries, economic considerations hinder the use of the non-calcium-based phosphate binders, calcium-sensing receptor agonists (calcimimetics), or vitamin D receptor agonists. Therefore prolonged or additional dialysis sessions are frequently prescribed to control hyperphosphatemia in patients undergoing hemodialysis.^{159,179,440,455}

The results of one survey have suggested that practicing Saudi nephrologists have an adequate perception of the morbidity in patients with CKD-MBD.⁴⁵⁶ However, the results of this survey also identified a need for creating national guidelines because the physicians' assessment of the prevalence, patterns, and results of therapeutic interventions in patients with CKD was relatively inadequate. Approximately 25% of their patients had hyperphosphatemia (serum phosphorus levels >6 mg/dL), and 20% had hypocalcemia (serum calcium levels <8.4 mg/dL), although vitamin D was administered orally to most patients. In view of the status of CKD-MBD in the ME and the associated economic and logistic constraints, a committee of experienced nephrologists from the region

was created. Its members examined the requisite KDIGO guidelines and formulated recommendations that can be implemented practically for the management of CKD-MBD in the ME.⁴⁵⁷ GCC HD population is significantly younger than in western European countries, Japan, and North America.^{186,187,438} During the 2012–2018 time period, the GCC Dialysis Outcomes and Practice Patterns Study (DOPPS) showed a U-shaped relation between PTH and mortality, approximately—2- and 1.5-folds higher mortality for patients with PTH above 700 pg/mL (observed in 20% of study cohort) and <300 pg/mL (observed in 23% of cohort), respectively, compared with patients with PTH measured between 301 and 450 pg/mL.¹⁸⁶ The GCC DOPPS also revealed high comorbidity burden among GCC patients with diabetes. After adjusting for covariates pertaining to demographics, years on dialysis, body mass index, comorbidities, single-pool Kt/V, catheter use, and laboratory values, the hazard ratio of mortality for DM versus non-DM patients was 1.72 (95% confidence interval 1.23–2.39).¹⁸⁷

MANAGEMENT OF END-STAGE KIDNEY DISEASE WITH VIRAL HEPATITIS

Available global data have indicated that the prevalence of chronic infection with hepatitis B virus (HBV) and HCV is high in African and ME populations.^{458–463} These two viral infections have considerable effects on morbidity and mortality among patients with ESKD, as well as on graft survival in kidney transplant recipients.^{464–468} However, surveys are still needed to estimate the incidence and prevalence correctly of these infections in patients with CKD and ESKF in each ME country. In addition, screening will identify patients who would benefit from a number of highly efficacious, direct-acting, antiviral oral therapies.^{468–470} The HCV epidemic is particularly devastating in Egypt, where its prevalence in the general population is estimated to be more than 18%. The start of the HCV epidemic in Egypt is attributed to the use of unsterilized needles and syringes during the mass antischistosomiasis treatment programs that were conducted during the 1960s and 1970s.⁴⁷¹ Several studies have reported a dismal prevalence of anti-HCV serologic findings among Egyptian patients undergoing hemodialysis and kidney transplant recipients that ranged between 46% and 100%.^{462,468,471–473} Implementation of an infection control program by 60 Egyptian dialysis facilities has 3 years later resulted in improvements in infection control practices among health care workers and a subsequent decrease in the annual incidence of HCV infection among patients undergoing dialysis from 28% to 6%.⁴⁷² Moreover, the elimination of HCV among patients receiving dialysis is now possible with effective and well-tolerated third-generation antiviral agents.^{114,474–476} Of note, responders should be carefully followed up for occult HCV infection (OCI). OCI is defined as the presence of HCV-RNA in hepatocytes or peripheral blood mononuclear cells (PBMCs) with no detectable HCV-RNA in the serum by the conventional PCR assays. After HCV treatment, OCI was found in 5% to 7% of Egyptian HD patients.^{477,478}

Among Saudi patients undergoing hemodialysis, according to the 2016 SCOT report,¹⁴³ the prevalence of HBV positivity is 4%, whereas HCV infection continued

to be problematic (12%). Nonetheless, it has decreased compared with findings in previous reports.⁴⁵⁰ However, there is considerable variability in the prevalence of HCV infection between the various regions of Saudi Arabia (9%–16%).¹⁴³ This variability is important to consider when patients who are undergoing hemodialysis travel to other units and emphasizes the importance of screening them for seroconversion after their return.

Clinical Relevance

Hepatitis C in Dialysis

Owing to the high prevalence of HCV infection in some countries in the ME, patients who travel to the ME and receive dialysis should be monitored for seroconversion after their return.

The annual rate of HCV seroconversion in Saudi patients undergoing hemodialysis is between 7% and 9%. The most common variant of the virus is HCV genotype 4, which is also prevalent in North African countries in the ME, followed by HCV genotypes 1a and 1b.^{464,466,479} Implementation of strict measures of infection control and isolation of HCV-positive patients and dialysis machines has resulted in a significant drop in the annual incidence of HCV infections from 2.4% to 0.2% in one Saudi hemodialysis center.⁴⁸⁰ A similar high prevalence of HCV was observed in neighboring GCC countries, which was associated with a higher risk of death, hospitalization, and worse quality of life scores.^{142,481}

In Iran, where genotypes 3a and 1a are the common variants, impressive drops in the rates of HCV and hepatitis B surface antigen (HBsAg) positivity have been reported in patients undergoing hemodialysis. These decreases (14.4% in 1999 to 4.5% in 2006 for HCV and 3.8% in 1999 to 2.6% in 2006 for HBV) have been attributed to the introduction of several measures, such as strict isolation policies, no reuse of dialyzers, compulsory HBV vaccination in patients, and early kidney transplantation.⁴⁵¹ Implementation of routine virologic testing, strict isolation measures, HBV vaccination, and ESA administration with few blood product transfusion requirements have resulted in a low prevalence of both HBV (1.7%–2%) and HCV (3.7%–5.7%) infection among Israeli patients undergoing dialysis between 2005 and 2010, according to the 2012 ICDC report.⁴⁸²

Using second-generation immunoassays to detect HCV infection in 30 Turkish dialysis centers, Köhler has found that the prevalence of HCV was 49.9% in 1995.⁴⁶⁶ Sayiner and colleagues⁴⁸³ have reported that the duration of dialysis, kidney transplantation history, and history of blood product transfusion were all related to HCV transmission and high prevalence of HCV among Turkish patients undergoing hemodialysis. Turkey is also endemic for HBV infection; 10% of patients undergoing hemodialysis and 2.9% of blood donors are HBsAg-positive. Of note is that the 2008 and 2016 registry of the Turkish Society of Nephrology^{159,484} has reported a progressive decline in the prevalence of HBV infection (4.5%–3.88%), HCV infection (12.7%–5.2%), and both infections (0.9%–0.3%) in Turkish patients undergoing hemodialysis. A similar trend was observed among hemodialysis patients in Lebanon, who have the lowest prevalence of HBV and HCV (1.6% and 4.7%, respectively).⁴⁸⁵ The results of a survey conducted among Jordanian patients undergoing hemodialysis during

2003 revealed HBV positivity in 4% and HCV positivity in 21%, with annual seroconversion rates of 0.34% for HBV and 2.6% for HCV.¹⁴⁴ In 2007, high seropositivities for HBV (8%) and HCV (22%) were also detected in Palestinian patients undergoing hemodialysis who were residing in the Gaza Strip.⁴⁸⁶

Alashek and colleagues¹⁷³ have found a remarkably high prevalence of seropositivity for HCV among Libyan patients undergoing hemodialysis (31.1%), which is 25-fold higher than the general population. The overall incidence of seroconversion during this 1-year prospective study was 7.1%, with wide variation between hemodialysis units. Conversely, the prevalence of HBsAg positivity was low (2.6%), with a seroconversion incidence of 0.6%. Factors associated with seroconversions, such as duration of dialysis, history of receiving dialysis in another center in Libya, and prior kidney transplantation, possibly suggest nosocomial transmission.¹⁷³ In Jordan, 1.2% and 7.8% of prevalent patients on maintenance dialysis are seropositive to HBV and HCV, respectively.¹⁶⁰ In comparison, the adjusted prevalence of HBV is 4.6% among patients in Germany who are undergoing hemodialysis, 4.3% in Italy, 3.7% in France, 2.1% in Spain, 2.1% in Japan, and 2.4% in the United States, and adjusted seroconversion rates range from 0.4 to 1.8/100 patient-years.⁴⁸⁷ The adjusted prevalence of HCV also varies among these countries—22.9% in Spain, 20.5% in Italy, 10.4% in France, 3.8% in Germany, 2.6% in the United Kingdom, 14.8% in Japan, and 14% in the United States. The adjusted seroconversion rates range from 1.2 to 3.9/100 patient-years and are highest in Italy and Spain.⁴⁸⁸

DIALYSIS-RELATED OUTCOMES

With regard to other dialysis-related outcomes, reports from Egypt have revealed significant center-specific effects (which were not defined) on the survival and quality of life of patients undergoing dialysis; these effects were dependent largely on funding.⁴⁴⁵ In Syria, the 3-year survival rate among patients undergoing hemodialysis was unsatisfactory, ranging from 26% to 64% in different hemodialysis centers.¹⁵⁰ Regrettably, during recent conflicts, death due to lack of dialysis or underdialysis has been common in Syria, Yemen, and Gaza, although the exact figures are difficult to extract from published literature.^{23,30} In Jordan, the 1-year mortality rate among patients undergoing hemodialysis was approximately 20%,¹⁴⁴ although it decreased to 7% in 2016.¹⁶⁰ In Libya, the 1-year mortality was similar to many other countries (21.2%).¹⁷⁵ However, this is a poor outcome when taking into consideration the young mean age of patients undergoing hemodialysis. Although not reported, it is expected that the survival of Libyan ESKF patients was also aggravated by the conflicts in their country.¹⁷⁵ According to the 2018 ICDC report,¹⁵⁸ the unadjusted 1-, 2-, 3-, and 10-year survival rates among Israeli patients undergoing dialysis were 86.6%, 73%, 61.3%, and 15%, respectively, from 1990 to 2010. Of note, there appears to be a survival advantage for Arab patients over Jewish patients on maintenance dialysis in Israel, in contrast to the life expectancy of Arabs in the general population, which is 3 to 4 years lower than that of the Jewish population.¹⁶⁹ The survival of patients undergoing hemodialysis was high in Qatar, with 1-year and 5-year survival rates of 84% and 53%,

respectively,¹⁵⁵ as well as in other GCC countries (annual mortality, 8%–14.65%).¹⁴²

It is of interest that a health disparity exists for both legal and undocumented immigrants from ME countries with CKD who live and work in Western industrialized countries and may need KRT. The ethnic and educational backgrounds of these patients are different from those of the local population. Accordingly, they face potential problems with communication, health insurance, and care.^{9,489,490} Fogazzi and Castelnovo⁴⁹¹ have reported their experience with these patients who were treated in an Italian dialysis unit. Although the number of patients was small, they reported that those undergoing hemodialysis who came from developing countries, which included some ME countries, were younger at the initiation of dialysis (38.2 ± 7.9 years vs. 63 ± 12.6 years), were referred at a later stage of their disease, and had more infections, such as tuberculosis and viral hepatitis, than did patients from the local population who were undergoing hemodialysis. Furthermore, they also reported that a visit to their native countries was usually associated with medical complications, such as worsening anemia.

KIDNEY TRANSPLANTATION

In ME countries, as in all other countries, kidney transplantation is recognized as the treatment of choice for ESKF. The beneficial effects of kidney transplantation on life expectancy, quality of life, and medical expenses are greater than those associated with maintenance dialysis. However, legislative obstacles, an underdeveloped and poorly funded health care infrastructure, poor public awareness of the importance of organ donation, lack of effective kidney transplantation programs, cultural and religious barriers, and lack of trained multidisciplinary medical teams are some of the obstacles that prevent the promotion of kidney transplantation in some ME countries.^{102,113,118,119,153,157,492-498}

Most ME countries are members of the Middle East Society for Organ Transplantation (MESOT). MESOT includes Iran, Turkey, and all the Arab countries in the ME, as well as Pakistan, Cyprus, and some countries of central Asia.^{152,157,492} There are few organ procurement centers in the MESOT member countries for overseeing the activities of organ donation, sharing, and transplantation at a national level. Israel is not a member of MESOT; it has its own national center for organ transplantation. The first successful living- and deceased-donor kidney transplants in the ME were performed in Israel (1964 and 1966, respectively)¹⁴⁸ and were reported to have been performed in Iran in 1967 and 2003, respectively.^{157,499} The first successful kidney transplantation in an Arab country was performed in Jordan in 1972, with a kidney from a deceased non-heart beating donor. Most other ME countries, such as Lebanon and Turkey, started their kidney transplantation programs during the early 1970s, and Egypt and Saudi Arabia commenced their programs in 1976 and 1979, respectively. Since then, the remaining ME countries have established their own kidney transplantation programs; Libya did so in 2004. Currently, kidney transplantation programs are active in most ME countries but are relatively limited in Algeria, Yemen, the UAE, and Bahrain (see [Tables 76.3 and 76.4](#)).^{102,152-154,157,500-503}

Most kidneys for transplantation are obtained from living donors. An important milestone that paved the way for organ donation for transplantation from deceased donors in Arab countries in the ME was the Amman declaration in 1986. In this declaration, Islamic theologians recognized that brain death was irreversible and could be used to declare a person legally dead, thereby making it permissible to disconnect that person from mechanical life support systems. This declaration was preceded in 1982 by a resolution of the Islamic Council in Saudi Arabia that permitted the use of organs from both living and deceased donors for transplantation.^{152,504,505} Similar declarations have since been made by religious authorities in Egypt, Turkey, and Iran.^{152,504} As a result of these declarations, most Arab countries in the ME, including Egypt, have now enacted laws for regulated organ donation for transplantation from both living and cadaveric donors, and transplantation programs for kidneys and other organs are beginning to expand.^{119,139,150,506} Programs for donation after cardiac death are limited in many Arab countries because of religious, legal, ethical, and social issues.⁵⁰⁷ Such a program began more than 8 years ago in Israel.⁵⁰⁸

In his code of Jewish law, the Mishneh Torah (Laws of Sanhedrin, 12:3), the 12th-century philosopher and physician Maimonides interpreted the Talmud as saying that someone who saves the life of one person is considered to have saved the entire world. This is similar to what is written in the Koran, [Chapter 5](#), verse 32: "... if any one saved a life, it would be as if he saved the life of all mankind." Therefore organ donation that saves a life is considered in Islam and Judaism to be a good deed. With regard to living related donors, Bulka has argued that organ donation is permissible because the danger to the donor is minimal, but it is not obligatory.⁵⁰⁹ Likewise, Christians believe that organ donation is an act of love and nobility (e.g., as interpreted from Corinthians and from the parable of the Good Samaritan in Luke, [Chapter 10](#), verses 25 to 37).⁵¹⁰

Active deceased-donor kidney transplantation programs exist in Iran, Israel, Turkey, and 10 Arab countries (Algeria, Bahrain, Jordan, Kuwait, Lebanon, Morocco, Oman, Qatar, Saudi Arabia, and Tunisia; see [Table 76.3](#)).^{102,119,153,511} However, deceased-donor kidney transplantation is inadequate to address the current need for allografts, and the number of patients on kidney transplantation waiting lists in these ME countries has been progressively increasing. The use of kidneys from living-related donors is increasing, but the gap between demand and supply is continuously widening in all ME countries. To meet the growing demand for kidneys and the shortfall in donors, some countries have initiated kidney transplantation programs involving organs from living unrelated donors. Using emotionally related donors extends donor eligibility to include individuals who are not genetically related to recipients. However, in Saudi Arabia, kidney transplantation from living unrelated donors is forbidden.

Kidney transplantation involving organs from deceased donors accounts for <21% of kidney transplantations in most ME countries, with large national variations.^{119,157,159,512,513} In fact, in 2006, Saudi Arabia had the highest reported rate of living-donor kidney transplantations worldwide, at 32 procedures pmp, followed by Jordan (29 procedures pmp), Iceland (26 procedures pmp), Iran (23 procedures

pmp), and the United States (21 procedures pmp).⁵¹⁴ However, the quoted Saudi statistic should be interpreted cautiously because the rate of living-related donor kidney transplantations is 10.1 pmp. According to Noël,⁵¹⁵ the report by Horvat and colleagues⁵¹⁴ probably included "transplant tourism" activity because it incorporated data on kidney transplantation in Saudi patients from living unrelated donors that was performed in other countries. Global epidemiologic data indicated that the incidence of kidney transplant was highest in the United States (77 pmp), followed by Mexico (57–58 pmp), and Israel (56 pmp). Most of the reporting European countries had rates of 30 to 40 pmp.¹⁶⁴

In Egypt, approximately 7% of Egyptian patients with ESKD are offered kidney transplantation in the year of diagnosis, and no kidneys are obtained from deceased donors—80% of the kidneys are obtained from living unrelated donors, and 20% are obtained from living-related donors.

Kidney transplantation in Syria commenced in 1976 with exclusive reliance on kidneys from living-related donors. Since 2003, transplantation of kidneys from deceased donors has been allowed. As of 2010, 13.2 kidney transplantations pmp were performed, of which 22% were from deceased donors.^{150,153} Unfortunately, the war in Syria destroyed the hospitals' infrastructure and decreased the kidney transplant programs from eight to four centers, along with difficulties in providing immunosuppressive therapies.⁵¹⁶

In Iran, the living unrelated donor transplantation program operates under the close supervision and scrutiny of the Ministry of Health and Medical Education, Iranian Scientific Society of Organ Transplantation, Foundation for Patients with Special Diseases, and Dialysis and Transplant Patients Association (DATPA).⁵¹⁷ In the Iranian model, patients who need kidney transplantation are referred to the DATPA, a charity founded in 1978 by Iranian patients with ESKF, which acts as a liaison agency between patients and potential donors. The altruistic volunteers are also registered by the DATPA and undergo evaluation in the foundation's clinics. Several important features characterize the Iranian model of living unrelated donors for kidney transplantation:⁵⁰¹

- No coercion is allowed.
- Written consent is obtained from the donor and the donor's parents or spouse.
- Donors are rewarded with gifts from the government.
- No commercialism is allowed.
- The medical teams receive no financial benefit.
- Recipients and donors must be Iranian citizens.
- Time on the waiting list is minimal.

Consequently, the annual number of kidney transplantations substantially increased in Iran, from 1421 in 2000 to 2285 in 2010.^{151,501} However, the 86% share of living unrelated kidney donations in 2000 (20.1 pmp) decreased to 75% in 2006 (23 pmp) and to 69% in 2010 (21.8 pmp). This change was mainly due to a substantial parallel increase in brain-dead kidney donation (2.2% in 2000, equivalent to 0.4 pmp, increasing to 26% and 7.9 pmp in 2010), after the Organ Transplantation and Brain Death Act in 2000, which legalized brain-dead organ donation.¹⁵¹ Of note, 10 years after the introduction of the controlled living unrelated

donor kidney transplantation program in Iran, the national waiting list for renal transplants was eliminated.^{518,519}

Simforoosh and associates⁵²⁰ have reported that patient and graft survival rates of 2155 Iranian recipients of kidneys from living unrelated donors after 15 years were 76.4%, and 53.2%, respectively. These results are comparable with survival data in the 2003 annual report of the U.S. Renal Data System on living donor kidney transplantations.⁴² However, the Iranian model has been criticized for ethical reasons because 85% of the vendors are poor, a fact that may increase the risk that potential donors will withhold relevant medical information.⁵²¹ A study from the Shiraz Transplant Center has revealed that Iranian-paid unrelated donors have a lower quality of life and a higher incidence of microalbuminuria compared with related donors.⁵²² Hence, the lack of long-term donor follow-up and the direct financial connection between donor and recipient are major weak points that should be ethically reviewed.^{151,523-525}

COMMERCIAL KIDNEY TRANSPLANTATION

The lack of deceased donors for kidney allograft transplantation, combined with other medical factors (prolonged graft survival and improved surgical techniques, including laparoscopic nephrectomy) and nonmedical factors (economic and cultural), has motivated many patients with ESKF to seek kidney transplants from living unrelated donors outside their home countries. This practice is often called commercial kidney transplantation or organ tourism⁵²⁶ because the donor sells his or her kidney for a certain amount of money. These kidney transplantations are performed in many countries worldwide and in some ME countries.^{494,527-529} The surgery is commonly performed in substandard conditions under the cover of secrecy. Many transplant recipients return to their homeland 1 to 2 weeks after transplantation without crucial information (e.g., a complete medical report, essential information about the donor, HLA typing, and details of the medical treatment, surgery, and postoperative care). In addition, the recipients are exposed to potential risk for infection, such as HBV, HCV, or human immunodeficiency virus infection, as well as tuberculosis from inadequately evaluated donors. Thus the results of such commercial kidney transplantations are reported to be substandard.^{526,528,529}

To undermine the practice of global commercial kidney transplantation, the Transplantation Society and the ISN convened an international summit meeting in Istanbul, Turkey, in April 2008. The outcome of this meeting was the “Declaration of Istanbul on Organ Trafficking and Transplant Tourism,” which suggested that strategies to increase the donor pool and encourage legitimate, lifesaving transplantation programs be developed by countries to prevent organ trafficking, transplant commercialism, and transplant tourism. In addition, strategies should be aimed at stopping and prohibiting unethical activities, as well as encouraging safe and accountable practices that both meet the needs of transplant recipients and protect donors.^{527,530,531} Many ME countries need to develop national self-sufficiency in organ donation to combat organ tourism and achieve global justice in transplantation.^{527,530}

In 2008, two laws were approved by the Israeli parliament, the Brain-Respiratory Death Law and the Organ Transplantation Law. As a result, health insurance reimbursement for transplantation performed abroad that

contravenes local Israeli laws was prohibited, and criminal penalties for brokering organ sales were introduced. Consequently, Israel has increased organ donations from living and deceased sources and has reduced the number of transplant candidates seeking transplantation abroad (from 150 in 2006 to 41 in 2021).^{511,532} In Egypt, however, in spite of the April 2010 organ transplantation law, which prohibits and penalizes organ trafficking and permits deceased donation in accord with the WHO Guiding Principles and the Declaration of Istanbul, some enforcement has been evident, but the problem of a black market in organs from living donors remains a challenge.^{494,506,527,533}

PHARMACOLOGIC TREATMENT OF KIDNEY TRANSPLANT RECIPIENTS

In general, all approved medications that are used worldwide to treat kidney transplant recipients are also used in the ME.^{113,118,119,151,154,500} In all ME countries with kidney transplantation programs, induction therapy with methylprednisolone, lymphocyte-depleting agents, or IL-2 receptor antagonists is widely used. In most commercial transplantation programs, induction therapy with lymphocyte-depleting agents is routinely given to reduce the frequency of acute graft rejection so that the recipient can be discharged early after the transplantation surgery. No reports have yet been published on the long-term complications of this routine treatment, such as bone marrow suppression, cytomegalovirus infections, and posttransplantation lymphoproliferative disorder.

Maintenance treatment after kidney transplantation in most ME countries consists of triple therapy with corticosteroids, azathioprine or mycophenolate mofetil, and calcineurin inhibitors. A small number of kidney transplantation centers have started prescribing rapamycin, a drug that inhibits a serine-threonine kinase called the mammalian target of rapamycin (mTOR), to prevent transplant rejection.^{151,504} Treatment of acute rejection in most ME countries consists mainly of methylprednisolone (Solu-Medrol) and rabbit antithymocyte globulin (Thymoglobulin). Hyperimmune globulins and plasmapheresis are used by some centers when indicated.

POSTTRANSPLANTATION COMPLICATIONS

Only a few reports in the literature address the outcomes, as well as patient and graft survival rates, after kidney transplantation in ME countries. In general, these reports claim that outcomes in Saudi Arabia, Kuwait, Egypt, and North African countries in the ME are in line with international standards, although infections remain the main reason for morbidity and mortality.^{159,181,184,498,520,529}

Distinctive infections are of special concern in some ME countries. Tuberculosis is prevalent among patients receiving dialysis in some countries such as Saudi Arabia, Yemen, Turkey, and the North African countries in the ME.^{484,495,534-537} Patients in these countries are at increased risk for developing active tuberculosis after kidney transplantation, and approximately 30% have symptoms of tuberculosis in extrapulmonary organs, particularly the lymph nodes, gastrointestinal tract, and peritoneal cavity. The Mantoux test often yields negative results in kidney transplant recipients, possibly because of the suppression of cellular immunity. Tuberculosis in the graft kidney tends to

manifest as granulomatous interstitial nephritis. Urinalysis results are often negative for bacilli. The diagnosis is usually made on kidney biopsy or after nephrectomy in transplant recipients, who often present with fever of unknown origin and deteriorating graft function.

Prophylactic treatment with isoniazid or rifampin for patients at high risk (Mantoux skin test reaction of >10 mm) has decreased the development of active tuberculosis. An important sequela of this treatment is the induction of cytochrome P450 enzymes by antituberculous drugs, which results in a severe drop in the circulating therapeutic levels of calcineurin inhibitors potentially precipitating severe acute rejection. Therefore increasing the dose of calcineurin inhibitors and frequent monitoring of their circulating levels are mandatory in such cases.^{484,495}

Viral hepatitis, as discussed previously, is also common in the ME, especially among patients undergoing dialysis. Eligible patients with HBV and HCV infections are offered transplantation in many ME kidney transplantation centers after appropriate presurgical workup and management.^{464,484,538}

The most common types of neoplasia among kidney transplant recipients are skin malignancies, lymphoproliferative disorders, and Kaposi sarcoma. The incidence of some uncommon tumors in the general population (e.g., Kaposi sarcoma) can be 400- to 500-fold higher among kidney transplant recipients. Kaposi sarcoma is most often seen in transplant recipients of Mediterranean, Jewish, and Arabic descent. Its reported incidence is 0.5% in most Western industrialized countries and as high as 5.3% in Saudi Arabia.^{484,539-542}

Clinical Relevance

Kaposi Sarcoma Posttransplant

Kaposi sarcoma is more common in patients from the ME, potentially linked to greater rates of human herpesvirus 8 seropositivity than in Europe and North America.

The preponderance of cases of Kaposi sarcoma in certain ethnic groups appears to be linked to the geographic distribution of human herpesvirus 8 infection, inasmuch as more than 80% of transplant recipients with Kaposi sarcoma are seropositive for human herpesvirus 8 before undergoing transplantation.⁵⁴²

SUMMARY

The ME has a strategic global location. Its fascinating geography is combined with rich national histories, cultures, and resources. In addition, the countries of the ME differ in their economics, political systems, culture, and bioecology, all of which eventually translate into disparities in their health care systems, as well as in disease epidemiologic features, causes, management, and outcomes. The burden of kidney diseases is high in all ME countries and is compounded by many risk factors in the region, although data from developing countries are possibly underestimates. Practicing nephrologists in the ME are familiar with the genetic background, social habits, and culture of their patients. However, this is not so for many individuals of ME

origin who now live in non-ME countries. Nephrologists in those countries need to take these considerations into account when diagnosing or treating kidney diseases and comorbid conditions of their patients of ME origin. Therefore the emerging epidemic of diabetes mellitus in many ME countries is pertinent, and the prevailing genetic kidney diseases in the various population groups who live in the ME have been reviewed in this chapter.

Moreover, the ME has experience in management of mass disasters and consequent kidney complications, which can be implemented in regions outside the ME. However, several issues still need to be addressed, and hurdles must be overcome to improve the overall management of kidney diseases in the ME. Well-conducted epidemiologic cohort studies are still needed, and regional and national registries must be established as sources of transparent and accurate data. The results of the studies should provide accurate estimates of the public health burden of AKI, CKD, and ESKD; their risk factors; and comorbid conditions. This information should also improve the quality of therapy provided to patients. In addition, these efforts should also focus on the special needs of refugees in countries where human-engendered and natural disasters have occurred.

The entire international nephrology community agrees that improving existing diagnostic methods and establishing preventive strategies for the detection and treatment of kidney diseases at the earliest possible stage is of utmost importance, especially in countries with limited resources or health care expenditures. Therefore to achieve these goals, ME countries need to invest in the following: 1. training qualified nephrologists and medical personnel; 2. public education and campaigns for lifestyle modification to combat obesity and diabetes mellitus, to explain proper use of over-the-counter drugs and early detection of CKD in high risk patients; 3. premarital genetic counseling to decrease the burden of genetic diseases; 4. vigorous treatment of comorbid conditions, which include infectious diseases, and strategies for preventing dehydration in tropical ME countries; and 5. use of low-cost, generic medications.^{61,88,102,534,543}

Because transplantation is the most cost-effective treatment for ESKD, ME countries also need to create a societal environment that motivates and encourages its population to support organ donation for transplantation in patients with ESKD. Encouraging organ donation through deceased donors, living-related donors, and paired-exchange kidney transplantation programs, as well as organ sharing among the MESOT member states, will enable ME countries to reach national sufficiency for organ transplantation. The establishment of such programs will then help reduce the long waiting lists in each ME country and combat unethical commercial kidney transplantations.^{88,102,492,544}

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KEY REFERENCES

1. Held CC, Cummings JT. *Middle East Patterns: Places, Peoples, and Politics*. 6th ed. Boulder, Co: Westview Press; 2014.
9. Van Biesen W, Vanholder R, Hernandez T, et al. Caring for migrants and refugees with end-stage kidney disease in Europe. *Am J Kidney Dis*. 2018;71:701–709.
20. AlSahow A, AlYousef A. Global dialysis perspective: Kuwait. *Kidney360*. 2021;2:1015–1020.
23. Sekkarie MA, Zanabli AR, Rifai AO, et al. The Syrian conflict: assessment of the ESRD system and response to hemodialysis needs during a humanitarian and medical crisis. *Kidney Int*. 2015;87:262–265.
50. Zarbock A, Weiss R, Albert F, et al. Epidemiology of Surgery associated Acute Kidney Injury (EPIS-AKI): a prospective international observational multi-center clinical study. *Intensive Care Med*. 2023;49:1441–1455.
53. Bnaya A, Abu-Amer N, Beckerman P, et al. Acute kidney injury and hair-straightening products: a case series. *Am J Kidney Dis*. 2023;82:43–52.
57. Barsoum RS. Parasitic kidney disease: milestones in the evolution of our knowledge. *Am J Kidney Dis*. 2013;61:501–513.
63. Raimann JG, Riella MC, Levin NW. International Society of Nephrology's 0by25 initiative (zero preventable deaths from acute kidney injury by 2025): focus on diagnosis of acute kidney injury in low-income countries. *Clin Kidney J*. 2018;11:12–19.
72. Kovesdy CP. Epidemiology of chronic kidney disease: an update 2022. *Kidney Int Suppl*. 2022;12:7–11.
73. Brauer M, Roth GA, Aravkin AY, et al. Global burden and strength of evidence for 88 risk factors in 204 countries and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet*. 2024;403:2162–2203.
76. Collaboration GBDCKD. Global, regional, and national burden of chronic kidney disease, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet*. 2020;395:709–733.
78. Alashek WA, McIntyre CW, Taal MW. Epidemiology and aetiology of dialysis-treated end-stage kidney disease in Libya. *BMC Nephrol*. 2012;13:33.
83. Jaber LA, Brown MB, Hammad A, et al. The prevalence of the metabolic syndrome among arab americans. *Diabetes Care*. 2004;27:234–238.
87. Suleymanlar G, Utas C, Arinsoy T, et al. A population-based survey of Chronic RENal Disease in Turkey—the CREDIT study. *Nephrol Dial Transplant*. 2011;26:1862–1871.
89. Benghanem GM, Elseviers M, Zamd M, et al. Chronic kidney disease, hypertension, diabetes, and obesity in the adult population of Morocco: how to avoid “over”- and “under”-diagnosis of CKD. *Kidney Int*. 2016;89:1363–1371.
91. Karam S, Amouzegar A, Alshamsi IR, et al. Capacity for the management of kidney failure in the international society of nephrology Middle East region: report from the 2023 ISN Global Kidney Health Atlas (ISN-GKHA). *Kidney Int Suppl*. 2024;13:57–70.
96. Karnib HH, Gharavi AG, Aftimos G, et al. A 5-year survey of biopsy proven kidney diseases in Lebanon: significant variation in prevalence of primary glomerular diseases by age, population structure and consanguinity. *Nephrol Dial Transplant*. 2010;25:3962–3969.
97. Ozen S. Renal amyloidosis in familial Mediterranean fever. *Kidney Int*. 2004;65:1118–1127.
100. Sadowski CE, Lovric S, Ashraf S, et al. A single-gene cause in 29.5% of cases of steroid-resistant nephrotic syndrome. *J Am Soc Nephrol*. 2015;26:1279–1289.
105. Bello AK, Okpechi IG, Levin A, et al. An update on the global disparities in kidney disease burden and care across world countries and regions. *Lancet Glob Health*. 2024;12:e382–e395.
106. Lunney M, Bello AK, Levin A, et al. Availability, accessibility, and quality of conservative kidney management worldwide. *Clin J Am Soc Nephrol*. 2021;16:79–87.
107. Pippias M, Alfano G, Kelly DM, et al. Capacity for the management of kidney failure in the international society of nephrology western Europe region: report from the 2023 ISN Global Kidney Health Atlas (ISN-GKHA). *Kidney Int Suppl*. 2024;13:136–151.
110. Nelson RG, Grams ME, Ballew SH, et al. Development of risk prediction equations for incident chronic kidney disease. *JAMA*. 2019;322:2104–2114.
114. Farag YMK, El-Sayed E. Global dialysis perspective: Egypt. *Kidney360*. 2022;3:1263–1268.
116. Haviv YS, Golan E. Global dialysis perspective: Israel. *Kidney360*. 2020;1:119–122.
117. Aoun M, Koubar S. Global dialysis perspective: Lebanon. *Kidney360*. 2023;4:e1308–e1313.
118. Andacoglu O, Aki FT. Global perspective on kidney transplantation: Turkey. *Kidney360*. 2021;2:1160–1162.
119. Hafez MH. Global perspective: kidney transplantation in North Africa. *Kidney360*. 2024. <https://doi.org/10.34067/KID.0000000000000497>.
122. Calderon-Margalit R, Golan E, Twig G, et al. History of childhood kidney disease and risk of adult end-stage renal disease. *N Engl J Med*. 2018;378:428–438.
124. Gouda Z, Mashaal G, Bello AK, et al. Egypt Information, Prevention, and Treatment of Chronic Kidney Disease (EGIPT-CKD) programme: prevalence and risk factors for microalbuminuria among the relatives of patients with CKD in Egypt. *Saudi J Kidney Dis Transpl*. 2011;22:1055–1063.
130. Tohidi M, Hashemini M, Mohebi R, et al. Incidence of chronic kidney disease and its risk factors, results of over 10 year follow up in an Iranian cohort. *PLoS One*. 2012;7:e45304.
139. El MA. History of nephrology in mediterranean arab countries. *J Nephrol*. 2013;26:170–174.
167. Seyahi N, Kocyigit I, Eren N, et al. Current status of kidney replacement therapy in Türkiye: a summary of 2022 Turkish Society of Nephrology registry report. *Turk J Nephrol*. 2024;33:134–139.
169. Kalantar-Zadeh K, Golan E, Shohat T, et al. Survival disparities within American and Israeli dialysis populations: learning from similarities and distinctions across race and ethnicity. *Semin Dial*. 2010;23:586–594.
170. Council E, Cherni N, Kharrat M, et al. Trends of incident dialysis patients in Tunisia between 1992 and 2001. *Am J Kidney Dis*. 2008;51:463–470.
172. Alashek WA, McIntyre CW, Taal MW. Provision and quality of dialysis services in Libya. *Hemodial Int*. 2011;15:444–452.
187. Al-Ghamdi SMG, Bieber B, AlRukhaimi M, et al. Diabetes prevalence, treatment, control, and outcomes among hemodialysis patients in the Gulf Cooperation council countries. *Kidney Int Rep*. 2022;7:1093–1102.
219. Frishberg Y, Rinat C, Megged O, et al. Mutations in NPHS2 encoding podocin are a prevalent cause of steroid-resistant nephrotic syndrome among Israeli-Arab children. *J Am Soc Nephrol*. 2002;13:400–405.
258. Garrelfs SF, Frishberg Y, Hulton SA, et al. Lumasiran, an RNAi therapeutic for primary hyperoxaluria type 1. *N Engl J Med*. 2021;384:1216–1226.
262. Michael M, Groothoff JW, Shasha-Lavsky H, et al. Lumasiran for advanced primary hyperoxaluria type 1: phase 3 ILLUMINATE-C trial. *Am J Kidney Dis*. 2023;81:145–155 e141.
265. Baum MA, Langman C, Cochat P, et al. PHYOX2: a pivotal randomized study of nedosiran in primary hyperoxaluria type 1 or 2. *Kidney Int*. 2023;103:207–217.
316. Manukyan G, Aminov R. Update on pyrin functions and mechanisms of familial mediterranean fever. *Front Microbiol*. 2016;7:456.
409. Stankovic Stojanovic K, Delmas Y, Torres PU, et al. Dramatic beneficial effect of interleukin-1 inhibitor treatment in patients with familial Mediterranean fever complicated with amyloidosis and renal failure. *Nephrol Dial Transplant*. 2012;27:1898–1901.
427. Al Sahlawi M, AlRukhaimi M, Al-Ghamdi SM, et al. Peritoneal dialysis in the Arabian Gulf Countries: challenges and opportunities. *Perit Dial Int*. 2024;44:171–176.
428. Ersoy FF. The TULIP story. *Perit Dial Int*. 2008;28:220–223.
475. Saadi G, Kalantar-Zadeh K, Almasio P, et al. Hepatitis C virus infection and global kidney health: the consensus proceedings of the International Federation of Kidney Foundations. *Afr J Nephrol*. 2020;23:159–168.
505. Albar M. Organ transplantation: a Sunni Islamic perspective. *Saudi J Kidney Dis Transpl*. 2012;23:817–822.
511. Lavee J, Ashkenazi T, Stoler A, et al. Preliminary marked increase in the national organ donation rate in Israel following implementation of a new organ transplantation law. *Am J Transplant*. 2013;13:780–785.
517. Haghighi AN, Ghahramani N. Living unrelated kidney donor transplantation in Iran. *Nat Clin Pract Nephrol*. 2006;2:E1–E1.
527. Danovitch GM, Chapman J, Capron AM, et al. Organ trafficking and transplant tourism: the role of global professional ethical standards—the 2008 Declaration of Istanbul. *Transplantation*. 2013;95:1306–1312.

REFERENCES

- Held CC, Cummings JT. *Middle East Patterns: Places, Peoples, and Politics*. 6th ed. Boulder, Co: Westview Press; 2014.
- Saleh SS, Alameddine MS, Natafqi NM, et al. The path towards universal health coverage in the Arab uprising countries Tunisia, Egypt, Libya, and Yemen. *Lancet*. 2014;383:368–381.
- World Bank. *How we classify countries*. Available at <https://data.worldbank.org/about/country-classifications>. Accessed August 20, 2023.
- World Bank. *Countries and economies*. Available at <https://data.worldbank.org/country>. Accessed August 20, 2023.
- Access GBDH, Quality Collaborators. Electronic address cue, Access GBDH, et al. Healthcare Access and Quality Index based on mortality from causes amenable to personal health care in 195 countries and territories, 1990–2015: a novel analysis from the Global Burden of Disease Study 2015. *Lancet*. 2017;390:231–266.
- Mokdad AH, Forouzanfar MH, Daoud F, et al. Health in times of uncertainty in the eastern Mediterranean region, 1990–2013: a systematic analysis for the Global Burden of Disease Study 2013. *Lancet Glob Health*. 2016;4:e704–e713.
- Barsoum RS. Burden of chronic kidney disease: North Africa. *Kidney Int Suppl*. 2013;3:164–166.
- Chao F, You D, Pedersen J, et al. National and regional under-5 mortality rate by economic status for low-income and middle-income countries: a systematic assessment. *Lancet Glob Health*. 2018;6:e535–e547.
- Van Biesen W, Vanholder R, Hernandez T, et al. Caring for migrants and refugees with end-stage kidney disease in Europe. *Am J Kidney Dis*. 2018;71:701–709.
- Perini W, Snijder MB, Peters RJG, et al. Ethnic disparities in estimated cardiovascular disease risk in Amsterdam, The Netherlands: the HELIUS study. *Neth Heart J*. 2018;26:252–262.
- Muhsen K, Green MS, Soskolne V, et al. Inequalities in non-communicable diseases between the major population groups in Israel: achievements and challenges. *Lancet*. 2017;389:2531–2541.
- Tinghög P, Malm A, Arwidson C, et al. Prevalence of mental ill health, traumas and postmigration stress among refugees from Syria resettled in Sweden after 2011: a population-based survey. *BMJ Open*. 2017;7:e018899.
- Belqacem S. Health issues in the Arab American community. Commentary: the growing risk factors for noncommunicable diseases in the Arab world. *Ethn Dis*. 2006;17:51–S53.
- Shah SM, Ali R, Loney T, et al. Prevalence of diabetes among migrant women and duration of residence in the United Arab Emirates: a cross sectional study. *PLoS One*. 2017;12:e0169949.
- Asfour LW, Stanley ZD, Weitzman M, et al. Uncovering risky behaviors of expatriate teenagers in the United Arab Emirates: a survey of tobacco use, nutrition and physical activity habits. *BMC Public Health*. 2015;15:944.
- AlFaleh HF, Al Shamiri MQ, Ullah A, et al. Disparities in health care delivery and hospital outcomes between non-Saudis and Saudi nationals presenting with acute coronary syndromes in Saudi Arabia. *PLoS One*. 2015;10:e0124012.
- Graetz V, Rechel B, Groot W, et al. Utilization of health care services by migrants in Europe—a systematic literature review. *Br Med Bull*. 2017;121:5–18.
- Hussin AH. Adjustment factors to per capita health-care indicators in countries with expatriate male-majority populations. *East Mediterr Health J*. 2014;20:690–697.
- Chernin G, Gal-Oz A, Schwartz IF, et al. Care of undocumented-undocumented immigrants in a large urban dialysis unit. *BMC Nephrol*. 2012;13:112.
- AlSahow A, AlYousef A. Global dialysis perspective: Kuwait. *Kidney360*. 2021;2:1015–1020.
- Devi S. Yemeni health under relentless pressure. *Lancet*. 2018;391:646.
- Butler D. What the numbers say about refugees. *Nature*. 2017;543:22–23.
- Sekkarie MA, Zanabli AR, Rifai AO, et al. The Syrian conflict: assessment of the ESRD system and response to hemodialysis needs during a humanitarian and medical crisis. *Kidney Int*. 2015;87:262–265.
- Mateen FJ, Carone M, Al-Saedy H, et al. Medical conditions among Iraqi refugees in Jordan: data from the united Nations refugee assistance information system. *Bull World Health Organ*. 2012;90:444–451.
- Marshall M. Why was the Morocco earthquake so deadly? *Nature*. 2023.
- Scandoni R, Cingolani M, Tambone V, et al. Operational health pavilions in mass disasters: lessons learned from the 2023 earthquake in Turkey and Syria. *Healthcare*. 2023;11.
- Marshall M. Libya floods: how climate change intensified the death and devastation. *Nature*. 2023;621:452–453.
- Cohen-Hagai K, Goldman S, Wald R, et al. Delivering dialysis during wartime: the Israeli experience. *Kidney360*. 2024;5:615–617.
- Alasfar S, Alashavi H, Nasan KH, et al. Providing hemodialysis in unstable areas: an assessment and framework for effective care. *Kidney Int Rep*. 2024;9:580–588.
- Tuglular S, Luyckx V. Maintaining kidney replacement therapy during armed conflicts. *Nephrol Dial Transplant*. 2024;39:735–738.
- The L. Syria suffers as the world watches. *Lancet*. 2017;389:1075.
- Fouad FM, Sparrow A, Tarakji A, et al. Health workers and the weaponisation of health care in Syria: a preliminary inquiry for the Lancet–American University of Beirut Commission on Syria. *Lancet*. 2017;390:2516–2526.
- Zimmerman C, Kiss L, Hossain M. Migration and health: a framework for 21st century policy-making. *PLoS Med*. 2011;8:e1001034.
- Lameire N, Sever MS, Van Biesen W, et al. Role of the international and national renal organizations in natural disasters: strategies for renal rescue. *Semin Nephrol*. 2020;40:393–407.
- Sever MS, Luyckx V, Tonelli M, et al. Disasters and kidney care: pitfalls and solutions. *Nat Rev Nephrol*. 2023;19:672–686.
- Musani A, Shaikh IA. The humanitarian consequences and actions in the Eastern Mediterranean Region over the last 60 years—a health perspective. *East Mediterr Health J*. 2008;14(suppl):S150–S156.
- Lameire NH, Bagga A, Cruz D, et al. Acute kidney injury: an increasing global concern. *Lancet*. 2013;382:170–179.
- Hoste EA, Bagshaw SM, Bellomo R, et al. Epidemiology of acute kidney injury in critically ill patients: the multinational AKI-EPI study. *Intensive Care Med*. 2015;41:1411–1423.
- Negi S, Koreeda D, Kobayashi S, et al. Acute kidney injury: epidemiology, outcomes, complications, and therapeutic strategies. *Semin Dial*. 2018;31:519–527. Wiley Online Library.
- Hadidy S, Asfari R, Shammaa MZ, et al. Acute renal failure among a Syrian population. Incidence, aetiology, treatment and outcome. *Int Urol Nephrol*. 1989;21:455–461.
- Al-Azzam SI, Al-Husein BA, Abu-Dahoud EY, et al. Etiologies of acute renal failure in a sample of hospitalized Jordanian patients. *Ren Fail*. 2008;30:373–376.
- Al-Homrany M. Epidemiology of acute renal failure in hospitalized patients: experience from southern Saudi Arabia. *East Mediterr Health J*. 2003;9:1061–1067.
- Shema L, Ore L, Geron R, et al. Hospital-acquired acute kidney injury in Israel. *Isr Med Assoc J*. 2009;11:269–274.
- Balushi F, Khan S, Riyami D, et al. Acute kidney injury in a teaching hospital in Oman. *Saudi J Kidney Dis Transpl*. 2011;22:825–828.
- Kavaz A, Ozcarar ZB, Kendirli T, et al. Acute kidney injury in a paediatric intensive care unit: comparison of the pRIFLE and AKIN criteria. *Acta Paediatr*. 2012;101:e126–e129.
- Oweis AO, Alshelleh SA. Incidence and outcomes of acute kidney injury in octogenarians in Jordan. *BMC Res Notes*. 2018;11:279.
- Arabi YM, Arifi AA, Balkhy HH, et al. Clinical course and outcomes of critically ill patients with Middle East respiratory syndrome coronavirus infection. *Ann Intern Med*. 2014;160:389–397.
- Fowler RA, Lapinsky SE, Hallett D, et al. Critically ill patients with severe acute respiratory syndrome. *JAMA*. 2003;290:367–373.
- KDIGO clinical practice guideline for acute kidney injury. *Kidney Int Suppl*. 2012;2:1–138.
- Zarbock A, Weiss R, Albert F, et al. Epidemiology of Surgery associated Acute Kidney Injury (EPIS-AKI): a prospective international observational multi-center clinical study. *Intensive Care Med*. 2023;49:1441–1455.
- Duzova A, Bakkaloglu A, Kalyoncu M, et al. Etiology and outcome of acute kidney injury in children. *Pediatr Nephrol*. 2010;25:1453–1461.
- Said R. Acute renal failure in Jordan. *Saudi J Kidney Dis Transpl*. 1998;9:301–305.

53. Bnaya A, Abu-Amer N, Beckerman P, et al. Acute kidney injury and hair-straightening products: a case series. *Am J Kidney Dis*. 2023;82:43–52.e41.
54. Al-Malki H, Sadek M, Rashed A, et al. Acute renal failure in the State of Qatar: presentation and outcome. *Transplant Proc*. 2009;41:1530–1532.
55. Al Rohani M, Aljawshaei H, Aduolimi E. Acute renal failure in Yemeni patients. *Saudi J Kidney Dis Transpl*. 2011;22:829–833.
56. Barsoum RS. Malarial acute renal failure. *J Am Soc Nephrol*. 2000;11:2147–2154.
57. Barsoum RS. Parasitic kidney disease: milestones in the evolution of our knowledge. *Am J Kidney Dis*. 2013;61:501–513.
58. Sitprija V. Altered fluid, electrolyte and mineral status in tropical disease, with an emphasis on malaria and leptospirosis. *Nat Clin Pract Nephrol*. 2008;4:91–101.
59. Bernieh B, Al Hakim M, Boobes Y, et al. Pattern of acute renal failure in a tertiary hospital in the United Arab Emirates. *Transplant Proc*. 2004;36:1780–1783.
60. Alsultan MA. The renal recovery of critically ill patients with acute renal failure requiring dialysis. *Saudi J Kidney Dis Transpl*. 2013;24:1175–1179.
61. Mehta RL, Burdmann EA, Cerda J, et al. Recognition and management of acute kidney injury in the International Society of Nephrology 0by25 Global Snapshot: a multinational cross-sectional study. *Lancet*. 2016;387:2017–2025.
62. Mehta RL, Cerda J, Burdmann EA, et al. International Society of Nephrology's 0by25 initiative for acute kidney injury (zero preventable deaths by 2025): a human rights case for nephrology. *Lancet*. 2015;385:2616–2643.
63. Raimann JG, Riella MC, Levin NW, International Society of Nephrology's 0by25 initiative. (zero preventable deaths from acute kidney injury by 2025): focus on diagnosis of acute kidney injury in low-income countries. *Clin Kidney J*. 2018;11:12–19.
64. Hatamizadeh P, Najafi I, Vanholder R, et al. Epidemiologic aspects of the Bam earthquake in Iran: the nephrologic perspective. *Am J Kidney Dis*. 2006;47:428–438.
65. Kantarci G, Vanholder R, Tuglular S, et al. Acute renal failure due to crush syndrome during Marmara earthquake. *Am J Kidney Dis*. 2002;40:682–689.
66. Gibney RT, Sever MS, Vanholder RC. Disaster nephrology: crush injury and beyond. *Kidney Int*. 2014;85:1049–1057.
67. Better OS, Rubinstein I, Reis DN. Muscle crush compartment syndrome: fulminant local edema with threatening systemic effects. *Kidney Int*. 2003;63:1155–1157.
68. Correa-Rotter R, Naicker S, Katz JJ, et al. Demographic and epidemiologic transition in the developing world: role of albuminuria in the early diagnosis and prevention of renal and cardiovascular disease. *Kidney Int*. 2004;66:S32–S37.
69. Farag YM, Kari JA, Singh AK. Chronic kidney disease in the Arab world: a call for action. *Nephron Clin Pract*. 2012;121:c120–c123.
70. Hijazi S. Major health issues in Jordan. *Ethn Dis*. 2005;15:S1–77–82.
71. Khogali M:B. Health and disease in a changing arab world 2000/2025/2050. *Ethn Dis*. 2005;15:74–75.
72. Kovesdy CP. Epidemiology of chronic kidney disease: an update 2022. *Kidney Int Suppl*. 2022;12:7–11.
73. Brauer M, Roth GA, Aravkin AY, et al. Global burden and strength of evidence for 88 risk factors in 204 countries and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet*. 2024;403:2162–2203.
74. Ferrari AJ, Santomauro DF, Aali A, et al. Global incidence, prevalence, years lived with disability (YLDs), disability-adjusted life-years (DALYs), and healthy life expectancy (HALE) for 371 diseases and injuries in 204 countries and territories and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet*. 2024;403:2133–2161.
75. GBDCoD C. Global burden of 288 causes of death and life expectancy decomposition in 204 countries and territories and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet*. 2024;403:2100–2132.
76. Collaboration GBDCKD. Global, regional, and national burden of chronic kidney disease, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet*. 2020;395:709–733.
77. Federation ID. *IDF Diabetes Atlas*. 10th ed. International diabetes federation; 2021.
78. Alashek WA, McIntyre CW, Taal MW. Epidemiology and aetiology of dialysis-treated end-stage kidney disease in Libya. *BMC Nephrol*. 2012;13:33.
79. Al-Azzam SI, Abu-Dahoud EY, El-Khatib HA, et al. Etiologies of chronic renal failure in Jordanian population. *J Nephrol*. 2007;20:336–339.
80. Alhyas L, McKay A, Majeed A. Prevalence of type 2 diabetes in the States of the co-operation council for the Arab States of the Gulf: a systematic review. *PLoS One*. 2012;7:e40948.
81. Bennet L, Groop L, Lindblad U, et al. Ethnicity is an independent risk indicator when estimating diabetes risk with FINDRISC scores: a cross sectional study comparing immigrants from the Middle East and native Swedes. *Primary care diabetes*. 2014;8:231–238.
82. Farag YM, Al Wakeel JS. Diabetic nephropathy in the arab Gulf countries. *Nephron Clin Pract*. 2011;119:c317–c322; discussion c322–313.
83. Jaber LA, Brown MB, Hammad A, et al. The prevalence of the metabolic syndrome among arab americans. *Diabetes Care*. 2004;27:234–238.
84. Kridli SA-O, Herman WH, Brown MB, et al. The epidemiology of diabetes and its risk factors among Chaldean Americans. *Ethn Dis*. 2006;16:351–356.
85. Safarinejad MR. The epidemiology of adult chronic kidney disease in a population-based study in Iran: prevalence and associated risk factors. *J Nephrol*. 2009;22:99–108.
86. Suleiman K, Ghattas B, Makhlood C. The health status of the Palestinian Arab community in Israel in relation to the Jewish community in Israel. *Ethn Dis*. 2005;15.
87. Suleymanlar G, Utas C, Arinsoy T, et al. A population-based survey of Chronic RENal Disease in Turkey—the CREDIT study. *Nephrol Dial Transplant*. 2011;26:1862–1871.
88. Bello AK, Johnson DW, Feehally J, et al. Global Kidney Health Atlas (GKHA): design and methods. *Kidney Int Suppl*. 2017;7:145–153.
89. Benghanem GM, Elseviers M, Zamd M, et al. Chronic kidney disease, hypertension, diabetes, and obesity in the adult population of Morocco: how to avoid “over”- and “under”-diagnosis of CKD. *Kidney Int*. 2016;89:1363–1371.
90. Thomas B, Matsushita K, Abate KH, et al. Global cardiovascular and renal outcomes of reduced GFR. *J Am Soc Nephrol*. 2017;28:2167–2179.
91. Karam S, Amouzegar A, Alshamsi IR, et al. Capacity for the management of kidney failure in the international society of nephrology Middle East region: report from the 2023 ISN Global Kidney Health Atlas (ISN-GKHA). *Kidney Int Suppl*. 2024;13:57–70.
92. Al-Eisa AA, Samhan M, Naseef M. End-stage renal disease in Kuwaiti children: an 8-year experience. *Transplant Proc*. 2004;36:1788–1791.
93. Barbari A, Stephan A, Masri M, et al. Consanguinity-associated kidney diseases in Lebanon: an epidemiological study. *Mol Immunol*. 2003;39:1109–1114.
94. Bener A, Hussain R, Teebi AS. Consanguineous marriages and their effects on common adult diseases: studies from an endogamous population. *Med Princ Pract*. 2007;16:262–267.
95. Finer G, Shalev H, Landau D. Genetic kidney diseases in the pediatric population of southern Israel. *Pediatr Nephrol*. 2006;21:910–916.
96. Karnib HH, Gharavi AG, Aftimos G, et al. A 5-year survey of biopsy proven kidney diseases in Lebanon: significant variation in prevalence of primary glomerular diseases by age, population structure and consanguinity. *Nephrol Dial Transplant*. 2010;25:3962–3969.
97. Ozen S. Renal amyloidosis in familial Mediterranean fever. *Kidney Int*. 2004;65:1118–1127.
98. Zlotogora J, Shalev SA. Marriage patterns and reproductive decision-making in the inhabitants of a single Muslim village during a 50-year period. *Hum Hered*. 2014;77:10–15.
99. Balat A, Kilic BD, Aksu B, et al. Kidney disease profile and encountered problems during follow-up in Syrian refugee children: a multicenter retrospective study. *Pediatr Nephrol*. 2022;37:393–402.
100. Sadowski CE, Lovric S, Ashraf S, et al. A single-gene cause in 29.5% of cases of steroid-resistant nephrotic syndrome. *J Am Soc Nephrol*. 2015;26:1279–1289.
101. Taal MW. Screening for chronic kidney disease: preventing harm or harming the healthy? *PLoS Med*. 2012;9:e1001345.

102. Bello AK, Levin A, Tonelli M, et al. Assessment of global kidney health care status. *JAMA*. 2017;317:1864–1881.
103. Hassanien AA, Al-Shaikh F, Vamos EP, et al. Epidemiology of end-stage renal disease in the countries of the Gulf Cooperation Council: a systematic review. *JRSM Short Rep*. 2012;3(38).
104. Tawhari M, Alhamadh M, Alhabeeb A, et al. Establishing the Kidney disease in the National guard (KIND) registry: an opportunity for epidemiological and clinical research in Saudi Arabia. *BMC Nephrol*. 2024;25(59).
105. Bello AK, Okpechi IG, Levin A, et al. An update on the global disparities in kidney disease burden and care across world countries and regions. *Lancet Glob Health*. 2024;12:e382–e395.
106. Lunney M, Bello AK, Levin A, et al. Availability, accessibility, and quality of conservative kidney management worldwide. *Clin J Am Soc Nephrol*. 2021;16:79–87.
107. Pippias M, Alfano G, Kelly DM, et al. Capacity for the management of kidney failure in the international society of nephrology western Europe region: report from the 2023 ISN Global Kidney Health Atlas (ISN-GKHA). *Kidney Int Suppl*. 2024;13:136–151.
108. International Society of Nephrology—ISN-Global Kidney Health Atlas. Available at <https://www.theisn.org/initiatives/global-kidney-health-atlas/>. Accessed August 8, 2024.
109. Grams ME, Sang Y, Ballew SH, et al. Predicting timing of clinical outcomes in patients with chronic kidney disease and severely decreased glomerular filtration rate. *Kidney Int*. 2018;93:1442–1451.
110. Nelson RG, Grams ME, Ballew SH, et al. Development of risk prediction equations for incident chronic kidney disease. *JAMA*. 2019;322:2104–2114.
111. Grams ME, Brunskill NJ, Ballew SH, et al. Development and validation of prediction models of adverse kidney outcomes in the population with and without diabetes. *Diabetes Care*. 2022;45:2055–2063.
112. Shin JI, Chang AR, Grams ME, et al. Albuminuria testing in hypertension and diabetes: an individual-participant data meta-analysis in a global Consortium. *Hypertension*. 2021;78:1042–1052.
113. *Nephrology Worldwide*. Springer Nature Switzerland AG; 2021.
114. Farag YMK, El-Sayed E. Global dialysis perspective: Egypt. *Kidney360*. 2022;3:1263–1268.
115. Regev-Epstein LC, Frishberg Y, Davidovits M, et al. Dialysis in Israeli children between 1990 and 2020: trends and international comparisons. *Clin J Am Soc Nephrol*. 2023;18:363–373.
116. Haviv YS, Golan E. Global dialysis perspective: Israel. *Kidney360*. 2020;1:119–122.
117. Aoun M, Koubar S. Global dialysis perspective: Lebanon. *Kidney360*. 2023;4:e1308–e1313.
118. Andacoglu O, Aki FT. Global perspective on kidney transplantation: Turkey. *Kidney360*. 2021;2:1160–1162.
119. Hafez MH. Global perspective: kidney transplantation in North Africa. *Kidney360*. 2024. <https://doi.org/10.34067/KID.0000000000000497>.
120. Vivante A, Golan E, Tzur D, et al. Body mass index in 1.2 million adolescents and risk for end-stage renal disease. *Arch Intern Med*. 2012;172:1644–1650.
121. Vivante A, Afek A, Frenkel-Nir Y, et al. Persistent asymptomatic isolated microscopic hematuria in Israeli adolescents and young adults and risk for end-stage renal disease. *JAMA*. 2011;306:729–736.
122. Calderon-Margalit R, Golan E, Twig G, et al. History of childhood kidney disease and risk of adult end-stage renal disease. *N Engl J Med*. 2018;378:428–438.
123. Perico N, Bravo RF, De Leon FR, et al. Screening for chronic kidney disease in emerging countries: feasibility and hurdles. *Nephrol Dial Transplant*. 2009;24:1355–1358.
124. Gouda Z, Mashaal G, Bello AK, et al. Egypt Information, Prevention, and Treatment of Chronic Kidney Disease (EGIPT-CKD) programme: prevalence and risk factors for microalbuminuria among the relatives of patients with CKD in Egypt. *Saudi J Kidney Dis Transpl*. 2011;22:1055–1063.
125. Eknoyan G, Lameire N, Eckardt K, et al. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int*. 2013;3:5–14.
126. Alsuwaidia AO, Farag YM, Al Sayyari AA, et al. Epidemiology of chronic kidney disease in the Kingdom of Saudi Arabia (SEEK-Saudi investigators)—a pilot study. *Saudi J Kidney Dis Transpl*. 2010;21:1066–1072.
127. El-Reshaid W, El-Reshaid K, Kapoor M, et al. Chronic renal disease in Kuwaiti nationals: a prospective study during the past 4 years. *Ren Fail*. 2005;27:227–233.
128. Mahdavi-Mazdeh M, Saeed Hashemi Nazri S, Hajghasemi E, et al. Screening for decreased renal function in taxi drivers in Tehran, Iran. *Ren Fail*. 2010;32:62–68.
129. Hosseinpah F, Kasraei F, Nassiri AA, et al. High prevalence of chronic kidney disease in Iran: a large population-based study. *BMC Public Health*. 2009;9:44.
130. Tohidi M, Hasheminia M, Mohebi R, et al. Incidence of chronic kidney disease and its risk factors, results of over 10 year follow up in an Iranian cohort. *PLoS One*. 2012;7:e45304.
131. Asghari G, Farhadnejad H, Mirmiran P, et al. Adherence to the Mediterranean diet is associated with reduced risk of incident chronic kidney diseases among Tehranian adults. *Hypertens Res*. 2017;40:96–102.
132. Yuzbashian E, Asghari G, Mirmiran P, et al. Adherence to low-sodium Dietary Approaches to Stop Hypertension-style diet may decrease the risk of incident chronic kidney disease among high-risk patients: a secondary prevention in prospective cohort study. *Nephrol Dial Transplant*. 2018;33:1159–1168.
133. Asghari G, Yuzbashian E, Mirmiran P, et al. The association between dietary Approaches to Stop hypertension and incidence of chronic kidney disease in adults: the Tehran lipid and glucose study. *Nephrol Dial Transplant*. 2017;32:ii224–ii230.
134. Perico N, Remuzzi G. Prevention programs for chronic kidney disease in low-income countries. *Intern Emerg Med*. 2016;11:385–389.
135. Elkeraie AF, Al-Ghamdi S, Abu-Alfa AK, et al. Impact of sodium-glucose cotransporter-2 inhibitors in the management of chronic kidney disease: a Middle East and Africa perspective. *Int J Nephrol Renovasc Dis*. 2024;17:1–16.
136. Kidney disease: improving global outcomes hepatitis CWG: KDIGO 2022 clinical practice guideline for the prevention, diagnosis, evaluation, and treatment of hepatitis C in chronic kidney disease. *Kidney Int*. 2022;102:S129–S205.
137. Aghighi M, Mahdavi-Mazdeh M, Zamyadi M, et al. Changing epidemiology of end-stage renal disease in last 10 years in Iran. *Iran J Kidney Dis*. 2009;3:192–196.
138. Najafi I, Alatab S, Atabak S, et al. Seventeen years' experience of peritoneal dialysis in Iran: first official report of the Iranian peritoneal dialysis registry. *Perit Dial Int*. 2014;34:636–642.
139. El Matri A. History of nephrology in mediterranean arab countries. *J Nephrol*. 2013;26:170–174.
140. Al AS. Renal replacement therapy in Bahrain. *Saudi J Kidney Dis Transpl*. 1998;9:457–458.
141. Soliman AR, Fathy A, Roshd D. The growing burden of end-stage renal disease in Egypt. *Ren Fail*. 2012;34:425–428.
142. AlSahow A, AlRukhaimi M, Al Wakeel J, et al. Demographics and key clinical characteristics of hemodialysis patients from the Gulf Cooperation Council countries enrolled in the dialysis outcomes and practice patterns study phase 5 (2012–2015). *Saudi J Kidney Dis Transpl*. 2016;27:S12–S23.
143. Data SCOT. Dialysis in the Kingdom of Saudi Arabia. *Saudi J Kidney Dis Transpl*. 2017;28:949–957.
144. Batieha A, Abdallah S, Maghaireh M, et al. Epidemiology and cost of haemodialysis in Jordan. *East Mediterr Health J*. 2007;13:654–663.
145. Khader MI, Snouber S, Alkhatib A, et al. Prevalence of patients with end-stage renal disease on dialysis in the West Bank, Palestine. *Saudi J Kidney Dis Transpl*. 2013;24:832–837.
146. Younis M, Jabr S, Al-Khatib A, et al. A cost analysis of kidney replacement therapy options in Palestine. *Inquiry: The Journal of Health Care Organization, Provision, and Financing*. 2015;52:0046958015573494.
147. Ereke E, Suleymanlar G, Serdengeci K. Nephrology, dialysis and transplantation in Turkey. *Nephrol Dial Transplant*. 2002;17:2087–2093.
148. Erlik D, Better OS, Shramek A, et al. Cadaveric renal transplantation in man: first report from Israel. *Isr J Med Sci*. 1967;3:88–92.
149. Abomelha MS. Renal failure and transplantation activity in the arab world. Arab society of nephrology and renal transplantation. *Nephrol Dial Transplant*. 1996;11:28–29.
150. Saeed B, Derani R, Hajibrahim M, et al. Organ failure in Syria: initiating a national deceased donation program. *Saudi J Kidney Dis Transpl*. 2007;18:270–276.
151. Mahdavi-Mazdeh M. The Iranian model of living renal transplantation. *Kidney Int*. 2012;82:627–634.

152. Al Sayyari AA. The history of renal transplantation in the Arab world: a view from Saudi Arabia. *Am J Kidney Dis.* 2008;51:1033–1046.
153. Saeed B. Pediatric versus adult kidney transplantation activity in Arab countries. *Saudi J Kidney Dis Transpl.* 2013;24:1031–1038.
154. Shaheen FA, Salmeen NM, Al-Muzairy IA, et al. Renal transplantation in the Gulf Co-operation council countries. *Saudi J Kidney Dis Transpl.* 1995;6:5–8.
155. Shigidi MM, Ramachandiran G, Rashed AH, et al. Demographic data and hemodialysis population dynamics in Qatar: a five year survey. *Saudi J Kidney Dis Transpl.* 2009;20:493–500.
156. Najafi I. Peritoneal dialysis in Iran and the middle East. *Perit Dial Int.* 2009;29(suppl 2):S217–S221.
157. Masri M, Haberal M. Solid-organ transplant activity in MESOT countries. *Exp Clin Transplant.* 2013;11:93–98.
158. Renal replacement therapy in Israel, 1990–2015. *Israel Ministry of Health and the Israel Society for Nephrology and Hypertension [ICDC Publication No. 367, January 2018, in Hebrew].* Available at https://www.health.gov.il/PublicationsFiles/RRT1990_2015.pdf. Accessed
159. Turkish Society of Nephrology: *Registry of the nephrology, dialysis and transplantation in Turkey: registry 2016.* Available at http://www.nefroloji.org.tr/folders/file/2016_REGISTRY.pdf. Accessed
160. The Hashemite Kingdom of Jordan Ministry of Health, Non-Communicable Diseases Directorate: *National Registry of End Stage Renal Disease (ESRD), 13th annual report, 2020.* Available at https://moh.gov.jo/ebv4.0/root_storage/ar/eb_list_page/kidney_report_2020_finale.pdf. Accessed August 8, 2024.
161. *The 2012 report of the Lebanese kidney registry.* Available at <http://kidneyregistrylb.com/national-registry-2/annual-report-2012/>. Accessed
162. Kramer A, Pippias M, Noordzij M, et al. The European renal association - European dialysis and transplant association (ERA-EDTA) registry annual report 2015: a summary. *Clin Kidney J.* 2018;11:108–122.
163. Boerstra BA, Boenink R, Astley ME, et al. The ERA registry annual report 2021: a summary. *Clinical Kidney Journal.* 2023;17.
164. US Renal Data System 2023 Annual Data Report: *International Comparisons.* Available at <https://usrdp.adr.niddk.nih.gov/2023/end-stage-renal-disease/11-international-comparisons>. Accessed August 8, 2024.
165. Saran R, Robinson B, Abbott KC, et al. US renal data system 2017 annual data report: epidemiology of kidney disease in the United States. *Am J Kidney Dis.* 2018;71:A7.
166. Suleymanlar G, Serdengeci K, Altıparmak MR, et al. Trends in renal replacement therapy in Turkey, 1996–2008. *Am J Kidney Dis.* 2011;57:456–465.
167. Seyahi N, Kocyigit I, Eren N, et al. Current status of kidney replacement therapy in Türkiye: a summary of 2022 Turkish Society of Nephrology registry report. *Turk J Nephrol.* 2024;33:134–139.
168. Utaş C. The development of PD in Turkey. *Perit Dial Int.* 2008;28:217–219.
169. Kalantar-Zadeh K, Golan E, Shohat T, et al. Survival disparities within American and Israeli dialysis populations: learning from similarities and distinctions across race and ethnicity. *Semin Dial.* 2010;23:586–594.
170. Counil E, Cherni N, Kharrat M, et al. Trends of incident dialysis patients in Tunisia between 1992 and 2001. *Am J Kidney Dis.* 2008;51:463–470.
171. Bouguerra R, Alberti H, Salem LB, et al. The global diabetes pandemic: the Tunisian experience. *Eur J Clin Nutr.* 2007;61:160–165.
172. Alashek WA, McIntyre CW, Taal MW. Provision and quality of dialysis services in Libya. *Hemodial Int.* 2011;15:444–452.
173. Alashek WA, McIntyre CW, Taal MW. Hepatitis B and C infection in haemodialysis patients in Libya: prevalence, incidence and risk factors. *BMC Infect Dis.* 2012;12:265.
174. Alashek WA, McIntyre CW, Taal MW. Vascular access in patients receiving hemodialysis in Libya. *J Vasc Access.* 2012;13:468–474.
175. Alashek WA, McIntyre CW, Taal MW. Determinants of survival in patients receiving dialysis in Libya. *Hemodial Int.* 2013;17:249–255.
176. Al AB. Renal replacement therapy in the Kingdom of Saudi Arabia. *Saudi J Kidney Dis Transpl.* 2021;32:1188–1200.
177. Health indicators for renal replacement therapy in Saudi Arabia. *Saudi J Kidney Dis Transpl.* 2018;29:225–231.
178. Haghighi AN, Broumand B, D'Amico M, et al. The epidemiology of end-stage renal disease in Iran in an international perspective. *Nephrol Dial Transplant.* 2002;17:28–32.
179. Mahdavi-Mazdeh M, Zamyadi M, Nafar M. Assessment of management and treatment responses in haemodialysis patients from Tehran province, Iran. *Nephrol Dial Transplant.* 2008;23:288–293.
180. Ghods AJ, Nasrollahzadeh D, Kazemeini M. Afghan refugees in Iran model renal transplantation program: ethical considerations. *Transplant Proc.* 2005;37:565–567.
181. Barsoum RS. End-stage renal disease in North Africa. *Kidney Int.* 2003;S111–S114.
182. Mortada WI, Sobh MA, El-Defrawy MM, et al. Study of lead exposure from automobile exhaust as a risk for nephrotoxicity among traffic policemen. *Am J Nephrol.* 2001;21:274–279.
183. Tuglular S, Yalcinkaya F, Paydas S, et al. A retrospective analysis for aetiology and clinical findings of 287 secondary amyloidosis cases in Turkey. *Nephrol Dial Transplant.* 2002;17:2003–2005.
184. Abboud O. Incidence, prevalence, and treatment of end-stage renal disease in the Middle East. *Ethn Dis.* 2006;16:S2–2–4.
185. Abumwais JQ. Etiology of chronic renal failure in Jenin district, Palestine. *Saudi J Kidney Dis Transpl.* 2012;23:158–161.
186. Al SI, Bieber B, Al RM, et al. Parathyroid hormone serum levels and mortality among hemodialysis patients in the Gulf Cooperation council countries: results from the DOPPS (2012–2018). *Kidney* 2020;1:1083–1090.
187. Al-Ghamdi SMG, Bieber B, AlRukhaimi M, et al. Diabetes prevalence, treatment, control, and outcomes among hemodialysis patients in the Gulf Cooperation council countries. *Kidney Int Rep.* 2022;7:1093–1102.
188. Badheeb AM. Causes of chronic renal failure in hemodialysis unit: a single center experience in Yemen. *Saudi J Kidney Dis Transpl.* 2006;17:66–69.
189. Teebi AS, Teebi SA. Genetic diversity among the Arabs. *Community Genet.* 2005;8:21–26.
190. Zlotogora J. Molecular basis of autosomal recessive diseases among the Palestinian Arabs. *Am J Med Genet.* 2002;109:176–182.
191. Raz AE, Atar M, Rodnay M, et al. Between acculturation and ambivalence: knowledge of genetics and attitudes towards genetic testing in a consanguineous bedouin community. *Community Genet.* 2003;6:88–95.
192. Cohen T, Vardi-Saliternik R, Friedlander Y. Consanguinity, intra-community and intercommunity marriages in a population sample of Israeli Jews. *Ann Hum Biol.* 2004;31:38–48.
193. Mattoo TK. Genetically transmitted renal diseases in children: a Saudi perspective. *Saudi J Kidney Dis Transpl.* 1998;9:105–109.
194. Al-Eisa A, Naseef M, Al-Hamad N, et al. Chronic renal failure in Kuwaiti children: an eight-year experience. *Pediatr Nephrol.* 2005;20:1781–1785.
195. Jaouad IC, Elalaoui SC, Sbiti A, et al. Consanguineous marriages in Morocco and the consequence for the incidence of autosomal recessive disorders. *J Biosoc Sci.* 2009;41:575–581.
196. Hassan MO, Jaju D, Albarwani S, et al. Non-dipping blood pressure in the metabolic syndrome among Arabs of the Oman family study. *Obesity.* 2007;15:2445–2453.
197. Vogel F, Motulsky AG. *Human Genetics: Problems and Approaches.* New York: Springer-Verlag; 1997.
198. Ben-Moshe Y, Shlomovitz O, Atias-Varon D, et al. Diagnostic utility of exome sequencing among Israeli children with kidney failure. *Kidney Int Rep.* 2023;8:2126–2135.
199. Frishberg Y, Ben-Neria Z, Suvanto M, et al. Misleading findings of homozygosity mapping resulting from three novel mutations in NPHS1 encoding nephrin in a highly inbred community. *Genet Med.* 2007;9:180–184.
200. Saeed MB. The major causes of chronic renal insufficiency in Syrian children: a one-year, single-center experience. *Saudi J Kidney Dis Transpl.* 2005;16:84–88.
201. Yousri NA, Fakhro KA, Robay A, et al. Whole-exome sequencing identifies common and rare variant metabolic QTLs in a Middle Eastern population. *Nat Commun.* 2018;9:333.
202. Mohamed SA, Fernandez-Tajes J, Franks PW, et al. GWAS in people of Middle Eastern descent reveals a locus protective of kidney function: a cross-sectional study. *BMC Med.* 2022;20(76).
203. Bourquia A. Societe marocaine des maladies r: [Autosomal dominant polycystic kidney disease (ADPKD). in Morocco. Multicenter study about 308 families]. *Nephrologie.* 2002;23:93–96.

204. Kazancioglu R, Ecder T, Altintepe L, et al. Demographic and clinical characteristics of patients with autosomal dominant polycystic kidney disease: a multicenter experience. *Nephron Clin Pract.* 2011;117:c270–c275.
205. Roozbeh J, Razmkon AR, Jalaiean H, et al. Outcome of kidney transplantation in patients with polycystic kidney disease: a single center study. *Saudi J Kidney Dis Transpl.* 2008;19:72–75.
206. Ward CJ, Yuan D, Masyuk TV, et al. Cellular and subcellular localization of the ARPKD protein; fibrocystin is expressed on primary cilia. *Hum Mol Genet.* 2003;12:2703–2710.
207. Department of Human Genetics. Available at <http://www.humgen.rwth-aachen.de/index.php?page=database>. Accessed
208. Bergmann C, Senderek J, Sedlacek B, et al. Spectrum of mutations in the gene for autosomal recessive polycystic kidney disease (ARPKD/PKHD1). *J Am Soc Nephrol.* 2003;14:76–89.
209. Bergmann C, Senderek J, Windelen E, et al. Clinical consequences of PKHD1 mutations in 164 patients with autosomal-recessive polycystic kidney disease (ARPKD). *Kidney Int.* 2005;67:829–848.
210. Quint A, Sagi M, Carmi S, et al. An Ashkenazi founder mutation in the PKHD1 gene. *Eur J Med Genet.* 2016;59:86–90.
211. Al-Belushi M, Al Ibrahim A, Ahmed M, et al. A review of Meckel-Gruber syndrome—incidence and outcome in the state of Qatar. *J Matern Fetal Neonatal Med.* 2016;29:2013–2016.
212. Aalimi U, Spiegel E, Chervinsky I, et al. [High incidence and broad genetic variability of Meckel-Gruber syndrome in the arab population residing in North-East Israel]. *Harefuah.* 2015;154:632–636, 676.
213. Landau D, Shalev H. Childhood genetic renal diseases in southern Israel]. *Harefuah.* 2010;149:180–185, 193.
214. Soliman NA, Hildebrandt F, Otto EA, et al. Clinical characterization and NPHP1 mutations in nephronophthisis and associated ciliopathies: a single center experience. *Saudi J Kidney Dis Transpl.* 2012;23:1090–1098.
215. Tory K, Rousset-Rouviere C, Gubler MC, et al. Mutations of NPHP2 and NPHP3 in infantile nephronophthisis. *Kidney Int.* 2009;75:839–847.
216. Dehghani M, Mojarad M, Ghayoor Karimiani E, et al. A common ancestral Asn242Ser mutation in TMEM67 identified in multiple Iranian families with Joubert syndrome. *Public Health Genom.* 2017;20:188–193.
217. Karnib HH, Sanna-Cherchi S, Zalloua PA, et al. Characterization of a large Lebanese family segregating IgA nephropathy. *Nephrol Dial Transplant.* 2007;22:772–777.
218. Boute N, Gribouval O, Roselli S, et al. NPHS2, encoding the glomerular protein podocin, is mutated in autosomal recessive steroid-resistant nephrotic syndrome. *Nat Genet.* 2000;24:349–354.
219. Frishberg Y, Rinat C, Megged O, et al. Mutations in NPHS2 encoding podocin are a prevalent cause of steroid-resistant nephrotic syndrome among Israeli-Arab children. *J Am Soc Nephrol.* 2002;13:400–405.
220. Frishberg Y, Feinstein S, Rinat C, et al. The heart of children with steroid-resistant nephrotic syndrome: is it all podocin? *J Am Soc Nephrol.* 2006;17:227–231.
221. Berdeli A, Mir S, Yavascan O, et al. NPHS2 (podocin) mutations in Turkish children with idiopathic nephrotic syndrome. *Pediatr Nephrol.* 2007;22:2031–2040.
222. Ozcakar ZB, Cengiz FB, Cakar N, et al. Analysis of NPHS2 mutations in Turkish steroid-resistant nephrotic syndrome patients. *Pediatr Nephrol.* 2006;21:1093–1096.
223. Basiratnia M, Yavarian M, Torabinezhad S, et al. NPHS2 gene in steroid-resistant nephrotic syndrome: prevalence, clinical course, and mutational spectrum in South-West Iranian children. *Iran J Kidney Dis.* 2013;7:357–362.
224. Landau D, Oved T, Geiger D, et al. Familial steroid-sensitive nephrotic syndrome in Southern Israel: clinical and genetic observations. *Pediatr Nephrol.* 2007;22:661–669.
225. Al-Hamed MH, Al-Sabban E, Al-Mojalli H, et al. A molecular genetic analysis of childhood nephrotic syndrome in a cohort of Saudi Arabian families. *J Hum Genet.* 2013;58:480–489.
226. Trautmann A, Bodria M, Ozaltin F, et al. Spectrum of steroid-resistant and congenital nephrotic syndrome in children: the PoDoNet registry cohort. *Clin J Am Soc Nephrol.* 2015;10:592–600.
227. Kari JA, El-Desoky SM, Gari M, et al. Steroid-resistant nephrotic syndrome: impact of genetic testing. *Ann Saudi Med.* 2013;33:533–538.
228. Mir S, Yavascan O, Berdeli A, et al. TRPC6 gene variants in Turkish children with steroid-resistant nephrotic syndrome. *Nephrol Dial Transplant.* 2012;27:205–209.
229. Braun DA, Rao J, Mollet G, et al. Mutations in KEOPS-complex genes cause nephrotic syndrome with primary microcephaly. *Nat Genet.* 2017;49:1529–1538.
230. Ashraf S, Gee HY, Woerner S, et al. ADCK4 mutations promote steroid-resistant nephrotic syndrome through CoQ10 biosynthesis disruption. *J Clin Invest.* 2013;123:5179–5189.
231. Atmaca M, Gulhan B, Korkmaz E, et al. Follow-up results of patients with ADCK4 mutations and the efficacy of CoQ10 treatment. *Pediatr Nephrol.* 2017;32:1369–1375.
232. Wang L, Ellis MJ, Gomez JA, et al. Mechanisms of the proteinuria induced by Rho GTPases. *Kidney Int.* 2012;81:1075–1085.
233. Gee HY, Saisawat P, Ashraf S, et al. ARHGDI mutations cause nephrotic syndrome via defective RHO GTPase signaling. *J Clin Invest.* 2013;123:3243–3253.
234. Beck BB, Hoyer-Kuhn H, Gobel H, et al. Hyperoxaluria and systemic oxalosis: an update on current therapy and future directions. *Expert Opin Invest Drugs.* 2013;22:117–129.
235. Hoppe B. An update on primary hyperoxaluria. *Nat Rev Nephrol.* 2012;8:467–475.
236. Ben-Shalom E, Frishberg Y. Primary hyperoxalurias: diagnosis and treatment. *Pediatr Nephrol.* 2015;30:1781–1791.
237. Belostotsky R, Seboun E, Idelson GH, et al. Mutations in DHDPSL are responsible for primary hyperoxaluria type III. *Am J Hum Genet.* 2010;87:392–399.
238. Belostotsky R, Pitt JJ, Frishberg Y. Primary hyperoxaluria type III—a model for studying perturbations in glyoxylate metabolism. *J Mol Med (Berl).* 2012;90:1497–1504.
239. Mandrile G, Beck B, Acquaviva C, et al. Genetic assessment in primary hyperoxaluria: why it matters. *Pediatr Nephrol.* 2023;38:625–634.
240. Singh P, Viehman JK, Mehta RA, et al. Clinical characterization of primary hyperoxaluria type 3 in comparison with types 1 and 2. *Nephrol Dial Transplant.* 2022;37:869–875.
241. Hopp K, Cogal AG, Bergstralh EJ, et al. Phenotype-genotype correlations and estimated carrier frequencies of primary hyperoxaluria. *J Am Soc Nephrol.* 2015;26:2559–2570.
242. Bar R, Ben-Shalom E, Duvdevani M, et al. Mutations in HOGA1 do not confer a dominant phenotype manifesting as kidney stone disease. *J Urol.* 2021;205:1394–1399.
243. Frishberg Y, Rinat C, Shalata A, et al. Intra-familial clinical heterogeneity: absence of genotype-phenotype correlation in primary hyperoxaluria type 1 in Israel. *Am J Nephrol.* 2005;25:269–275.
244. Levin-Iaina N, Dinour D, Romero L, et al. Late diagnosis of primary hyperoxaluria type 2 in the adult: effect of a novel mutation in GRHPR gene on enzymatic activity and molecular modeling. *J Urol.* 2009;181:2146–2151.
245. Rinat C, Wanders RJ, Drukker A, et al. Primary hyperoxaluria type I: a model for multiple mutations in a monogenic disease within a distinct ethnic group. *J Am Soc Nephrol.* 1999;10:2352–2358.
246. Soliman NA, Nabhan MM, Abdelrahman SM, et al. Clinical spectrum of primary hyperoxaluria type 1: experience of a tertiary center. *Nephrol Ther.* 2017;13:176–182.
247. Soliman NA, Elmonem MA, Abdelrahman SM, et al. Clinical and molecular characterization of primary hyperoxaluria in Egypt. *Sci Rep.* 2022;12:15886.
248. Wannous H. Primary hyperoxaluria type 1 in children: clinical and laboratory manifestations and outcome. *Pediatr Nephrol.* 2023;38:2643–2648.
249. Harambat J, Fargue S, Acquaviva C, et al. Genotype-phenotype correlation in primary hyperoxaluria type 1: the p.Gly170Arg AGXT mutation is associated with a better outcome. *Kidney Int.* 2010;77:443–449.
250. Abu-Ghanem Y, Kleinmann N, Winkler HZ, et al. Nephrolithiasis in Israel: epidemiological characteristics of return patients in a tertiary care center. *Isr Med Assoc J.* 2016;18:725–728.
251. Almardini RI, Alfarah MG, Salaita GM. The clinical pattern of primary hyperoxaluria in pediatric patient at Queen Rania Abdulla Children Hospital. *Arab J Nephrol Transplant.* 2014;7:119–123.
252. Elmaci AM, Ece A, Akin F. Pediatric urolithiasis: metabolic risk factors and follow-up results in a Turkish region with endemic stone disease. *Urolithiasis.* 2014;42:421–426.

253. Al-Eisa AA, Al-Hunayyan A, Gupta R. Pediatric urolithiasis in Kuwait. *Int Urol Nephrol*. 2002;33:3–6.
254. Sanjad SA, Al-Abbad A, Al-Sabban E. Primary hyperoxaluria type 1: an underestimated cause of nephrocalcinosis and chronic renal failure in Saudi Arabian children. *Ann Saudi Med*. 1999;19:4–7.
255. Liebow A, Li X, Racie T, et al. An investigational RNAi therapeutic targeting glycolate oxidase reduces oxalate production in models of primary hyperoxaluria. *J Am Soc Nephrol*. 2017;28:494–503.
256. Frishberg Y, Zeharia A, Lyakhovetsky R, et al. Mutations in HAO1 encoding glycolate oxidase cause isolated glycolic aciduria. *J Med Genet*. 2014;51:526–529.
257. Frishberg Y, Deschenes G, Groothoff JW, et al. Phase 1/2 study of lumasiran for treatment of primary hyperoxaluria type 1: a placebo-controlled randomized clinical trial. *Clin J Am Soc Nephrol*. 2021;16:1025–1036.
258. Garrelfs SF, Frishberg Y, Hulton SA, et al. Lumasiran, an RNAi therapeutic for primary hyperoxaluria type 1. *N Engl J Med*. 2021;384:1216–1226.
259. Hayes W, Sas DJ, Magen D, et al. Efficacy and safety of lumasiran for infants and young children with primary hyperoxaluria type 1: 12-month analysis of the phase 3 ILLUMINATE-B trial. *Pediatr Nephrol*. 2023;38:1075–1086.
260. Sas DJ, Magen D, Hayes W, et al. Phase 3 trial of lumasiran for primary hyperoxaluria type 1: a new RNAi therapeutic in infants and young children. *Genet Med*. 2022;24:654–662.
261. Hulton SA, Groothoff JW, Frishberg Y, et al. Randomized clinical trial on the long-term efficacy and safety of lumasiran in patients with primary hyperoxaluria type 1. *Kidney Int Rep*. 2022;7:494–506.
262. Michael M, Groothoff JW, Shasha-Lavsky H, et al. Lumasiran for advanced primary hyperoxaluria type 1: phase 3 ILLUMINATE-C trial. *Am J Kidney Dis*. 2023;81:145–155 e141.
263. Sellier-Leclerc AL, Metry E, Clave S, et al. Isolated kidney transplantation under lumasiran therapy in primary hyperoxaluria type 1: a report of five cases. *Nephrol Dial Transplant*. 2023;38:517–521.
264. Goldfarb DS, Lieske JC, Groothoff J, et al. Nedosiran in primary hyperoxaluria subtype 3: results from a phase I, single-dose study (PHYOX4). *Urolithiasis*. 2023;51(80).
265. Baum MA, Langman C, Cochat P, et al. PHYOX2: a pivotal randomized study of nedosiran in primary hyperoxaluria type 1 or 2. *Kidney Int*. 2023;103:207–217.
266. Aydogdu SD, Kirel B, Coskun T, et al. Prevalence of cystinuria among elementary schoolchildren in Eskisehir, Turkey. *Scand J Urol Nephrol*. 2009;43:138–141.
267. Pras E, Raben N, Golomb E, et al. Mutations in the SLC3A1 transporter gene in cystinuria. *Am J Hum Genet*. 1995;56:1297–1303.
268. Al-Riyami AA, Suleiman AJ, Afifi M, et al. A community-based study of common hereditary blood disorders in Oman. *East Mediterr Health J*. 2001;7:1004–1011.
269. Tanzer F, Ozgur A, Bardakci F. Type I cystinuria and its genetic basis in a population of Turkish school children. *Int J Urol*. 2007;14:914–917.
270. Colombo R. Dating the origin of the V170M mutation causing non-type I cystinuria in Libyan Jews by linkage disequilibrium and physical mapping of the SLC7A9 gene. *Genomics*. 2000;69:131–134.
271. Sidi R, Levy-Nissenbaum E, Kreiss Y, et al. Clinical manifestations in Israeli cystinuria patients and molecular assessment of carrier rates in Libyan Jewish controls. *Isr Med Assoc J*. 2003;5:439–442.
272. Lotan D, Yoskovitz G, Bisceglia L, et al. A combined approach to the molecular analysis of cystinuria: from urinalysis to sequencing via genotyping. *Isr Med Assoc J*. 2007;9:513–516.
273. Aldahmesh MA, Abu-Safieh L, Khan AO, et al. Allelic heterogeneity in inbred populations: the Saudi experience with Alstrom syndrome as an illustrative example. *Am J Med Genet*. 2009;149A:662–665.
274. Najafi M, Tamandani DMK, Azarfar A, et al. A 57 kb genomic deletion causing CTNS loss of function contributes to the CTNS mutational spectrum in the Middle East. *Front Pediatr*. 2019;7(89).
275. Friedlaender MM, Kopolovic J, Rubinger D, et al. Renal biopsy in Fabry's disease eight years after successful renal transplantation. *Clin Nephrol*. 1987;27:206–211.
276. Raas-Rothschild A, Friedlaender MM, Pizov G, et al. The kidney in Fabry disease. *J Pediatr*. 2005;146:148.
277. Raas-Rothschild A. Pediatrics: implementing the promise of early intervention for Fabry disease. *Clin Ther*. 2007;29(suppl A):S6.
278. El-Reshaid W, El-Reshaid K, Kapoor MM, et al. Glomerulopathy in Kuwait: the spectrum over the past 7 years. *Ren Fail*. 2003;25:619–630.
279. Ozkara HA, Topcu M. Sphingolipidoses in Turkey. *Brain Dev*. 2004;26:363–366.
280. Okur I, Ezgu F, Biberoglu G, et al. Screening for Fabry disease in patients undergoing dialysis for chronic renal failure in Turkey: identification of new case with novel mutation. *Gene*. 2013;527:42–47.
281. Kabalan SN, Abbas S, Tawil L. A search for Fabry disease among male end-stage renal disease patients in Lebanon and a review of the literature. *J Med Liban*. 2013;61:144–147.
282. Guzel T, Caglar FNT, Ekici B, et al. Prevalence of Fabry Disease in patients with left ventricular hypertrophy in Turkey: multicenter study (LVH-TR subgroup analysis). *Int J Cardiovasc Imag*. 2023;39:1143–1155.
283. Abdel-al YK, Badawi MH, Yaeesh SA, et al. Bartter's syndrome in Arabic children: review of 13 cases. *Pediatr Int*. 1999;41:299–303.
284. Zelkovic I, Szargel R, Hawash A, et al. A novel mutation in the chloride channel gene, CLCNKB, as a cause of Gitelman and Bartter syndromes. *Kidney Int*. 2003;63:24–32.
285. Vivante A, Mark-Danieli M, Davidovits M, et al. Renal hypodysplasia associates with a WNT4 variant that causes aberrant canonical WNT signaling. *J Am Soc Nephrol*. 2013;24:550–558.
286. Vivante A, Mann N, Yonath H, et al. A dominant mutation in nuclear receptor interacting protein 1 causes urinary tract malformations via dysregulation of retinoic acid signaling. *J Am Soc Nephrol*. 2017;28:2364–2376.
287. Aldahmesh MA, Humeidan A, Almojalli HA, et al. Characterization of CTNS mutations in Arab patients with cystinosis. *Ophthalmol Genet*. 2009;30:185–189.
288. Bahat H, Dinour D, Ganon L, et al. Non-urate transporter 1-related renal hypouricemia and acute renal failure in an Israeli-Arab family. *Pediatr Nephrol*. 2009;24:999–1003.
289. Besbas N, Ozaltin F, Jeck N, et al. CLCN5 mutation (R347X) associated with hypokalaemic metabolic alkalosis in a Turkish child: an unusual presentation of Dent's disease. *Nephrol Dial Transplant*. 2005;20:1476–1479.
290. Dinour D, Chang MH, Satoh J, et al. A novel missense mutation in the sodium bicarbonate cotransporter (NBCe1/SLC4A4) causes proximal tubular acidosis and glaucoma through ion transport defects. *J Biol Chem*. 2004;279:52238–52246.
291. Dinour D, Davidovitz M, Levin-Iaina N, et al. Truncating mutations in the chloride/proton ClC-5 antiporter gene in Seven Jewish Israeli families with Dent's 1 disease. *Nephron Clin Pract*. 2009;112:c262–c267.
292. Dinour D, Gray NK, Campbell S, et al. Homozygous SLC2A9 mutations cause severe renal hypouricemia. *J Am Soc Nephrol*. 2010;21:64–72.
293. Jahanzad I, Amouei S, Attaranzadeh A. Familial lecithin-cholesterol acyltransferase deficiency. *Arch Iran Med*. 2009;12:179–181.
294. Magen D, Sprecher E, Zelkovic I, et al. A novel missense mutation in SLC5A2 encoding SGLT2 underlies autosomal-recessive renal glucosuria and aminoaciduria. *Kidney Int*. 2005;67:34–41.
295. Al-Haggar M. Cystinosis as a lysosomal storage disease with multiple mutant alleles: phenotypic-genotypic correlations. *World J Nephrol*. 2013;2:94–102.
296. Al-Haggar M, Bakr A, Tajima T, et al. Familial hypomagnesemia with hypercalciuria and nephrocalcinosis: unusual clinical associations and novel claudin 16 mutation in an Egyptian family. *Clin Exp Nephrol*. 2009;13:288–294.
297. Dinour D, Bahn A, Ganon L, et al. URAT1 mutations cause renal hypouricemia type 1 in Iraqi Jews. *Nephrol Dial Transplant*. 2011;26:2175–2181.
298. Dinour D, Gray NK, Ganon L, et al. Two novel homozygous SLC2A9 mutations cause renal hypouricemia type 2. *Nephrol Dial Transplant*. 2012;27:1035–1041.
299. Frishberg Y, Dinour D, Belostotsky R, et al. Dent's disease manifesting as focal glomerulosclerosis: is it the tip of the iceberg? *Pediatr Nephrol*. 2009;24:2369–2373.
300. Magen D, Berger L, Coady MJ, et al. A loss-of-function mutation in NaPi-IIa and renal Fanconi's syndrome. *N Engl J Med*. 2010;362:1102–1109.
301. Mannstadt M, Magen D, Segawa H, et al. Fanconi-Bickel syndrome and autosomal recessive proximal tubulopathy with hypercalciuria

- (ARPTH) are allelic variants caused by GLUT2 mutations. *J Clin Endocrinol Metab.* 2012;97:E1978–E1986.
302. Ozgul RK, Satman I, Collin GB, et al. Molecular analysis and long-term clinical evaluation of three siblings with Alstrom syndrome. *Clin Genet.* 2007;72:351–356.
 303. Soliman NA, Elmonem MA, van den Heuvel L, et al. Mutational spectrum of the CTNS gene in Egyptian patients with nephropathic cystinosis. *JIMD Rep.* 2014;14:87–97.
 304. Topaloglu R, Vilboux T, Coskun T, et al. Genetic basis of cystinosis in Turkish patients: a single-center experience. *Pediatr Nephrol.* 2012;27:115–121.
 305. Vivante A, Lotan D, Pode-Shakked N, et al. Familial autosomal recessive renal tubular acidosis: importance of early diagnosis. *Nephron Physiol.* 2011;119:p31–p39.
 306. Jaradat S, Al-Rababah B, Hazza I, et al. Molecular analysis of the CTNS gene in Jordanian families with nephropathic cystinosis. *Nefrologia.* 2015;35:547–553.
 307. Aleem A. Renal abnormalities in patients with sickle cell disease: a single center report from Saudi Arabia. *Saudi J Kidney Dis Transpl.* 2008;19:194–199.
 308. Steensma DP, Hoyer JD, Fairbanks VF. Hereditary red blood cell disorders in middle eastern patients. *Mayo Clin Proc.* 2001;76:285–293.
 309. Eliakim M, Levy M, Ehrenfeld M. *Recurrent Polyserositis: Familial Mediterranean Fever, Periodic Disease.* Elsevier-North-Holland Biomedical Press; 1981.
 310. French FMFC. A candidate gene for familial Mediterranean fever. *Nat Genet.* 1997;17:25–31.
 311. Consortium TIF. Ancient missense mutations in a new member of the RoRet gene family are likely to cause familial Mediterranean fever. *Cell.* 1997;90:797–807.
 312. Van Gorp H, Huang L, Saavedra P, et al. Blood-based test for diagnosis and functional subtyping of familial Mediterranean fever. *Ann Rheum Dis.* 2020;79:960–968.
 313. Stehlik C, Reed JC. The PYRIN connection: novel players in innate immunity and inflammation. *J Exp Med.* 2004;200:551–558.
 314. Xu H, Yang J, Gao W, et al. Innate immune sensing of bacterial modifications of Rho GTPases by the Pyrin inflammasome. *Nature.* 2014;513:237–241.
 315. Gao W, Yang J, Liu W, et al. Site-specific phosphorylation and microtubule dynamics control Pyrin inflammasome activation. *Proc Natl Acad Sci U S A.* 2016;113:E4857–E4866.
 316. Manukyan G, Aminov R. Update on pyrin functions and mechanisms of familial mediterranean fever. *Front Microbiol.* 2016;7:456.
 317. Bakkaloglu A. Familial mediterranean fever. *Pediatr Nephrol.* 2003;18:853–859.
 318. Tóitout I. Standardized testing for mutations in familial Mediterranean fever. *Clin Chem.* 2003;49:1781–1782.
 319. Kumpfel T, Gerdes LA, Wacker T, et al. Familial Mediterranean fever-associated mutation pyrin E148Q as a potential risk factor for multiple sclerosis. *Mult Scler.* 2012;18:1229–1238.
 320. el-Garf A, Salah S, Iskander I, et al. MEFV mutations in Egyptian patients suffering from familial Mediterranean fever: analysis of 12 gene mutations. *Rheumatol Int.* 2010;30:1293–1298.
 321. Kasifoglu T, Bilge SY, Sari I, et al. Amyloidosis and its related factors in Turkish patients with familial Mediterranean fever: a multicentre study. *Rheumatology.* 2014;53:741–745.
 322. Mohammadnejad L, Farajnia S. Mediterranean Fever gene analysis in the azeri turk population with familial mediterranean Fever: evidence for new mutations associated with disease. *Cell J.* 2013;15:152–159.
 323. Ozdemir O, Sezgin I, Kurtulgan HK, et al. Prevalence of known mutations in the MEFV gene in a population screening with high rate of carriers. *Mol Biol Rep.* 2011;38:3195–3200.
 324. Ait-Idir D, Khilan A, Djerdjouri B, et al. Spectrum of mutations and carrier frequency of familial Mediterranean fever gene in the Algerian population. *Rheumatology.* 2011;50:2306–2310.
 325. Al-Alami JR, Tayeh MK, Najib DA, et al. Familial Mediterranean fever mutation frequencies and carrier rates among a mixed Arabic population. *Saudi Med J.* 2003;24:1055–1059.
 326. Ayesh SK, Nassar SM, Al-Sharef WA, et al. Genetic screening of familial Mediterranean fever mutations in the Palestinian population. *Saudi Med J.* 2005;26:732–737.
 327. Belmahi L, Sefiani A, Fouveau C, et al. Prevalence and distribution of MEFV mutations among Arabs from the Maghreb patients suffering from familial Mediterranean fever. *C R Biol.* 2006;329:71–74.
 328. Ben-Chetrit E, Backenroth R. Amyloidosis induced, end stage renal disease in patients with familial Mediterranean fever is highly associated with point mutations in the MEFV gene. *Ann Rheum Dis.* 2001;60:146–149.
 329. Ben-Chetrit E, Urieli-Shoval S, Calko S, et al. Molecular diagnosis of FMF: lessons from a study of 446 unrelated individuals. *Clin Exp Rheumatol.* 2002;20:S25–S29.
 330. Bidari A, Ghavidel-Parsa B, Najmabadi H, et al. Common MEFV mutation analysis in 36 Iranian patients with familial Mediterranean fever: clinical and demographic significance. *Mod Rheumatol.* 2010;20:566–572.
 331. Brik R, Shinawi M, Kepten I, et al. Familial Mediterranean fever: clinical and genetic characterization in a mixed pediatric population of Jewish and Arab patients. *Pediatrics.* 1999;103:e70.
 332. Caglayan AO, Demiryilmaz F, Ozyazgan I, et al. MEFV gene compound heterozygous mutations in familial Mediterranean fever phenotype: a retrospective clinical and molecular study. *Nephrol Dial Transplant.* 2010;25:2520–2523.
 333. El Gezery DA, Abou-Zeid AA, Hashad DI, et al. MEFV gene mutations in Egyptian patients with familial Mediterranean fever. *Genet Test Mol Biomark.* 2010;14:263–268.
 334. Chaabouni HB, Ksantini M, M'Rad R, et al. MEFV mutations in Tunisian patients suffering from familial Mediterranean fever. *Semin Arthritis Rheum.* 2007;36:397–401.
 335. Esmaeili M, Bonyadi M, Rafeey M, et al. Common MEFV mutation analysis in Iranian Azeri Turkish patients with familial Mediterranean fever. *Semin Arthritis Rheum.* 2008;37:334–338.
 336. Langevitz P, Zemer D, Livneh A, et al. Protracted febrile myalgia in patients with familial Mediterranean fever. *J Rheumatol.* 1994;21:1708–1709.
 337. Livneh A, Langevitz P, Zemer D, et al. Criteria for the diagnosis of familial Mediterranean fever. *Arthritis Rheum.* 1997;40:1879–1885.
 338. Majeed HA, El-Khateeb M, El-Shanti H, et al. The spectrum of familial Mediterranean fever gene mutations in Arabs: report of a large series. *Semin Arthritis Rheum.* 2005;34:813–818.
 339. Mattit H, Joma M, Al-Cheikh S, et al. Familial Mediterranean fever in the Syrian population: gene mutation frequencies, carrier rates and phenotype-genotype correlation. *Eur J Med Genet.* 2006;49:481–486.
 340. Medlej-Hashim M, Rawashdeh M, Chouery E, et al. Genetic screening of fourteen mutations in Jordanian familial Mediterranean fever patients. *Hum Mutat.* 2000;15:384–384.
 341. Sabbagh AS, Ghasham M, Abdel KR, et al. MEFV gene mutations spectrum among Lebanese patients referred for Familial Mediterranean Fever work-up: experience of a major tertiary care center. *Mol Biol Rep.* 2008;35:447–451.
 342. Settin A, El-Baz R, Abd Rasool M, et al. Clinical and molecular diagnosis of familial mediterranean fever in Egyptian children. *J Gastrointest Liver Dis.* 2007;16:141–145.
 343. Sharkia R, Mahajnah M, Zalan A, et al. Comparative screening of FMF mutations in various communities of the Israeli society. *Eur J Med Genet.* 2013;56:351–355.
 344. Shinawi M, Brik R, Berant M, et al. Familial Mediterranean fever: high gene frequency and heterogeneous disease among an Israeli-Arab population. *J Rheumatol.* 2000;27:1492–1495.
 345. Solak M, Yildiz H, Koken R, et al. Analysis of familial Mediterranean fever gene mutations in 202 patients with familial Mediterranean fever. *Genet Test.* 2008;12:341–344.
 346. Tunca MAS, Onen F, et al. Familial Mediterranean fever (FMF) in Turkey: results of a nationwide multicenter study. *Medicine (Baltim).* 2005;84:1–11.
 347. Zaks N, Shinar Y, Padeh S, et al. Analysis of the three most common MEFV mutations in 412 patients with familial Mediterranean fever. *Isr Med Assoc J.* 2003;5:585–588.
 348. Koshy R, Sivasdas A, Scaria V. Genetic epidemiology of familial Mediterranean fever through integrative analysis of whole genome and exome sequences from Middle East and North Africa. *Clin Genet.* 2018;93:92–102.
 349. Dode C, Pecheux C, Cazeneuve C, et al. Mutations in the MEFV gene in a large series of patients with a clinical diagnosis of familial Mediterranean fever. *Am J Med Genet.* 2000;92:241–246.
 350. Sugiura T, Kawaguchi Y, Fujikawa S, et al. Familial Mediterranean fever in three Japanese patients, and a comparison of the

- frequency of MEFV gene mutations in Japanese and Mediterranean populations. *Mod Rheumatol*. 2008;18:57–59.
351. Chae JJ, Aksentijevich I, Kastner DL. Advances in the understanding of familial Mediterranean fever and possibilities for targeted therapy. *Br J Haematol*. 2009;146:467–478.
 352. El-Shanti H, Majeed HA, El-Khateeb M. Familial mediterranean fever in Arabs. *Lancet*. 2006;367:1016–1024.
 353. Guz G, Kanbay M, Ozturk MA. Current perspectives on familial Mediterranean fever. *Curr Opin Infect Dis*. 2009;22:309–315.
 354. Park H, Bourla AB, Kastner DL, et al. Lighting the fires within: the cell biology of autoinflammatory diseases. *Nat Rev Immunol*. 2012;12:570–580.
 355. Hashkes PJ, Spalding SJ, Giannini EH, et al. Rilonacet for colchicine-resistant or -intolerant familial Mediterranean fever: a randomized trial. *Ann Intern Med*. 2012;157:533–541.
 356. Richards N, Schaner P, Diaz A, et al. Interaction between pyrin and the apoptotic speck protein (ASC) modulates ASC-induced apoptosis. *J Biol Chem*. 2001;276:39320–39329.
 357. Gumucio DL, Diaz A, Schaner P, et al. Fire and ICE: the role of pyrin domain-containing proteins in inflammation and apoptosis. *Clin Exp Rheumatol*. 2002;20:S45–S53.
 358. Perfetto L, Gherardini PF, Davey NE, et al. Exploring the diversity of SPRY/B30.2-mediated interactions. *Trends Biochem Sci*. 2013;38:38–46.
 359. Kimura T, Jain A, Choi SW, et al. TRIM-mediated precision autophagy targets cytoplasmic regulators of innate immunity. *J Cell Biol*. 2015;210:973–989.
 360. Weinert C, Morger D, Djekic A, et al. Crystal structure of TRIM20 C-terminal coiled-coil/B30.2 fragment: implications for the recognition of higher order oligomers. *Sci Rep*. 2015;5:10819.
 361. Hoffman HM, Mueller JL, Broide DH, et al. Mutation of a new gene encoding a putative pyrin-like protein causes familial cold autoinflammatory syndrome and Muckle-Wells syndrome. *Nat Genet*. 2001;29:301–305.
 362. Gendelman O, Feifel AJ, Tsur AM, et al. Increased risk of venous thromboembolism among patients with familial Mediterranean fever. *J Thromb Thrombolysis*. 2022;54:669–674.
 363. Ben-Shabat N, Gendelman O, Fisher L, et al. Increased risk for stroke in patients with familial Mediterranean fever: results from a large population-based study. *Rheumatology*. 2023.
 364. Stoffels M, Szperl A, Simon A, et al. MEFV mutations affecting pyrin amino acid 577 cause autosomal dominant autoinflammatory disease. *Ann Rheum Dis*. 2014;73:455–461.
 365. Majeed HA, El-Shanti H, Al-Khateeb MS, et al. Genotype/phenotype correlations in Arab patients with familial Mediterranean fever. *Semin Arthritis Rheum*. 2002;31:371–376.
 366. Shohat M, Magal N, Shohat T, et al. Phenotype-genotype correlation in familial Mediterranean fever: evidence for an association between Met694Val and amyloidosis. *Eur J Hum Genet*. 1999;7:287–292.
 367. Gershoni-Baruch R, Brik R, Zacks N, et al. The contribution of genotypes at the MEFV and SAA1 loci to amyloidosis and disease severity in patients with familial Mediterranean fever. *Arthritis Rheum*. 2003;48:1149–1155.
 368. D'Cruz AA, Babon JJ, Norton RS, et al. Structure and function of the SPRY/B30. 2 domain proteins involved in innate immunity. *Protein Sci*. 2013;22:1–10.
 369. Kone-Paut I, Hentgen V, Guillaume-Czitrom S, et al. The clinical spectrum of 94 patients carrying a single mutated MEFV allele. *Rheumatology*. 2009;48:840–842.
 370. Sohar E, Gafni J, Pras M, et al. Familial Mediterranean fever. A survey of 470 cases and review of the literature. *Am J Med*. 1967;43:227–253.
 371. Rocken C, Shakespeare A. Pathology, diagnosis and pathogenesis of AA amyloidosis. *Virchows Arch*. 2002;440:111–122.
 372. Camus D, Shinar Y, Amar S, et al. Silent carriage of two familial Mediterranean fever gene mutations in large families with only a single identified patient. *Clin Genet*. 2012;82:288–291.
 373. Toutou I, Sarkisian T, Medlej-Hashim M, et al. Country as the primary risk factor for renal amyloidosis in familial Mediterranean fever. *Arthritis Rheum*. 2007;56:1706–1712.
 374. Ben-Zvi I, Brandt B, Berkun Y, et al. The relative contribution of environmental and genetic factors to phenotypic variation in familial Mediterranean fever (FMF). *Gene*. 2012;491:260–263.
 375. Samuels J, Aksentijevich I, Torosyan Y, et al. Familial Mediterranean fever at the millennium. Clinical spectrum, ancient mutations, and a survey of 100 American referrals to the National Institutes of Health. *Medicine (Baltim)*. 1998;77:268–297.
 376. Deveci K, Gokakin AK, Senel S, et al. Cystatin C in serum as an early marker of renal involvement in Familial Mediterranean Fever patients. *Eur Rev Med Pharmacol Sci*. 2013;17:253–260.
 377. Twig G, Livneh A, Vivante A, et al. Mortality risk factors associated with familial Mediterranean fever among a cohort of 1.25 million adolescents. *Ann Rheum Dis*. 2014;73:704–709.
 378. El-Shanti HI. Familial mediterranean Fever and renal disease. *Saudi J Kidney Dis Transpl*. 2003;14:378–385.
 379. Obici L, Perfetti V, Palladini G, et al. Clinical aspects of systemic amyloid diseases. *Biochim Biophys Acta*. 2005;1753:11–22.
 380. Dember LM. Amyloidosis-associated kidney disease. *J Am Soc Nephrol*. 2006;17:3458–3471.
 381. Libbey CA, Skinner M, Cohen AS. Use of abdominal fat tissue aspirate in the diagnosis of systemic amyloidosis. *Arch Intern Med*. 1983;143:1549–1552.
 382. Sungur C, Sungur A, Ruacan S, et al. Diagnostic value of bone marrow biopsy in patients with renal disease secondary to familial Mediterranean fever. *Kidney Int*. 1993;44:834–836.
 383. Eliakim M, Rachmilewitz M, Rosenmann E, et al. Renal manifestations in recurrent polyserositis (familial Mediterranean fever). *Isr J Med Sci*. 1970;6:228–245.
 384. Gok F, Sari E, Erdogan O, et al. Familial Mediterranean fever and IgA nephropathy: case report and review of the literature. *Clin Nephrol*. 2008;70:62–64.
 385. Said R, Hamzeh Y, Said S, et al. Spectrum of renal involvement in familial Mediterranean fever. *Kidney Int*. 1992;41:414–419.
 386. Sozeri B, Mir S, Ertan P, et al. Rapidly progressive glomerulonephritis in a child with Henoch-Schönlein Vasculitis and familial Mediterranean fever. *Pediatr Rheumatol Online J*. 2009;7(8).
 387. Yalcinkaya F, Tumer N. Glomerular lesions other than amyloidosis in patients with familial Mediterranean fever. *Nephrol Dial Transplant*. 1999;14:21–23.
 388. Goldfinger SE. Colchicine for familial Mediterranean fever. *N Engl J Med*. 1972;287:1302.
 389. Zemer D, Revach M, Pras M, et al. A controlled trial of colchicine in preventing attacks of familial mediterranean fever. *N Engl J Med*. 1974;291:932–934.
 390. Mansfield E, Chae JJ, Komarow HD, et al. The familial Mediterranean fever protein, pyrin, associates with microtubules and colocalizes with actin filaments. *Blood*. 2001;98:851–859.
 391. Kiraz S, Ertenli I, Arici M, et al. Effects of colchicine on inflammatory cytokines and selectins in familial Mediterranean fever. *Clin Exp Rheumatol*. 1998;16:721–724.
 392. Chae JJ, Wood G, Richard K, et al. The familial Mediterranean fever protein, pyrin, is cleaved by caspase-1 and activates NF-kappaB through its N-terminal fragment. *Blood*. 2008;112:1794–1803.
 393. Manukyan G, Petrek M, Tomankova T, et al. Colchicine modulates expression of pro-inflammatory genes in neutrophils from patients with familial Mediterranean fever and healthy subjects. *J Biol Regul Homeost Agents*. 2013;27:329–336.
 394. Taskiran EZ, Cetinkaya A, Balci-Peynircioglu B, et al. The effect of colchicine on pyrin and pyrin interacting proteins. *J Cell Biochem*. 2012;113:3536–3546.
 395. Nuki G. Colchicine: its mechanism of action and efficacy in crystal-induced inflammation. *Curr Rheumatol Rep*. 2008;10:218–227.
 396. Lidar M, Yonath H, Shechter N, et al. Incomplete response to colchicine in M694V homozygote FMF patients. *Autoimmun Rev*. 2012;12:72–76.
 397. Ben-Zvi I, Krichely-Vachdi T, Feld O, et al. Colchicine-free remission in familial Mediterranean fever: featuring a unique subset of the disease—a case control study. *Orphanet J Rare Dis*. 2014;9(3).
 398. Tufan A, Babaoglu MO, Akdogan A, et al. Association of drug transporter gene ABCB1 (MDR1) 3435C to T polymorphism with colchicine response in familial Mediterranean fever. *J Rheumatol*. 2007;34:1540–1544.
 399. Hoffman HM, Rosengren S, Boyle DL, et al. Prevention of cold-associated acute inflammation in familial cold autoinflammatory syndrome by interleukin-1 receptor antagonist. *Lancet*. 2004;364:1779–1785.
 400. Seyahi E, Ozdogan H, Celik S, et al. Treatment options in colchicine resistant familial Mediterranean fever patients: thalidomide and etanercept as adjunctive agents. *Clin Exp Rheumatol*. 2006;24:S99–S103.

401. Ter Haar N, Lachmann H, Ozen S, et al. Treatment of autoinflammatory diseases: results from the Eurofever Registry and a literature review. *Ann Rheum Dis*. 2013;72:678–685.
402. Tunca M, Tankurt E, Akbaylar Akpınar H, et al. The efficacy of interferon alpha on colchicine-resistant familial Mediterranean fever attacks: a pilot study. *Br J Rheumatol*. 1997;36:1005–1008.
403. Tunca M, Akar S, Soyuturk M, et al. The effect of interferon alpha administration on acute attacks of familial Mediterranean fever: a double-blind, placebo-controlled trial. *Clin Exp Rheumatol*. 2004;22:S37–S40.
404. Tweezer-Zaks N, Rabinovich E, Lidar M, et al. Interferon-alpha as a treatment modality for colchicine-resistant familial Mediterranean fever. *J Rheumatol*. 2008;35:1362–1365.
405. Mor A, Pilling MH, Kishimoto M, et al. Familial Mediterranean fever successfully treated with etanercept. *J Clin Rheumatol*. 2007;13:38–40.
406. Ozgocmen S, Ozcakar L, Ardicoglu O, et al. Familial Mediterranean fever responds well to infliximab: single case experience. *Clin Rheumatol*. 2006;25:83–87.
407. Belkhir R, Moulouguet-Doleris L, Hachulla E, et al. Treatment of familial Mediterranean fever with anakinra. *Ann Intern Med*. 2007;146:825–826.
408. Moser C, Pohl G, Haslinger I, et al. Successful treatment of familial Mediterranean fever with Anakinra and outcome after renal transplantation. *Nephrol Dial Transplant*. 2009;24:676–678.
409. Stankovic Stojanovic K, Delmas Y, Torres PU, et al. Dramatic beneficial effect of interleukin-1 inhibitor treatment in patients with familial Mediterranean fever complicated with amyloidosis and renal failure. *Nephrol Dial Transplant*. 2012;27:1898–1901.
410. Lachmann HJ, Kone-Paut I, Kuemmerle-Deschner JB, et al. Use of canakinumab in the cryopyrin-associated periodic syndrome. *N Engl J Med*. 2009;360:2416–2425.
411. van der Hilst J, Moutschen M, Messiaen PE, et al. Efficacy of anti-IL-1 treatment in familial Mediterranean fever: a systematic review of the literature. *Biologics*. 2016;10:75–80.
412. Hoffman HM, Throne ML, Amar NJ, et al. Efficacy and safety of rilonacept (interleukin-1 Trap) in patients with cryopyrin-associated periodic syndromes: results from two sequential placebo-controlled studies. *Arthritis Rheum*. 2008;58:2443–2452.
413. Bergesio F, Ciciani AM, Manganaro M, et al. Renal involvement in systemic amyloidosis: an Italian collaborative study on survival and renal outcome. *Nephrol Dial Transplant*. 2008;23:941–951.
414. Altıparmak MR, Pamuk ON, Ataman R, et al. Continuous ambulatory peritoneal dialysis in familial Mediterranean fever amyloidosis patients with end-stage renal failure: a single-centre experience from Turkey. *Nephron Clin Pract*. 2004;98:c119–c123.
415. Sahin S, Sahin GM, Ergin H, et al. The effect of dialytic modalities on clinical outcomes in ESRD patients with familial Mediterranean fever. *Ren Fail*. 2007;29:315–319.
416. Celik A, Saglam F, Dolek D, et al. Outcome of kidney transplantation for renal amyloidosis: a single-center experience. *Transplant Proc*. 2006;38:435–439.
417. Keven K, Sengul S, Kutlay S, et al. Long-term outcome of renal transplantation in patients with familial Mediterranean fever amyloidosis: a single-center experience. *Transplant Proc*. 2004;36:2632–2634.
418. Sherif AM, Refaie AF, Sobh MA, et al. Long-term outcome of live donor kidney transplantation for renal amyloidosis. *Am J Kidney Dis*. 2003;42:370–375.
419. Green H, Lichtenberg S, Rahamimov R, et al. Familial mediterranean fever is associated with increased mortality after kidney transplantation-A 19 Years' single center experience. *Transplantation*. 2017;101:2621–2626.
420. Heering P, Hetzel R, Grabensee B, et al. Renal transplantation in secondary systemic amyloidosis. *Clin Transplant*. 1998;12:159–164.
421. Ozdemir BH, Ozdemir FN, Sezer S, et al. Among therapy modalities of end-stage renal disease, renal transplantation improves survival in patients with amyloidosis. *Transplant Proc*. 2006;38:432–434.
422. Maes BD, Dalle I, Geboes K, et al. Erosive enterocolitis in mycophenolate mofetil-treated renal-transplant recipients with persistent afebrile diarrhea. *Transplantation*. 2003;75:665–672.
423. Alasfar S, Zakaria W, Nasan KH, et al. Dealing with dramatic health care problems during times of natural disaster and armed conflict. *Kidney Int*. 2023;104:221–224.
424. Thomas B, Wulf S, Bikbov B, et al. Maintenance dialysis throughout the world in years 1990 and 2010. *J Am Soc Nephrol*. 2015;26:2621–2633.
425. Shigidi MM, Fituri OM, Chandy SK, et al. Peritoneal dialysis, an expanding mode of renal replacement therapy in Qatar. *Saudi J Kidney Dis Transpl*. 2011;22:587–593.
426. Taher A, Enaya A, Sarsour A, et al. Growing a peritoneal dialysis program in Palestine from zero to 178 patients in 5 years: a single-center experience. *BMC Nephrol*. 2023;24(21).
427. Al Sahlawi M, AlRukhaimi M, Al-Ghamdi SM, et al. Peritoneal dialysis in the Arabian Gulf countries: challenges and opportunities. *Perit Dial Int*. 2024;44:171–176.
428. Ersoy FF. The TULIP story. *Perit Dial Int*. 2008;28:220–223.
429. Utaş C, Turkish Multicenter Peritoneal Dialysis Study G. Patient and technique survival on CAPD in Turkey. *Perit Dial Int*. 2001;21:602–606.
430. Akpolat T. Turkish Multicentre CSG: CAPD: a control strategy to prevent spread of HCV infection in end-stage renal disease. *Perit Dial Int*. 2001;21:77–79.
431. Akpolat T, Dilek M, Yavuz M, et al. Low seroconversion rates in CAPD patients compared to hemodialysis patients: potential advantages for transplant candidates. *Perit Dial Int*. 2002;22:520–523.
432. Al-wakeel J, Al-Ghonaim M, Al-Suwaidi A, et al. Peritoneal membrane characteristics in patients on peritoneal dialysis. *Saudi J Kidney Dis Transpl*. 2011;22:49–53.
433. Goldman S, Chan CT, Einbinder Y, et al. Nephrologists' perspectives on home dialysis utilization: a national survey from Israel. *Kidney Med*. 2023;5:100680.
434. Ghonimi T, Hamad A, Fouda T, et al. Cardiovascular calcification in hemodialysis patients: a Qatar-based prevalence and risk factors study. *Qatar Med J*. 2024;18:2024.
435. Riaz P, Caskey F, McIsaac M, et al. Workforce capacity for the care of patients with kidney failure across world countries and regions. *BMJ Glob Health*. 2021;6:e004014.
436. Hamadneh SA, Nueirat SA, Qadoomi J, et al. Vascular access mortality and hospitalization among hemodialysis patients in Palestine. *Saudi J Kidney Dis Transpl*. 2018;29:120–126.
437. Ureña-Torres P, Bieber B, Guebre-Egziabher F, et al. Citric acid-containing dialysate and survival rate in the dialysis outcomes and practice patterns study. *Kidney360*. 2021;2:666–673.
438. AlSahow A, Muenz D, Al-Ghonaim MA, et al. Kt/V: achievement, predictors and relationship to mortality in hemodialysis patients in the Gulf Cooperation Council countries: results from DOPPS (2012–18). *Clin Kidney J*. 2021;14:820–830.
439. Abouchacra S, Obaidli A, Al-Ghamdi SM, et al. Gulf Cooperation Council-dialysis outcomes and practice patterns study: an overview of anemia management trends at the regional and country specific levels in the Gulf Cooperation Council countries. *Saudi J Kidney Dis Transpl*. 2016;27:51–61.
440. Al SI, AlRukhaimi M, AlSahow A, et al. Mineral bone disorder and its management among hemodialysis patients in the Gulf Cooperation Council: initial findings from the dialysis outcomes and practice patterns study (2012–2015). *Saudi J Kidney Dis Transpl*. 2016;27:62–80.
441. Al-Ali FS, Bieber BA, Pisoni RL, et al. Nutritional status and outcomes in hemodialysis patients from the Gulf Cooperation Council countries enrolled in the dialysis outcome and practice patterns study phase 5 (2012–2015). *Saudi J Kidney Dis Transpl*. 2016;27:S31–S41.
442. AlYousef A, AlGhareeb S, Al Wakeel J, et al. Hemodialysis delivery, dialysis dose achievement, and vascular access types in hemodialysis patients from the Gulf Cooperation Council countries enrolled in the dialysis outcomes and practice patterns study phase 5 (2012–2015). *Saudi J Kidney Dis Transpl*. 2016;27:S42–S50.
443. Shaheen FA, Al WJ, Al-Ghamdi SM, et al. Cardiovascular and cerebrovascular comorbidities in hemodialysis patients from the Gulf Cooperation Council countries enrolled in the dialysis outcome and practice pattern study phase 5 (2012–2015). *Saudi J Kidney Dis Transpl*. 2016;27:S24–S30.
444. Karah N, Kurhani MN, Fadloun E, et al. Dialysis therapy for Syrian refugees in Lebanon: a call for action. *Bull World Health Organ*. 2018;96:370.
445. Barsoum RS. Overview: end-stage renal disease in the developing world. *Artif Organs*. 2002;26:737–746.
446. Barsoum RS. Chronic kidney disease in the developing world. *N Engl J Med*. 2006;354:997–999.

447. Bahadi A, El Farouki MR, Zajjari Y, et al. Initiating hemodialysis in Morocco: impact of late referral. *Nephrol Ther.* 2017;13:525–531.
448. Ahmed AM, Allam MF, Habil ES, et al. Compliance with haemodialysis practice guidelines in Egypt. *East Mediterr Health J.* 2013;19:4–9.
449. Al-Naseri SK, Mahdi ZM, Hashim MF. Quality of water in hemodialysis centers in Baghdad, Iraq. *Hemodial Int.* 2013;17:517–522.
450. Dialysis in the Kingdom of Saudi Arabia. *Saudi J Kidney Dis Transpl.* 2013;24:853–861.
451. Alavian SM, Bagheri-Lankarani K, Mahdavi-Mazdeh M, et al. Hepatitis B and C in dialysis units in Iran: changing the epidemiology. *Hemodial Int.* 2008;12:378–382.
452. Amini M, Aghighi M, Masoudkabar F, et al. Hemodialysis adequacy and treatment in Iranian patients: a national multicenter study. *Iran J Kidney Dis.* 2011;5:103–109.
453. Affi A. Renal osteodystrophy in developing countries. *Artif Organs.* 2002;26:767–769.
454. Schrooten I, Elseviers MM, Lamberts LV, et al. Increased serum strontium levels in dialysis patients: an epidemiological survey. *Kidney Int.* 1999;56:1886–1892.
455. Affi A, El-Sayed H, El-Setouhi M, et al. Hyperphosphatemia among end-stage renal disease patients in developing countries: a forgotten issue? *Hemodial Int.* 2005;9:409–415.
456. Souqiyeh MZ, Shaheen FA. Survey of attitudes of physicians toward the current evaluation and treatment of chronic kidney disease-mineral and bone disorder (CKD-MBD). *Saudi J Kidney Dis Transpl.* 2010;21:93–101.
457. Al RM, Al Sahow A, Boobes Y, et al. Adaptation and implementation of the “Kidney Disease: Improving Global Outcomes (KDIGO)” guidelines for evaluation and management of mineral and bone disorders in chronic kidney disease for practice in the Middle East countries. *Saudi J Kidney Dis Transpl.* 2014;25:133–148.
458. Messina JP, Humphreys I, Flaxman A, et al. Global distribution and prevalence of hepatitis C virus genotypes. *Hepatology.* 2015;61:77–87.
459. Lavanchy D. The global burden of hepatitis C. *Liver Int.* 2009;29(suppl 1):74–81.
460. Sunbul M. Hepatitis B virus genotypes: global distribution and clinical importance. *World J Gastroenterol.* 2014;20:5427–5434.
461. Te HS, Jensen DM. Epidemiology of hepatitis B and C viruses: a global overview. *Clin Liver Dis.* 2010;14:1–21, vii.
462. Mahmud S, Al-Kanaani Z, Chemaitelly H, et al. Hepatitis C virus genotypes in the Middle East and North Africa: distribution, diversity, and patterns. *J Med Virol.* 2018;90:131–141.
463. Harfouche M, Chemaitelly H, Kouyoumjian SP, et al. Hepatitis C virus viremic rate in the Middle East and North Africa: systematic synthesis, meta-analyses, and meta-regressions. *PLoS One.* 2017;12:e0187177.
464. Harfouche M, Chemaitelly H, Mahmud S, et al. Epidemiology of hepatitis C virus among hemodialysis patients in the Middle East and North Africa: systematic syntheses, meta-analyses, and meta-regressions. *Epidemiol Infect.* 2017;145:3243–3263.
465. Fehr T, Ambuhl PM. Chronic hepatitis virus infections in patients on renal replacement therapy. *Nephrol Dial Transplant.* 2004;19:1049–1053.
466. Köhler H. The prevalence of hepatitis C in different countries of the ERA/EDTA area. *Nephrol Dial Transplant.* 1995;10:468–469.
467. Rahnnavardi M, Hosseini Moghaddam SM, Alavian SM. Hepatitis C in hemodialysis patients: current global magnitude, natural history, diagnostic difficulties, and preventive measures. *Am J Nephrol.* 2008;28:628–640.
468. Barsoum RS, William EA, Khalil SS. Hepatitis C and kidney disease: a narrative review. *J Adv Res.* 2017;8:113–130.
469. Liang TJ, Ghany MG: current and future therapies for hepatitis C virus infection. *N Engl J Med.* 2013;368:1907–1917.
470. Chung RT, Baumert TF. Curing chronic hepatitis C—the arc of a medical triumph. *N Engl J Med.* 2014;370:1576–1578.
471. Frank C, Mohamed MK, Strickland GT, et al. The role of parenteral antischistosomal therapy in the spread of hepatitis C virus in Egypt. *Lancet.* 2000;355:887–891.
472. Centers for Disease C. Prevention: progress toward prevention and control of hepatitis C virus infection—Egypt, 2001–2012. *MMWR Morb Mortal Wkly Rep.* 2012;61:545–549.
473. Mohamoud YA, Mumtaz GR, Riome S, et al. The epidemiology of hepatitis C virus in Egypt: a systematic review and data synthesis. *BMC Infect Dis.* 2013;13:288.
474. Elmowafy AY, El Maghrabi HM, Eldahshan KF, et al. Treatment of hepatitis C infection among Egyptian hemodialysis patients: the dream becomes a reality. *Int Urol Nephrol.* 2019;51:1639–1647.
475. Saadi G, Kalantar-Zadeh K, Almasio P, et al. Hepatitis C virus infection and global kidney health: the consensus proceedings of the International Federation of Kidney Foundations. *Afr J Nephrol.* 2020;23:159–168.
476. Shiha G, Esmat G, Hassany M, et al. Ledipasvir/sofosbuvir with or without ribavirin for 8 or 12 weeks for the treatment of HCV genotype 4 infection: results from a randomised phase III study in Egypt. *Gut.* 2019;68:721–728.
477. Naguib H, Abouelnaga SF, Elsayed MM. Occult hepatitis C virus infection in hemodialysis patients who achieved a sustained virological response to directly acting antiviral drugs: is it a concern? *Int Urol Nephrol.* 2024;56:217–222.
478. Serwah MA, Omar SA, Khedr MS, et al. Occurrence and clinical characteristics of occult hepatitis C virus infection in hemodialysis units at Ismailia, Egypt. *Eur J Gastroenterol Hepatol.* 2021;33:1009–1014.
479. Al Balwi MA. Prevalence of mixed hepatitis C virus (HCV) genotypes among recently diagnosed dialysis patients with HCV infection. *Saudi J Kidney Dis Transpl.* 2011;22:712–716.
480. Karkar A. Hepatitis C in dialysis units: the Saudi experience. *Hemodial Int.* 2007;11:354–367.
481. Goodkin DA, Bieber B, Jadoul M, et al. Mortality, hospitalization, and quality of life among patients with hepatitis C infection on hemodialysis. *Clin J Am Soc Nephrol.* 2017;12:287–297.
482. Israel Ministry of Health and the Israel Society for Nephrology and Hypertension [ICDC Publication No. 350, December 2012, in Hebrew]. Available at http://www.isnh.org.il/RRTI1990_2010.pdf.
483. Sayiner AA, Zeytinoglu A, Ozkahya M, et al. HCV infection in haemodialysis and CAPD patients. *Nephrol Dial Transplant.* 1999;14:256–257.
484. Al-Khader AA, Shaheen FA. Posttransplant complications encountered in renal transplantation in the Middle East. *Transplant Proc.* 2004;36:180–183.
485. Abou Rached A, El Khoury L, El Imad T, et al. Incidence and prevalence of hepatitis B and hepatitis C viruses in hemodialysis patients in Lebanon. *World J Nephrol.* 2016;5:101–107.
486. El-kader YE-OA, Elmanama AA, Ayesha BM. Prevalence and risk factors of hepatitis B and C viruses among haemodialysis patients in Gaza strip, Palestine. *Virol J.* 2010;7:210.
487. Burdick RA, Bragg-Gresham JL, Woods JD, et al. Patterns of hepatitis B prevalence and seroconversion in hemodialysis units from three continents: the DOPPS. *Kidney Int.* 2003;63:2222–2229.
488. Fissell RB, Bragg-Gresham JL, Woods JD, et al. Patterns of hepatitis C prevalence and seroconversion in hemodialysis units from three continents: the DOPPS. *Kidney Int.* 2004;65:2335–2342.
489. Van Biesen W, Vanholder R, Vanderhaegen B, et al. Renal replacement therapy for refugees with end-stage kidney disease: an international survey of the nephrological community. *Kidney Int Suppl.* 2016;6:35–41.
490. Forzley M:I. Advancing the health of arab Americans. *Ethn Dis.* 2005;15:90–92.
491. Fogazzi GB, Castelnovo C. Maintenance dialysis in patients from developing countries: the experience of an Italian center. *J Nephrol.* 2004;17:552–558.
492. Shaheen FA. Organ donation in the Middle East countries. *Ethn Dis.* 2009;19:S1–16–17.
493. Mohsin N, Militsala E, Budruddin M, et al. Attitude of the Omani population toward organ transplantation. *Transplant Proc.* 2010;42:4305–4308.
494. Paris W, Nour B. Organ transplantation in Egypt. *Prog Transplant.* 2010;20:274–278.
495. Rizvi SANS, Hussain Z, et al. Renal transplantation in developing countries. *Kidney Int Suppl.* 2003;63:S96–S100.
496. Mohsin NB, Metry A. Reasons of preclusion of living-related donor renal transplants in Oman. *Exp Clin Transplant.* 2010;8:303–306.
497. Muralidharan A, White S. The need for kidney transplantation in low- and middle-income countries in 2012: an epidemiological perspective. *Transplantation.* 2015;99:476–481.
498. Abderrahim E, Zammouri A, Bacha MM, et al. Thirty years of experience at the first Tunisian kidney transplant center. *Exp Clin Transplant.* 2017;15:84–89.

499. Einollahi B. Iranian experience with the non-related renal transplantation. *Saudi J Kidney Dis Transpl.* 2004;15:421–428.
500. Ehtuish EF, Abouna GM, Shebani AH, et al. Kidney transplantation in Libya: a North African and middle eastern perspective. *Exp Clin Transplant.* 2006;4:425–428.
501. Einollahi B, Taheri S. Renal transplantation practice in Iran and the Middle East: report from Iran and a review of the literature. *Ann Transplant.* 2008;13:5–14.
502. El-Nono IH, Al-Ba'adani TH, Ghilan AM, et al. Adult-to-adult living related donor renal transplantation in Yemen: the first experience. *Saudi J Kidney Dis Transpl.* 2007;18:265–269.
503. El-Nono IH, Telha KA, Al-Alimy GM, et al. Challenges in renal transplantation in Yemen. *Ann Transplant.* 2015;20:92–96.
504. Masri MA, Haberal MA, Shaheen FA, et al. Middle East Society for Organ Transplantation (MESOT) transplant registry. *Exp Clin Transplant.* 2004;2:217–220.
505. Albar M. Organ transplantation: a Sunni Islamic perspective. *Saudi J Kidney Dis Transpl.* 2012;23:817–822.
506. Hamdy S. Not quite dead: why Egyptian doctors refuse the diagnosis of death by neurological criteria. *Theor Med Bioeth.* 2013;34:147–160.
507. Faraj W, Fakih H, Mukherji D, et al. Organ donation after cardiac death in the Middle East. *Transplant Proc.* 2010;42:713–715.
508. Cohen J, Rahamimov R, Hoffman A, et al. Derivation and implementation of a protocol in Israel for organ donation after cardio-circulatory death. *Isr Med Assoc J.* 2017;19:566–569.
509. Bulka RR. Jewish perspective on organ transplantation. *Transplant Proc.* 1990;22:945–946.
510. Ulshafer TR. A Catholic perspective on religion and organ transplantation. *Del Med J.* 1988;60:505–507.
511. Lavee J, Ashkenazi T, Stoler A, et al. Preliminary marked increase in the national organ donation rate in Israel following implementation of a new organ transplantation law. *Am J Transplant.* 2013;13:780–785.
512. Heidary Rouchi A, Mahdavi-Mazdeh M. Human tissue processing and transplantation in MESOT states: how to promote? *Int J Organ Transplant Med.* 2016;7:102–105.
513. SCOT Data. Organ donation and transplantation in the Kingdom of Saudi Arabia 2016. *Saudi J Kidney Dis Transpl.* 2017;28:1209–1214.
514. Horvat LD, Shariff SZ, Garg AX, et al. Global trends in the rates of living kidney donation. *Kidney Int.* 2009;75:1088–1098.
515. Noël L. The data on transplantation in Saudi Arabia are complex and require additional framework for interpretation. *Kidney Int.* 2009;76:914; author reply 914–915.
516. Saeed B. The effect of the Syrian crisis on organ transplantation in Syria. *Exp Clin Transplant.* 2015;13:206–208.
517. Haghighi AN, Ghahramani N. Living unrelated kidney donor transplantation in Iran. *Nat Clin Pract Nephrol.* 2006;2:E1–E1.
518. Ghods AJ. Renal transplantation in Iran. *Nephrol Dial Transplant.* 2002;17:222–228.
519. Ghods AJ, Savaj S. Iranian model of paid and regulated living-unrelated kidney donation. *Clin J Am Soc Nephrol.* 2006;1:1136–1145.
520. Simforoosh N, Basiri A, Fattahi MR, et al. Living unrelated versus living related kidney transplantation: 20 years' experience with 2155 cases. *Transplant Proc.* 2006;38:422–425.
521. Harmon W, Delmonico F. Payment for kidneys: a government-regulated system is not ethically achievable. *Clin J Am Soc Nephrol.* 2006;1:1146–1147.
522. Fallahzadeh MK, Jafari L, Roozbeh J, et al. Comparison of health status and quality of life of related versus paid unrelated living kidney donors. *Am J Transplant.* 2013;13:3210–3214.
523. Khatami SM, Mahdavi-Mazdeh M. Perfect is the enemy of good: the Iranian system of paid donation. *Am J Transplant.* 2014;14:1222–1223.
524. Randhawa G. Policy perspectives: international survey of nephrologists' perceptions of and attitudes towards rewards and compensation for kidney donation. *Nephrol Dial Transplant.* 2013;28:1343–1345.
525. Ghahramani N, Karparvar Z, Ghahramani M, et al. International survey of nephrologists' perceptions and attitudes about rewards and compensations for kidney donation. *Nephrol Dial Transplant.* 2013;28:1610–1621.
526. Shimazono Y. The state of the international organ trade: a provisional picture based on integration of available information. *Bull World Health Organ.* 2007;85:955–962.
527. Danovitch GM, Chapman J, Capron AM, et al. Organ trafficking and transplant tourism: the role of global professional ethical standards—the 2008 Declaration of Istanbul. *Transplantation.* 2013;95:1306–1312.
528. Delmonico FL, Dew MA. Living donor kidney transplantation in a global environment. *Kidney Int.* 2007;71:608–614.
529. Sever MS, Kazancioglu R, Yildiz A, et al. Outcome of living unrelated (commercial) renal transplantation. *Kidney Int.* 2001;60:1477–1483.
530. International Summit on Transplant T, Organ T. The declaration of Istanbul on organ trafficking and transplant tourism. *Clin J Am Soc Nephrol.* 2008;3:1227–1231.
531. International Summit on Transplant T, Organ T. The Declaration of Istanbul on organ trafficking and transplant tourism. *Kidney Int.* 2008;74:854–859.
532. Efrat A. The politics of combating the organ trade: lessons from the Israeli and Pakistani experience. *Am J Transplant.* 2013;13:1650–1654.
533. Moszynski P. Organ traffickers are targeting refugees in Egypt, warns charity. *Br Med J.* 2011;343:d8254.
534. Shaheen FA, Al-Khader AA. Preventive strategies of renal failure in the Arab world. *Kidney Int.* 2005;68:S37–S40.
535. Sen N, Turunc T, Karatasli M, et al. Tuberculosis in patients with end-stage renal disease undergoing dialysis in an endemic region of Turkey. *Transplant Proc.* 2008;40:81–84.
536. Naqvi A, Rizvi A, Hussain Z, et al. Developing world perspective of posttransplant tuberculosis: morbidity, mortality, and cost implications. *Transplant Proc.* 2001;33:1787–1788.
537. Abdelrahman M, Sinha AK, Karkar A. Tuberculosis in end-stage renal disease patients on hemodialysis. *Hemodial Int.* 2006;10:360–364.
538. Mahmoud IM, Sobh MA, El-Habashi AF, et al. Interferon therapy in hemodialysis patients with chronic hepatitis C: study of tolerance, efficacy and post-transplantation course. *Nephron Clin Pract.* 2005;100:c133–c139.
539. Samhan M, Al-Mousawi M, Donia F, et al. Malignancy in renal recipients. *Transplant Proc.* 2005;37:3068–3070.
540. Raeisi D, Payandeh M, Madani SH, et al. Kaposi's sarcoma after kidney transplantation: a 21-years experience. *Int J Hematol Oncol Stem Cell Res.* 2013;7:29–33.
541. Harzallah K, Abderrahim E, Chareffedine K, et al. Cancers after renal transplantation: multicenter experience. *Saudi J Kidney Dis Transpl.* 2008;19:825–830.
542. Euvrard S, Kanitakis J, Claudy A. Skin cancers after organ transplantation. *N Engl J Med.* 2003;348:1681–1691.
543. Levin A, Tonelli M, Bonventre J, et al. Global kidney health 2017 and beyond: a roadmap for closing gaps in care, research, and policy. *Lancet.* 2017;390:1888–1917.
544. Shaheen FA, Souqiyeh MZ. Kidney health in the Middle East. *Clin Nephrol.* 2010;74(suppl 1):S85–S88.

QUESTIONS

1. Which genetic kidney disease is likely to be prevalent in the Middle East?
 - a. Fabry disease
 - b. Alport syndrome due to *COL4A5* mutations
 - c. Congenital nephrotic syndrome of the Finnish type
 - d. Dent disease

Answer: c

Rationale: Congenital nephrotic syndrome of the Finnish type is a recessively inherited podocytopathy. Middle Eastern communities are characterized by a high rate of consanguinity and large families, resulting in rapid population growth. This setting increases the carrier rate of recessively inherited diseases and consequently the prevalence of affected individuals, but it has no bearing on X-linked inherited disorders (options a, b, and d).

2. Which statement describes correctly primary hyperoxaluria (PH) in the Middle East (ME)?
 - a. There is a 10-fold increase in the prevalence of PH2 in the ME compared with Europe.
 - b. There is a founder mutation in *HOGA1* leading to PH3 among Ashkenazi Jews.
 - c. The common phenotype of PH1 is of a first kidney stone during the sixth decade.
 - d. Most PH1 patients respond to pyridoxine and maintain preserved kidney function.

Answer: b

Rationale: PH1 may serve as a disease model for recessively inherited diseases in the ME. It is prevalent and quite often present with end-stage kidney disease (ESKD) during the first years of life. PH3 was described in Israel but subsequently identified in other communities around the world. There is a founder mutation among Ashkenazi Jews.